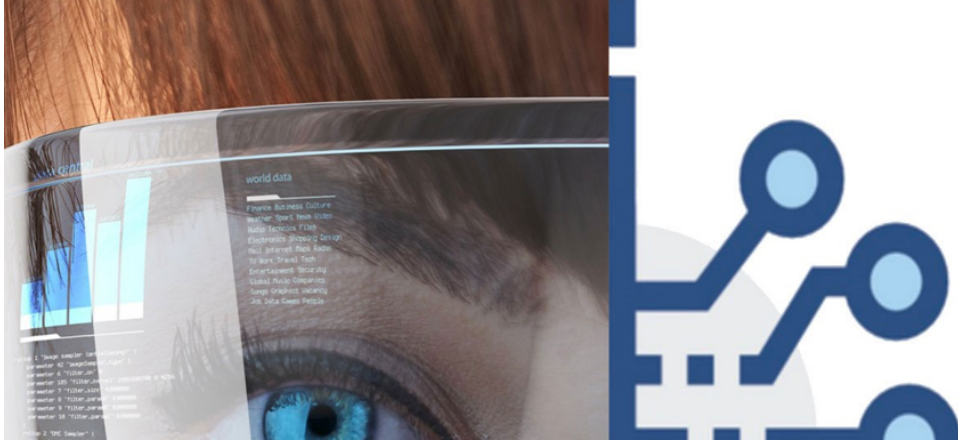




Multi-Asset Signal Generation

Principal Financial Group
Carnegie Mellon University

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1 Asset Classes

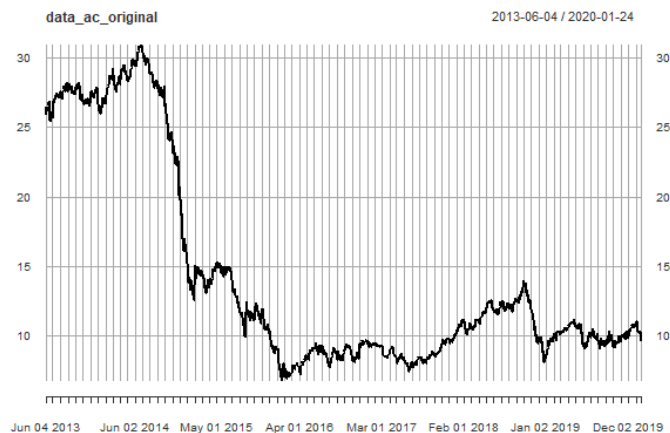
1.1 Summary of Asset Classes

Table 1: Commodity

	AssetClass	AC_ID	CSI_ID	Description	SOURCE	MOD_ID
14	Commodity	Commodity	CO_oil1	Commodities Oil 1	Principle	CO_oil1
15	Commodity	Commodity	CO_oil2	Commodities Oil 2	Principle	CO_oil2
16	Commodity	Commodity	CO_nat_gas	Commodities Natural Gas	Principle	CO_nat_gas
17	Commodity	Commodity	CO_gasoline	Commodities Gasoline	Principle	CO_gasoline
18	Commodity	Commodity	CO_gold	Commodities Gold	Principle	CO_gold
19	Commodity	Commodity	CO_silver	Commodities Silver	Principle	CO_silver
20	Commodity	Commodity	CO_grains	Commodities Grains	Principle	CO_grains

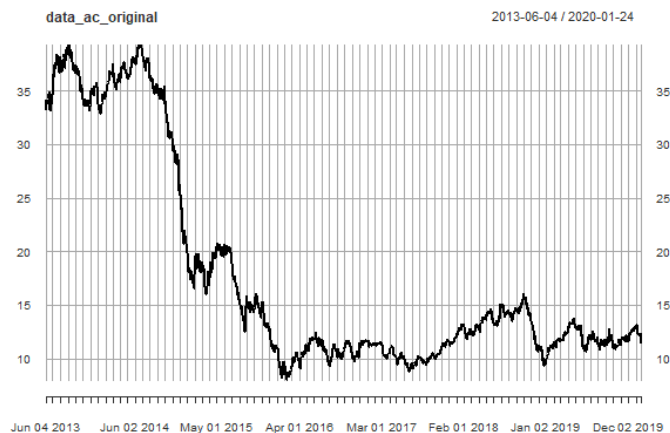
1.2 History: CO_{oil}

Current Version is $FI_{development.PriorVersionis}$.



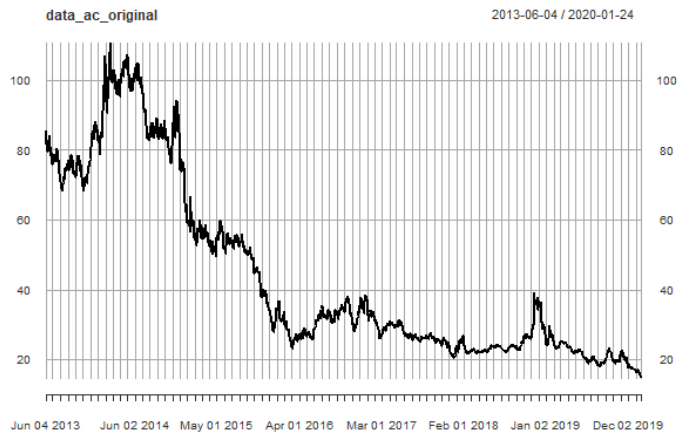
1.3 History: CO_{oil2}

Current Version is $FI_{development.PriorVersionis}$.



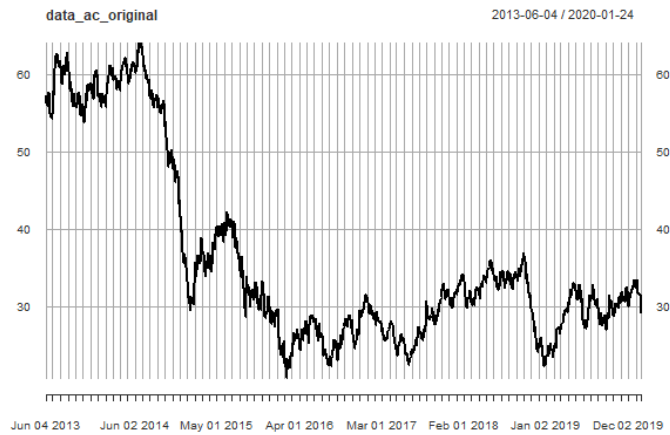
1.4 History: CO_{natgas}

Current Version is $FI_{development.PriorVersionis}$.



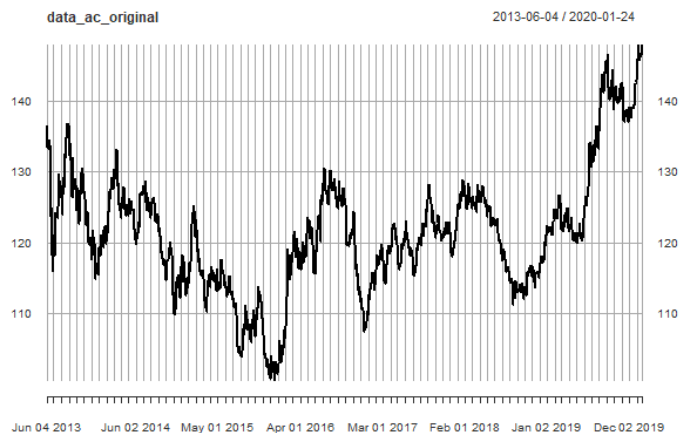
1.5 History: $CO_{gasoline}$

Current Version is $FI_{development.PriorVersionis}$.



1.6 History: CO_{gold}

Current Version is $FI_{development.PriorVersionis}$.



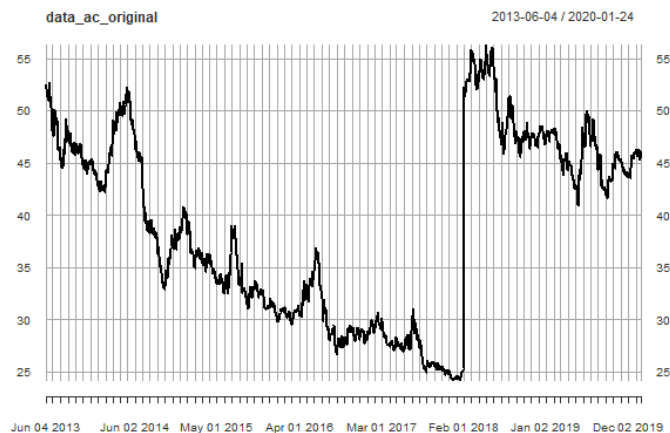
1.7 History: *CO_{silver}*

Current Version is *FI_{development.PriorVersionis}*.



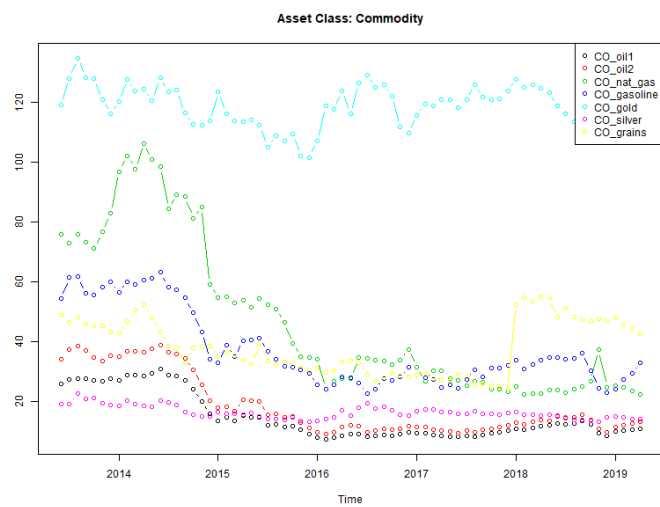
1.8 History: CO_{grains}

Current Version is $FI_{development.PriorVersionis}$.



2 Clusters Analysis of Asset Classes

2.1 Asset Class: *Commodity*



Commodity

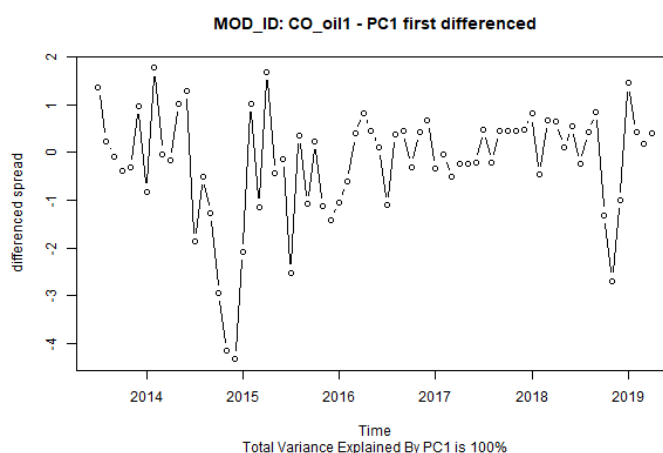
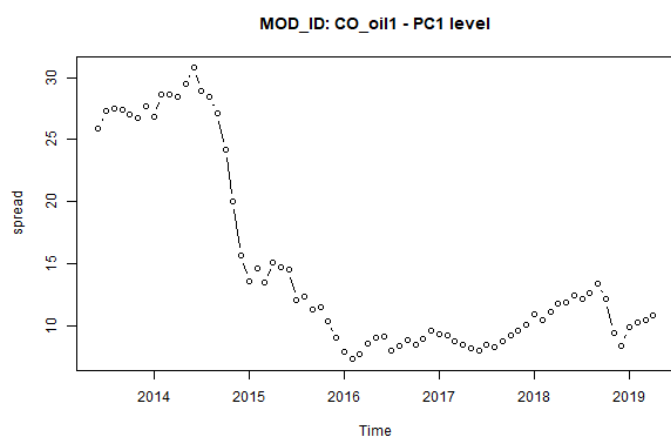
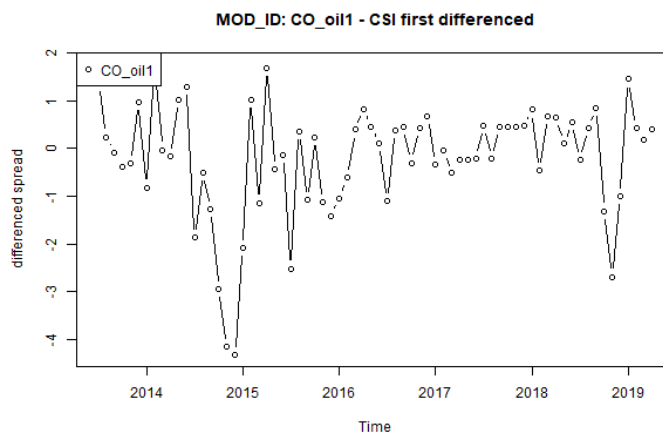
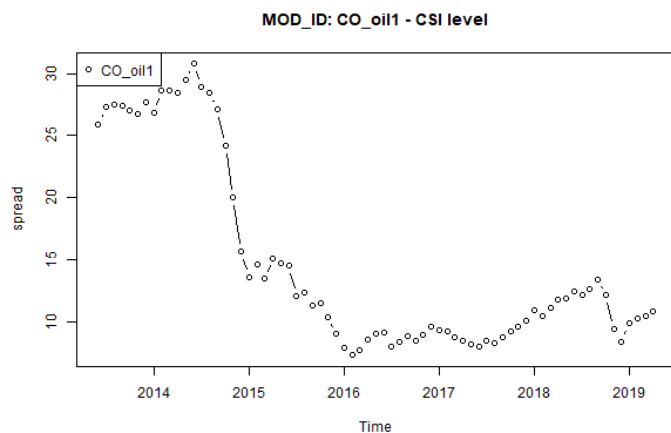
Variance%	CO_gasoline	CO_gold	CO_grains	CO_nat_gas	CO_oil1	CO_oil2	CO_silver	Spearman	Kendall
38.9%	1	1	1	1	1	1	1	-1%	-1%
53.7%	1	2	1	1	1	1	2	28%	20%
65.2%	1	3	2	2	1	1	3	31%	21%
91.6%	1	3	4	2	1	1	3	84%	65%
97.9%	3	4	5	2	1	1	4	84%	65%
97.9%	3	4	6	2	1	1	5	97%	87%
100%	4	5	7	3	1	2	6	100%	100%

Table 2: Cluster Analysis - Commodity

	1	2	3	4	5	6	7
CO_oil1	1	1	1	1	1	1	1
CO_oil2	1	1	1	1	1	1	2
CO_gasoline	1	1	1	1	3	3	4
CO_nat_gas	1	1	2	2	2	2	3
CO_grains	1	1	2	4	5	6	7
CO_gold	1	2	3	3	4	4	5
CO_silver	1	2	3	3	4	5	6
Variance% of PC1	38.9%	53.7%	65.2%	91.6%	97.9%	97.9%	100%
Retained Correlation	-1%	28%	31%	84%	84%	97%	100%

3 PCA for Clusters

3.1 Cluster: *CO_oil1*



	EIGENVECTOR	MULTIPLIER
CO_oil1	1.00	1.00

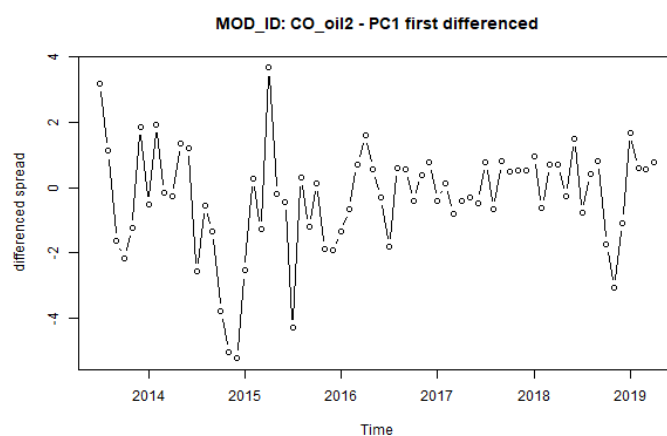
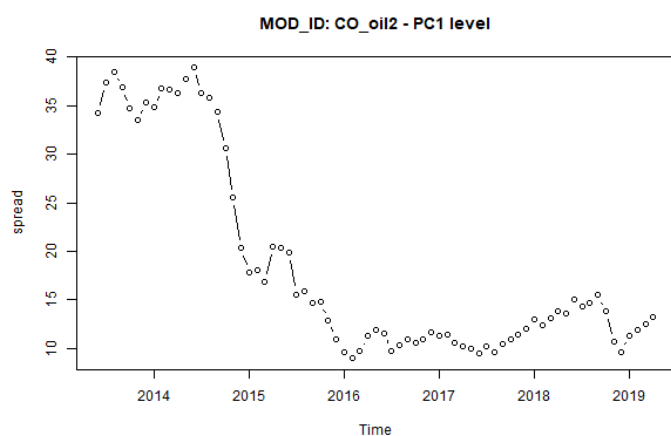
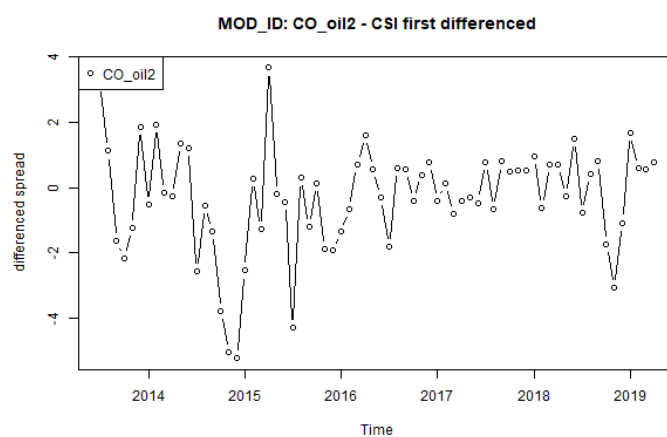
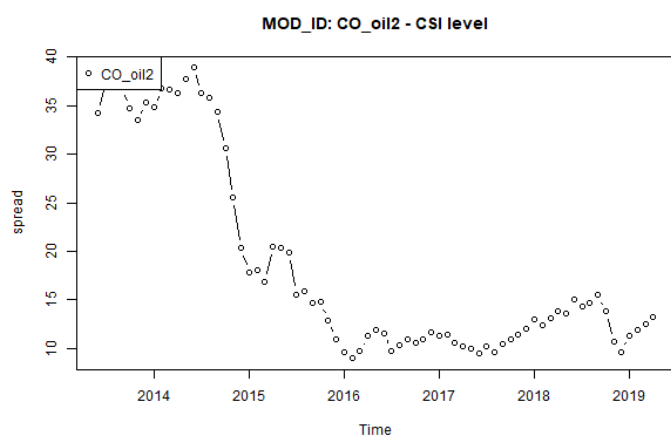
Kendall Correlation

	CO_oil1	pc1
CO_oil1	1	1
pc1	1	1

Spearman Correlation

	CO_oil1	pc1
CO_oil1	1	1
pc1	1	1

3.2 Cluster: *CO_oil2*



Total Variance Explained By PC1 is 100%

	EIGENVECTOR	MULTIPLIER
CO_oil2	1.00	1.00

Kendall Correlation

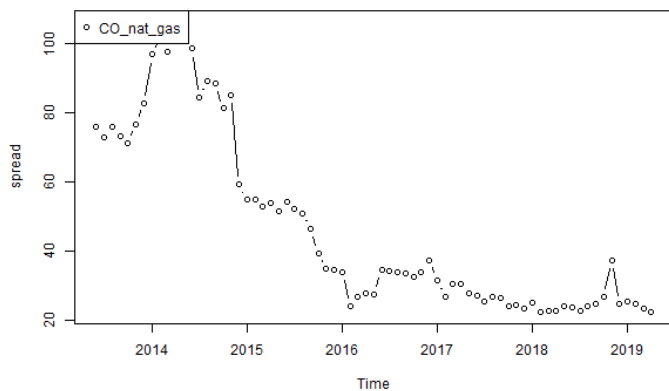
	CO_oil2	pc1
CO_oil2	1	1
pc1	1	1

Spearman Correlation

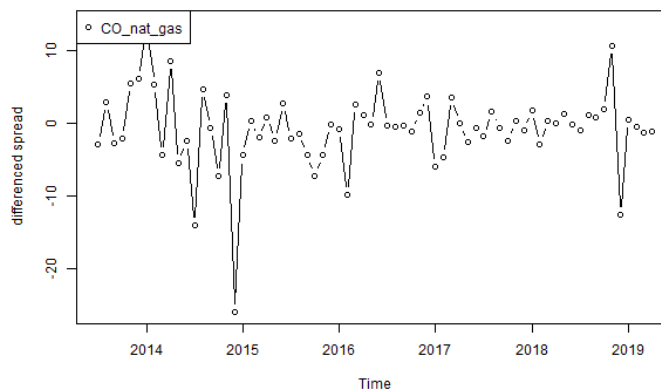
	CO_oil2	pc1
CO_oil2	1	1
pc1	1	1

3.3 Cluster: CO_{nat_gas}

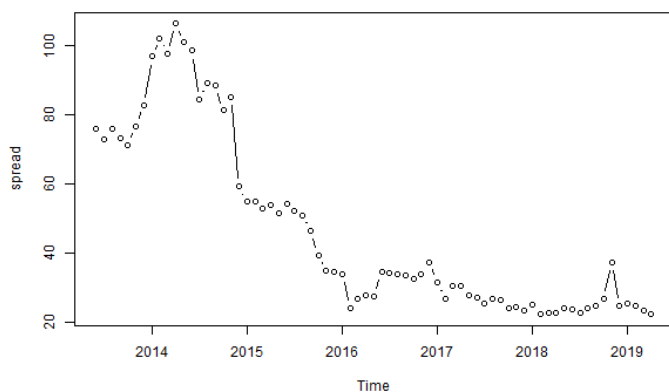
MOD_ID: CO_nat_gas - CSI level



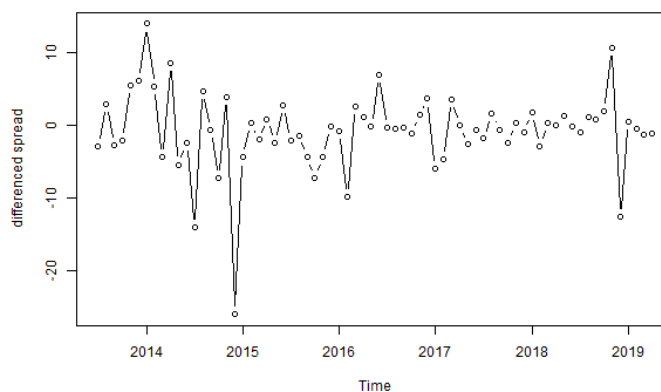
MOD_ID: CO_nat_gas - CSI first differenced



MOD_ID: CO_nat_gas - PC1 level



MOD_ID: CO_nat_gas - PC1 first differenced



Total Variance Explained By PC1 is 100%

	EIGENVECTOR	MULTIPLIER
CO_nat_gas	1.00	1.00

Kendall Correlation

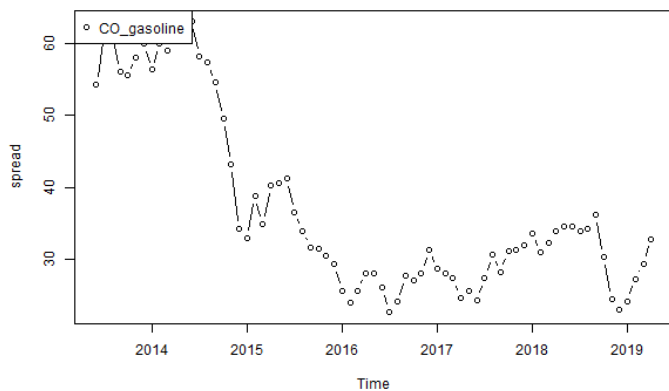
	CO_nat_gas	pc1
CO_nat_gas	1	1
pc1	1	1

Spearman Correlation

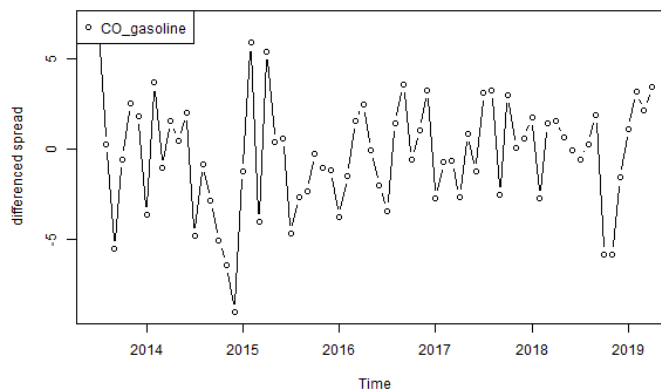
	CO_nat_gas	pc1
CO_nat_gas	1	1
pc1	1	1

3.4 Cluster: $CO_{gasoline}$

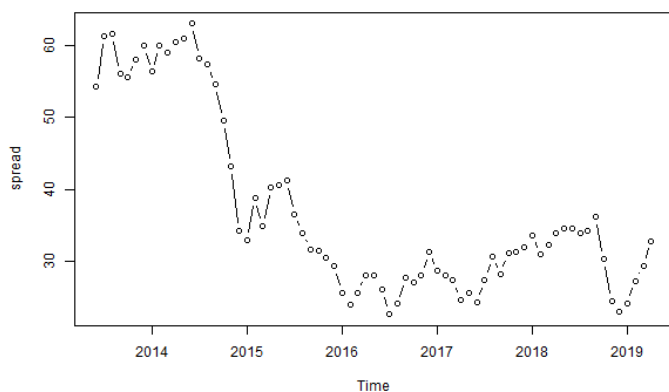
MOD_ID: CO_gasoline - CSI level



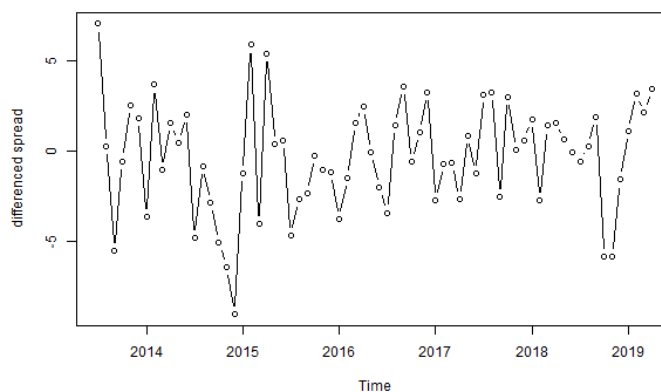
MOD_ID: CO_gasoline - CSI first differenced



MOD_ID: CO_gasoline - PC1 level



MOD_ID: CO_gasoline - PC1 first differenced



Total Variance Explained By PC1 is 100%

	EIGENVECTOR	MULTIPLIER
CO_gasoline	1.00	1.00

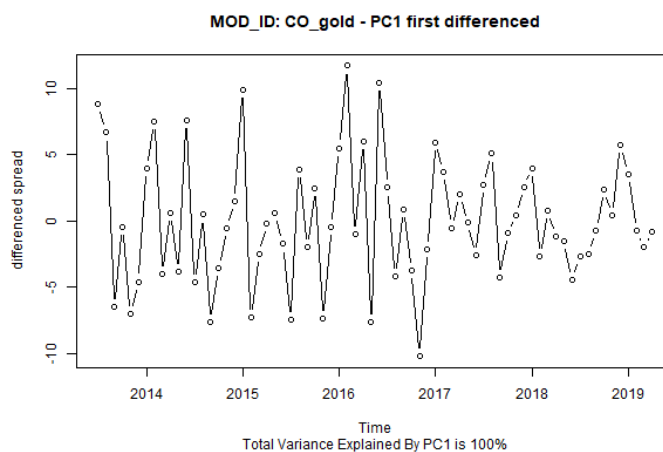
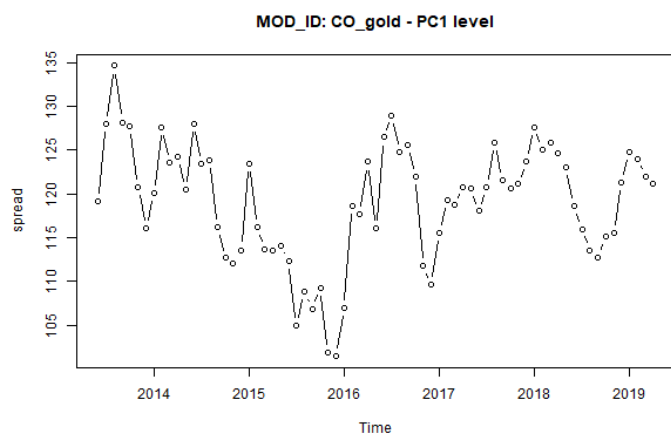
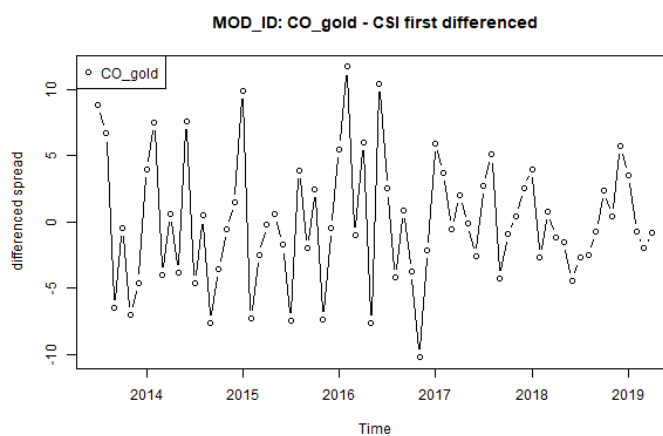
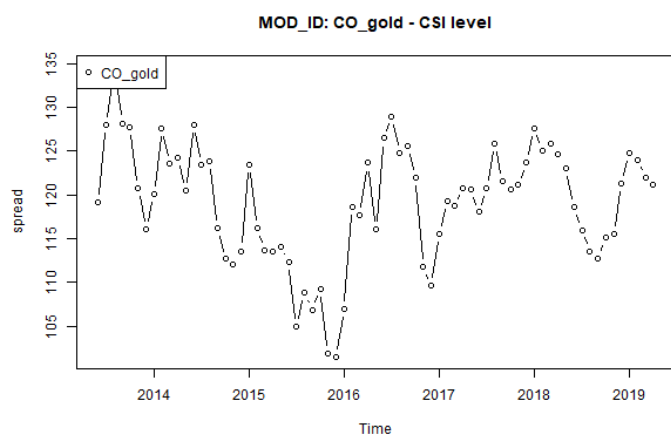
Kendall Correlation

	CO_gasoline	pc1
CO_gasoline	1	1
pc1	1	1

Spearman Correlation

	CO_gasoline	pc1
CO_gasoline	1	1
pc1	1	1

3.5 Cluster: CO_{gold}



	EIGENVECTOR	MULTIPLIER
CO_gold	1.00	1.00

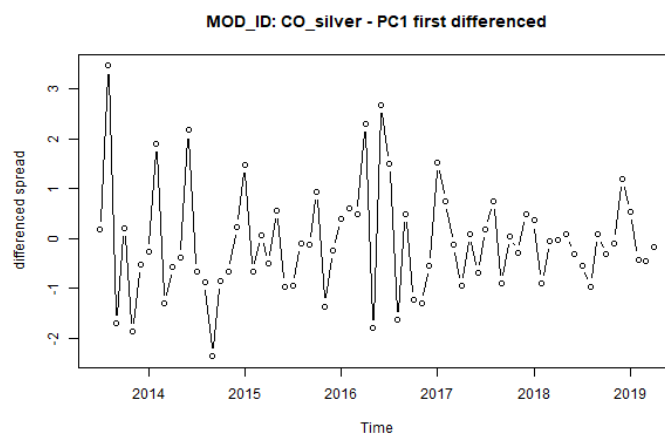
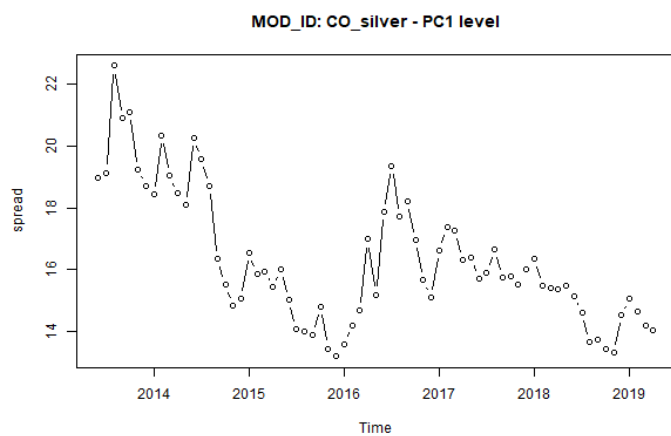
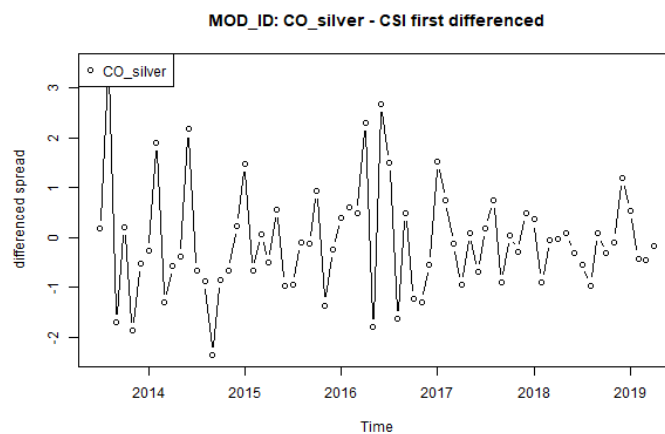
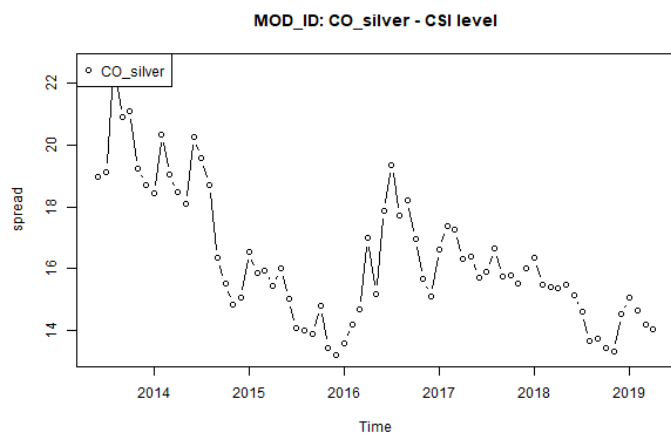
Kendall Correlation

	CO_gold	pc1
CO_gold	1	1
pc1	1	1

Spearman Correlation

	CO_gold	pc1
CO_gold	1	1
pc1	1	1

3.6 Cluster: *CO_{silver}*



Total Variance Explained By PC1 is 100%

	EIGENVECTOR	MULTIPLIER
CO_silver	1.00	1.00

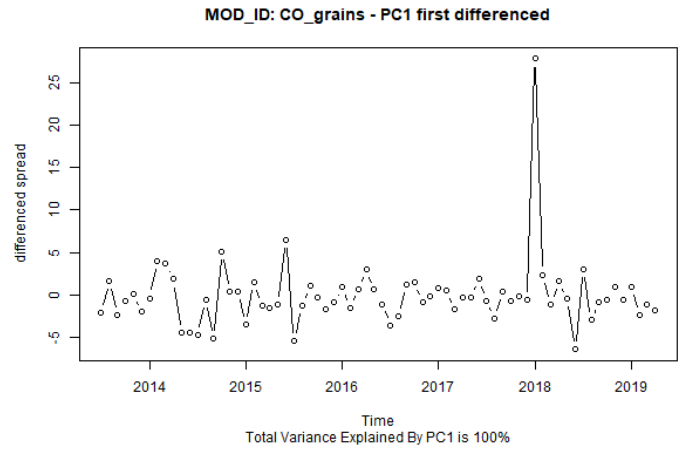
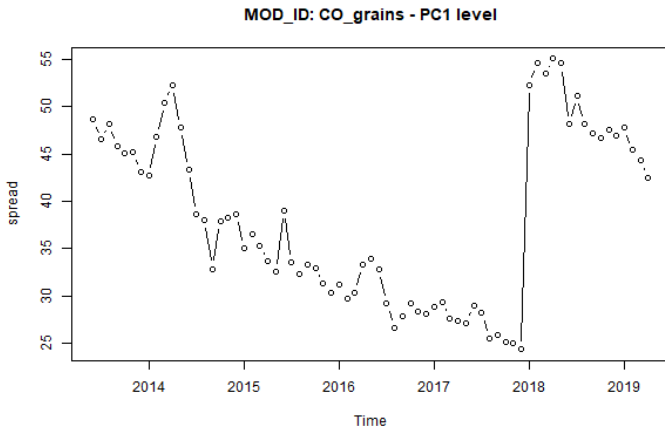
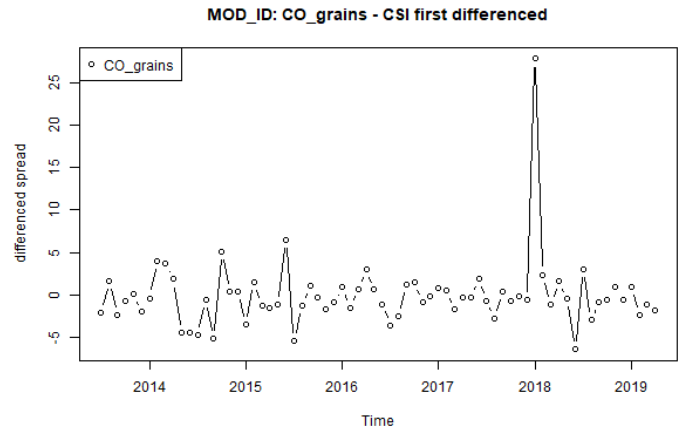
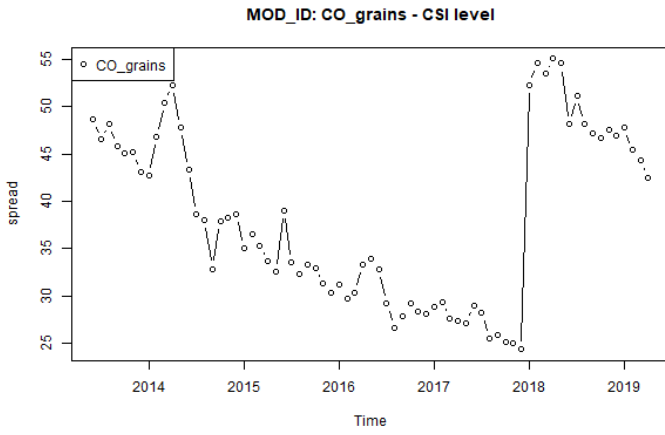
Kendall Correlation

	CO_silver	pc1
CO_silver	1	1
pc1	1	1

Spearman Correlation

	CO_silver	pc1
CO_silver	1	1
pc1	1	1

3.7 Cluster: CO_{grains}



	EIGENVECTOR	MULTIPLIER
CO_grains	1.00	1.00

Kendall Correlation

	CO_grains	pc1
CO_grains	1	1
pc1	1	1

Spearman Correlation

	CO_grains	pc1
CO_grains	1	1
pc1	1	1

4 Variable Selection

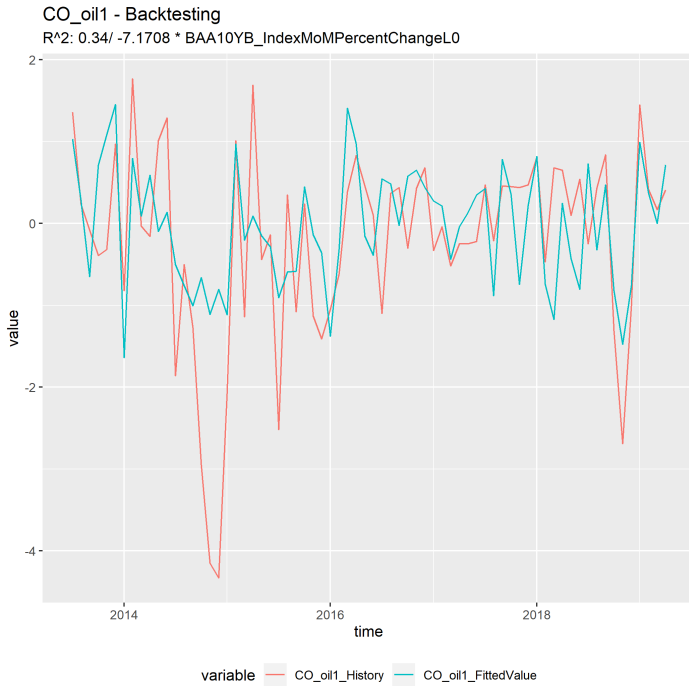
4.1 Single Factor Model for CO₂il1

	Variable	adjR2	AIC	Coef	Robst p	RMSE_OS	Criteria1	Criteria2	Criteria3
1	BAA10YB_IndexMoMPercentChangeL0	0.34	198.58	-7.17	0.00	0.71			
2	BAA10YB_IndexMoMDifferenceL0	0.29	203.93	-0.05	0.00	0.61			
3	USGG30YR_IndexMoMDifferenceL0	0.16	216.06	2.96	0.00	0.83			
4	WPSID61MoMDifferenceL0	0.16	216.17	0.36	0.03	0.94			
5	USGG30YR_IndexMoMPercentChangeL0	0.15	216.86	8.14	0.00	0.82			
6	WPSFD49502MoMDifferenceL0	0.13	218.55	0.26	0.04	0.91			
7	ISRATIOMoMPercentChangeL0	0.13	218.81	-63.13	0.04	0.84			
8	WPSFD49207MoMDifferenceL0	0.12	218.84	0.36	0.04	0.86			
9	ISRATIOMoMDifferenceL0	0.12	219.11	-46.25	0.04	0.84			
10	USGG10YR_IndexMoMDifferenceL0	0.12	219.11	2.41	0.00	0.76			
11	ism_man_pricesMoMDifferenceL0	0.12	219.37	0.09	0.05	0.84			
12	USGG10YR_IndexMoMPercentChangeL0	0.11	220.13	4.99	0.00	0.75			

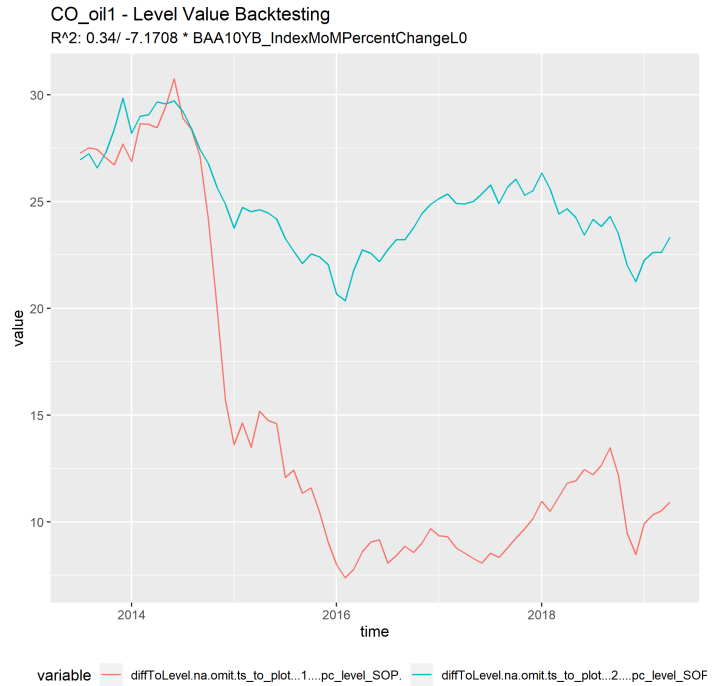
Table 3: Single Factor Analysis-CO₂il1

4.1.1 Model - 1

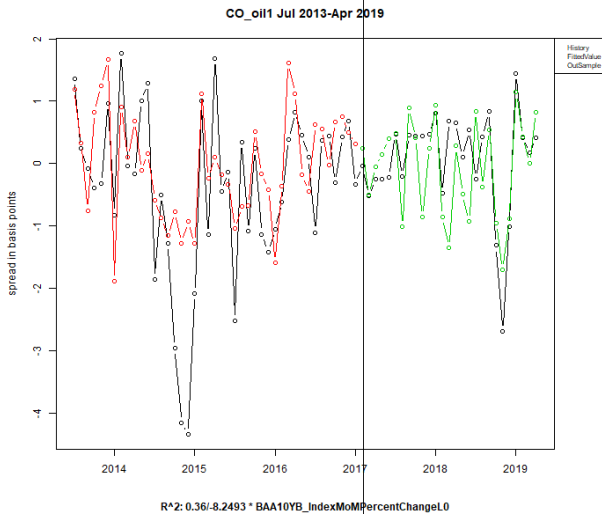
Backtesting on differenced values



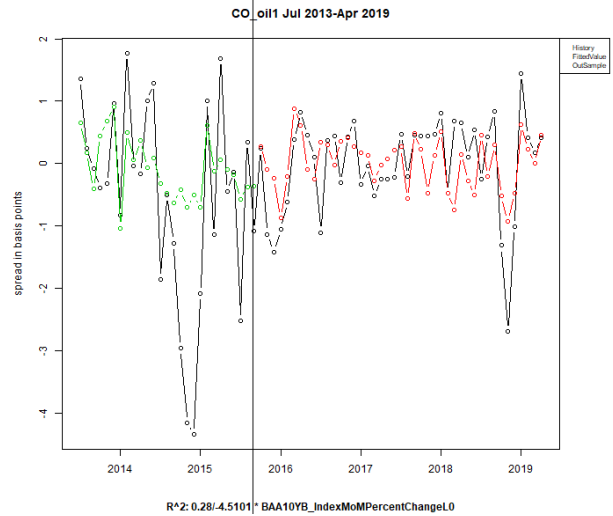
Backtesting on level values



Out-of-Sample testing

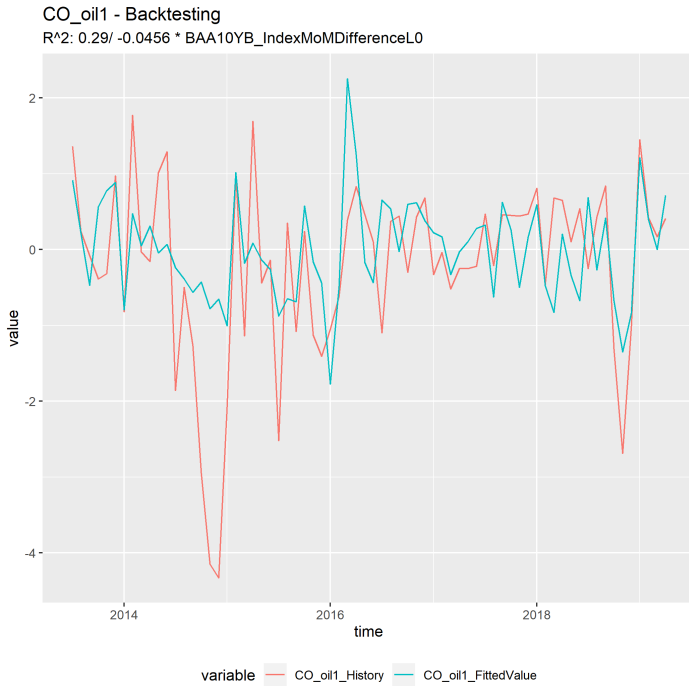


Out-of-Sample testing

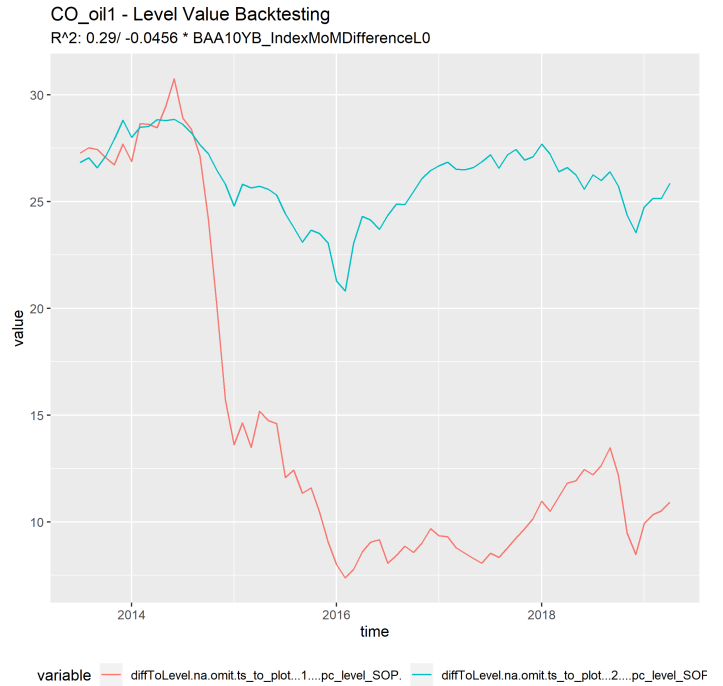


4.1.2 Model - 2

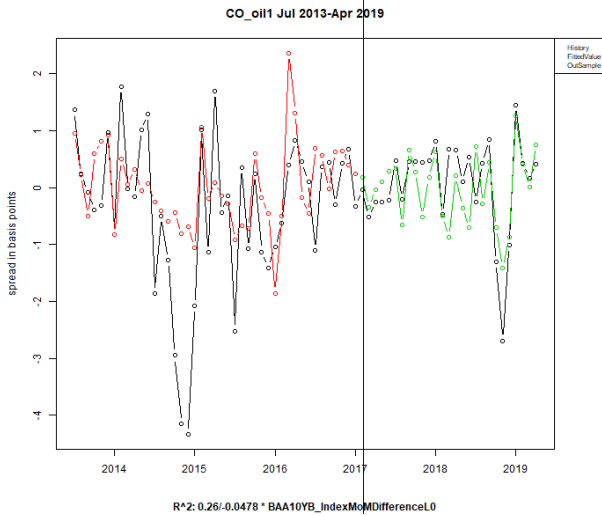
Backtesting on differenced values



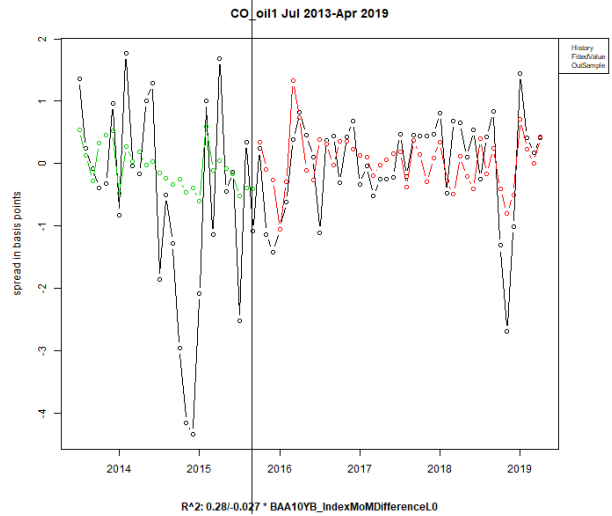
Backtesting on level values



Out-of-Sample testing

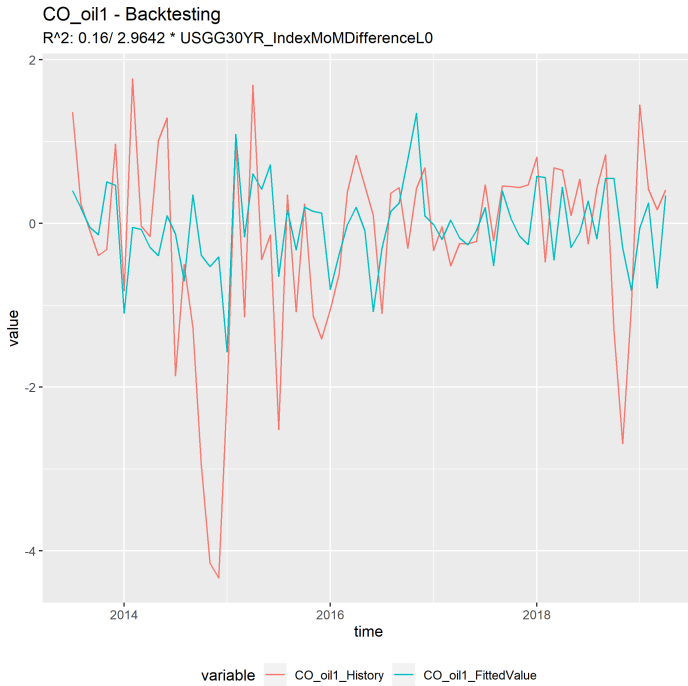


Out-of-Sample testing

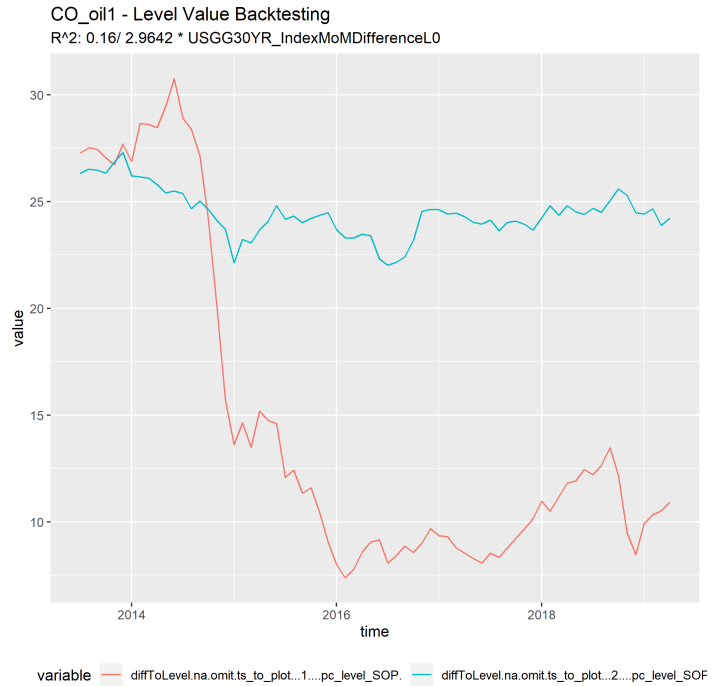


4.1.3 Model - 3

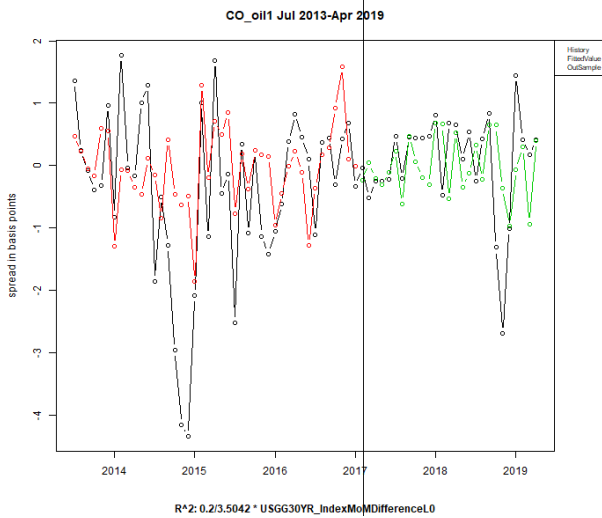
Backtesting on differenced values



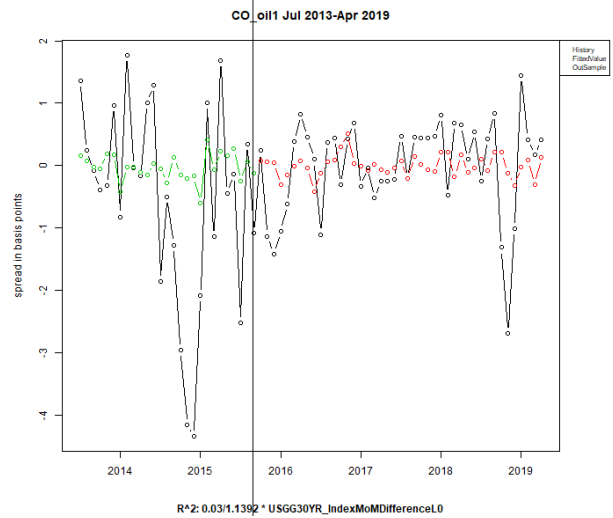
Backtesting on level values



Out-of-Sample testing

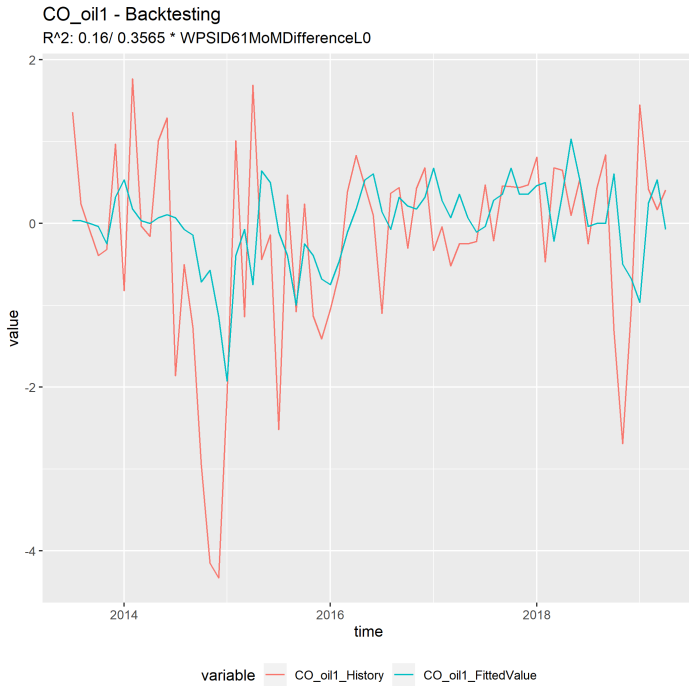


Out-of-Sample testing

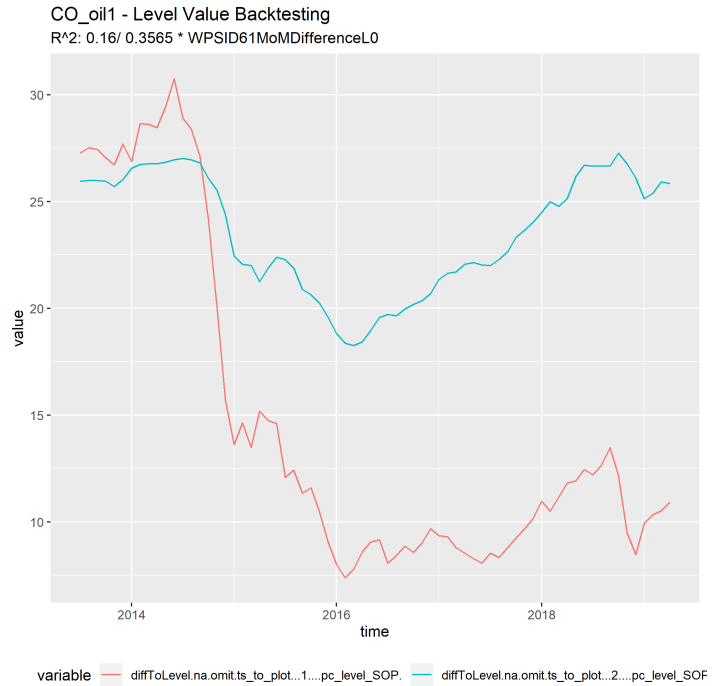


4.1.4 Model - 4

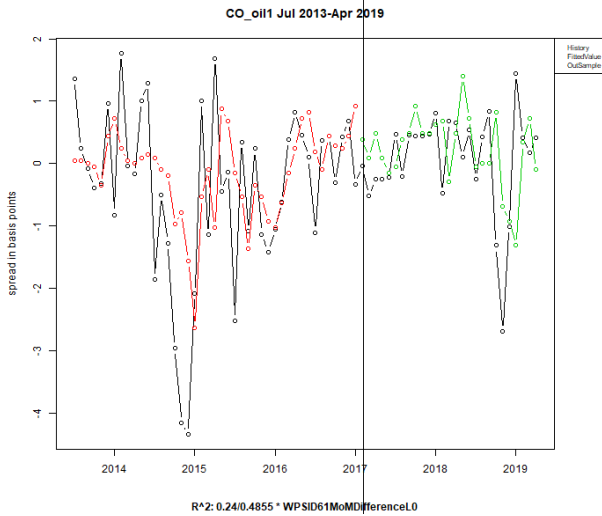
Backtesting on differenced values



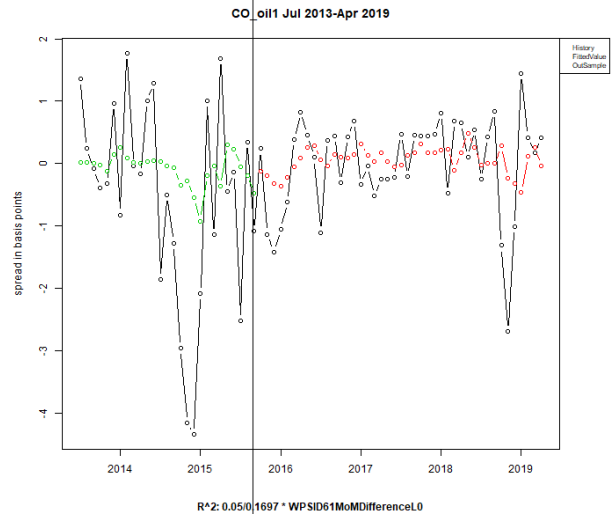
Backtesting on level values



Out-of-Sample testing

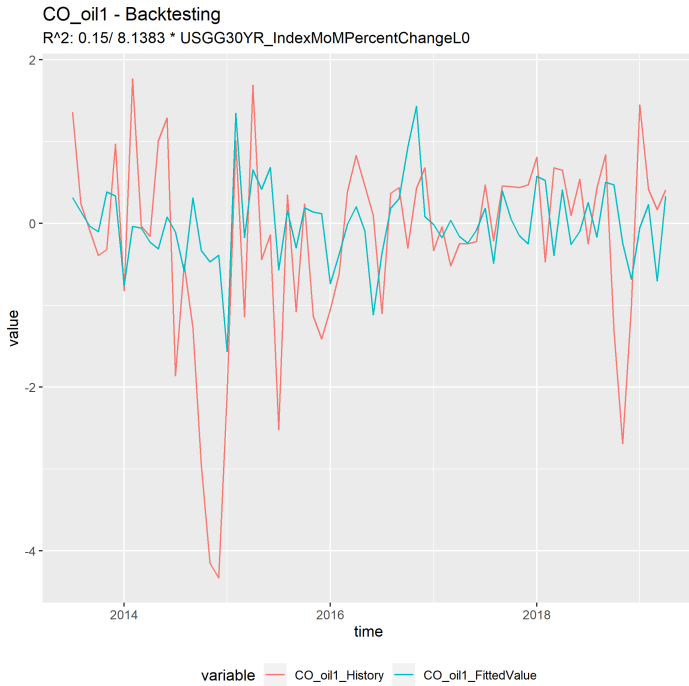


Out-of-Sample testing

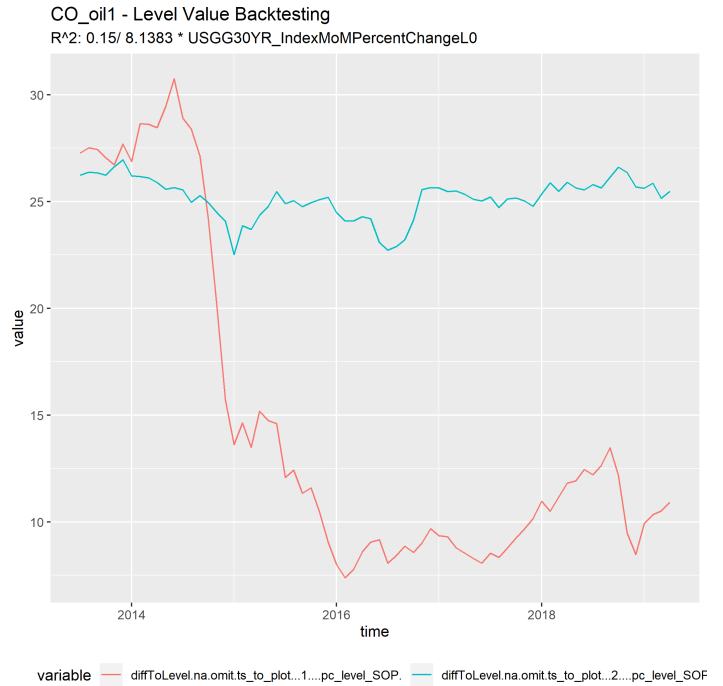


4.1.5 Model - 5

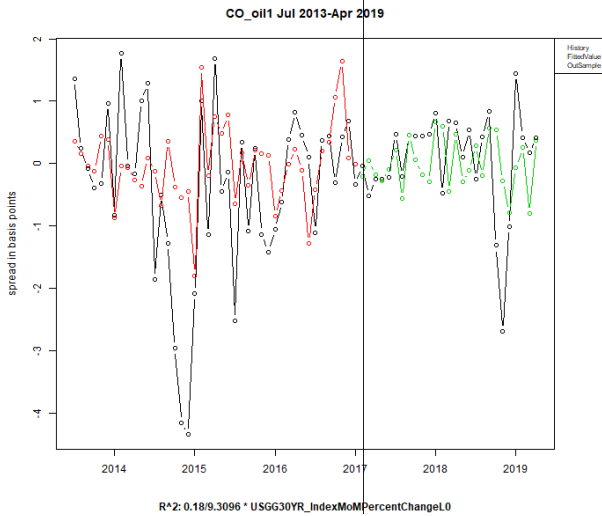
Backtesting on differenced values



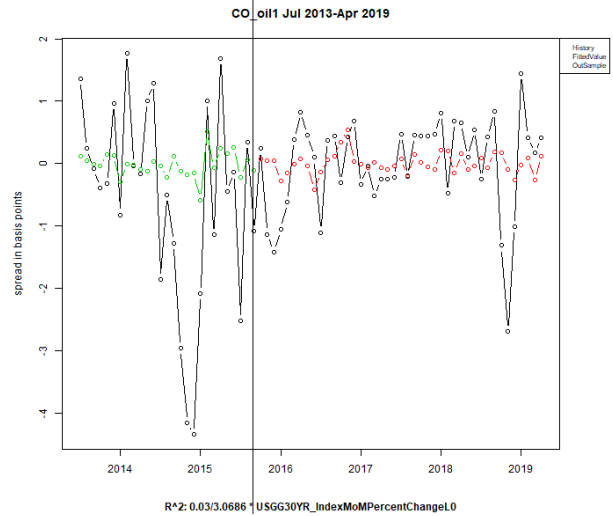
Backtesting on level values



Out-of-Sample testing

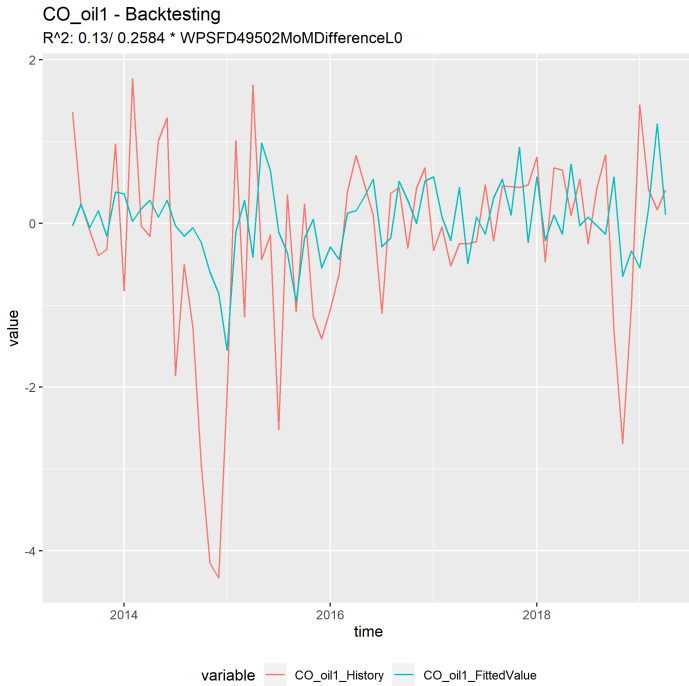


Out-of-Sample testing

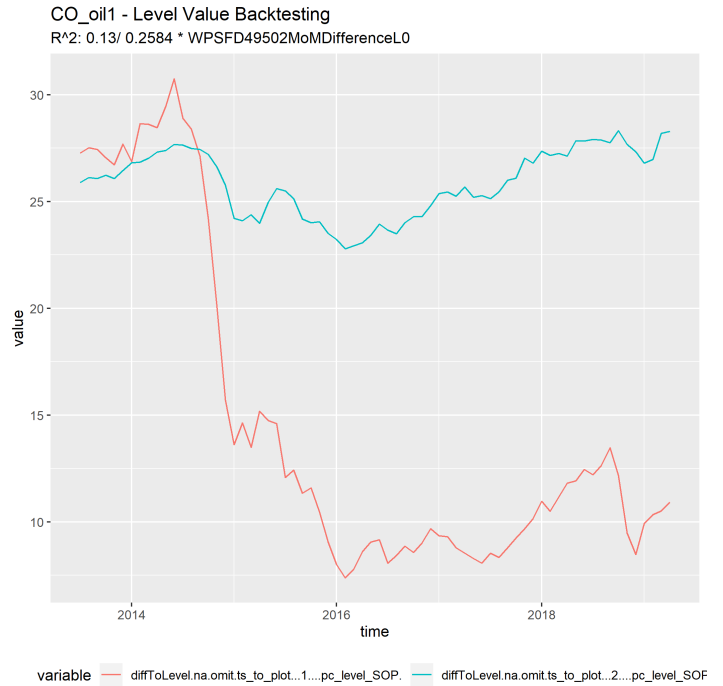


4.1.6 Model - 6

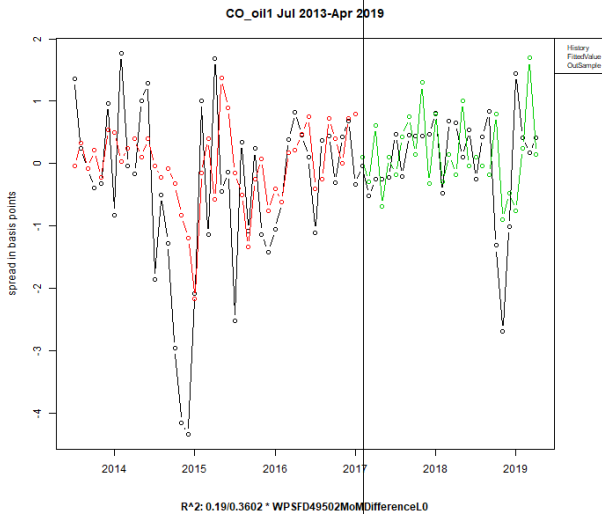
Backtesting on differenced values



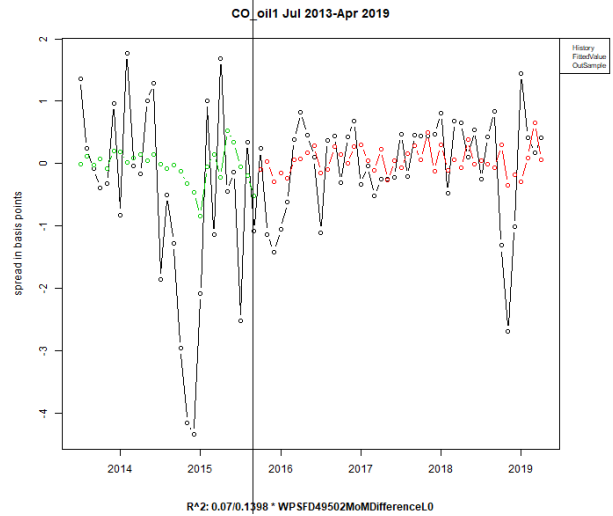
Backtesting on level values



Out-of-Sample testing

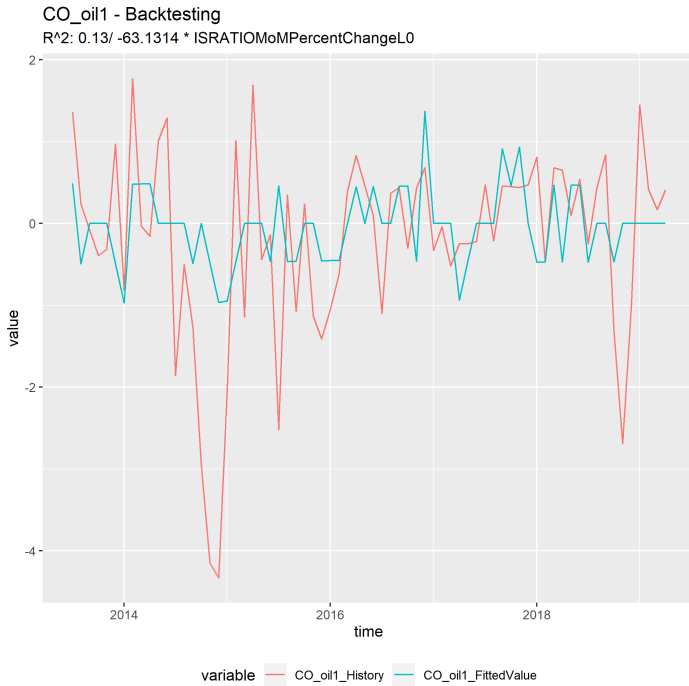


Out-of-Sample testing

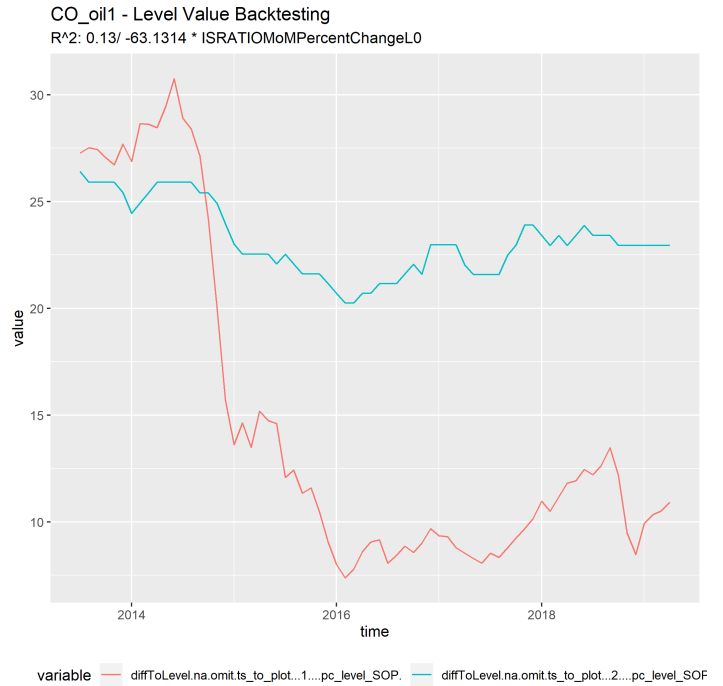


4.1.7 Model - 7

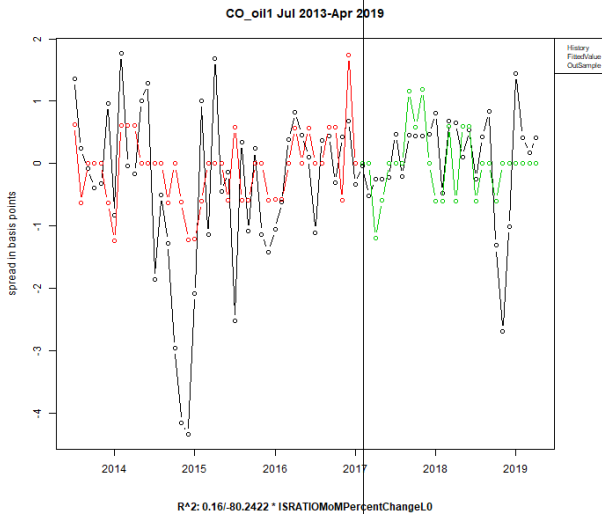
Backtesting on differenced values



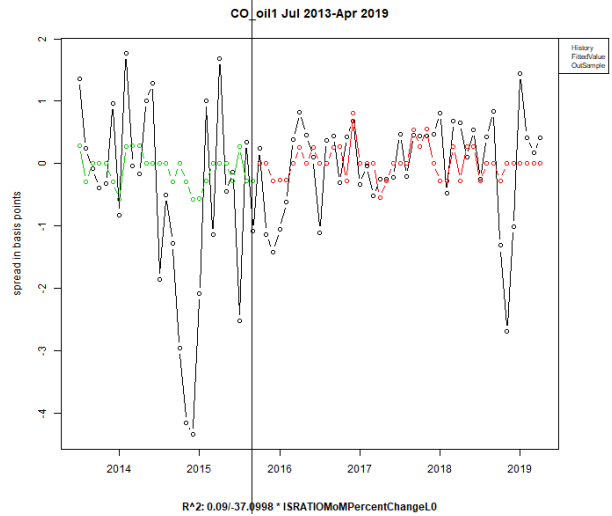
Backtesting on level values



Out-of-Sample testing

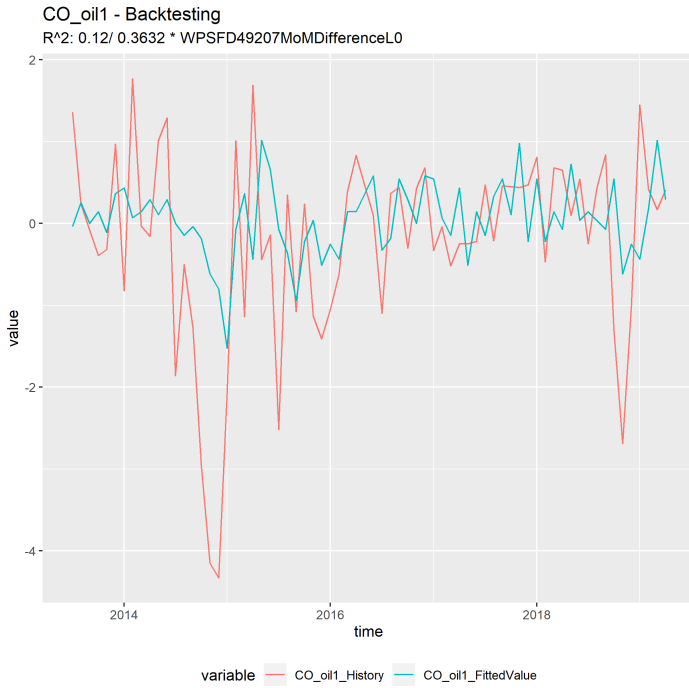


Out-of-Sample testing

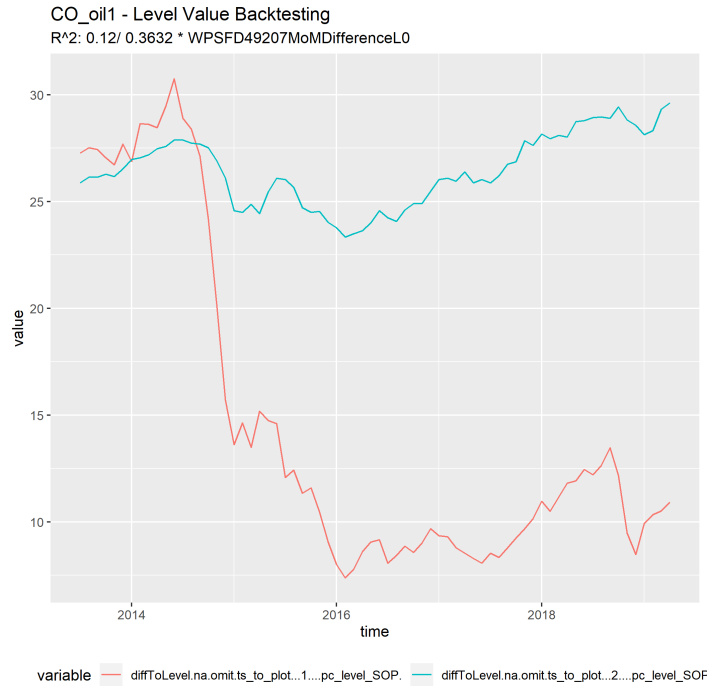


4.1.8 Model - 8

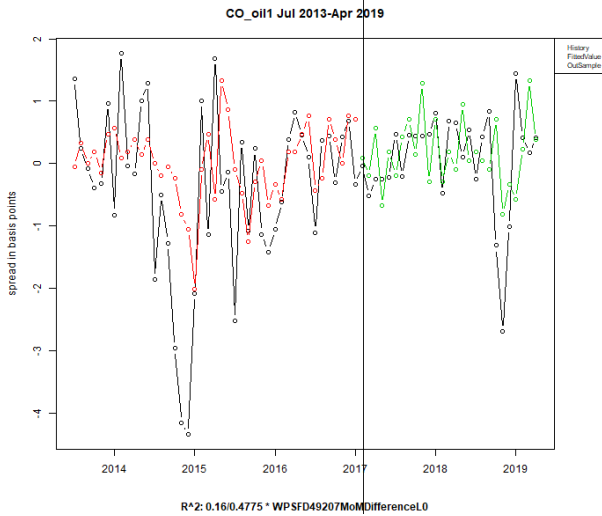
Backtesting on differenced values



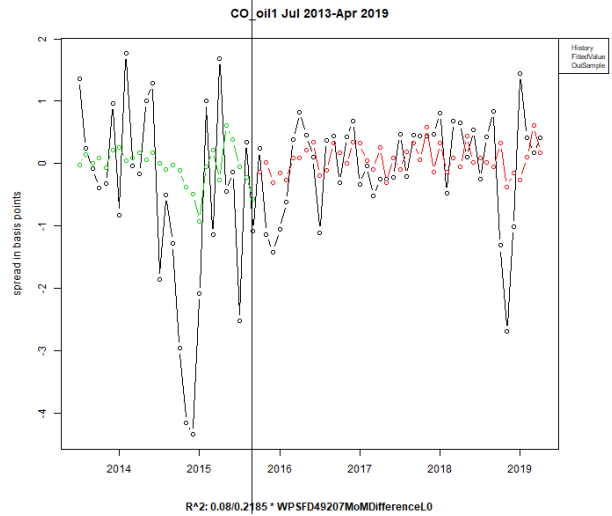
Backtesting on level values



Out-of-Sample testing

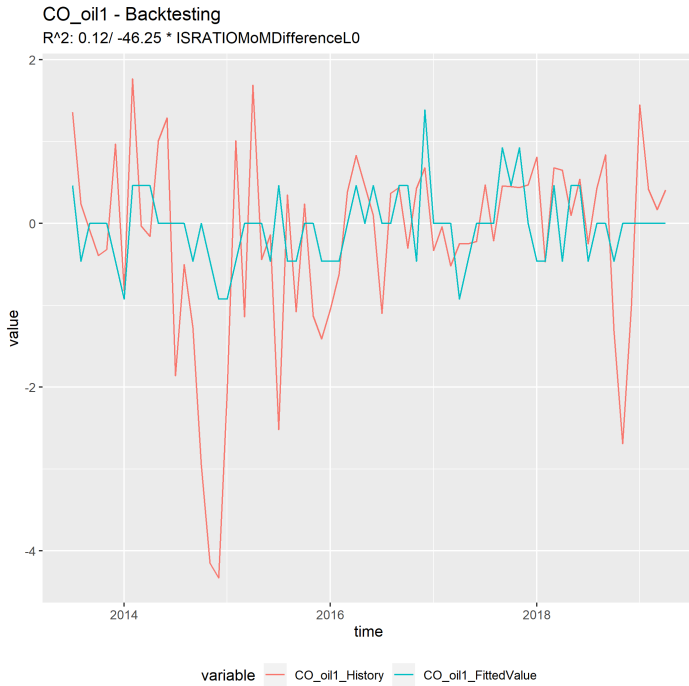


Out-of-Sample testing

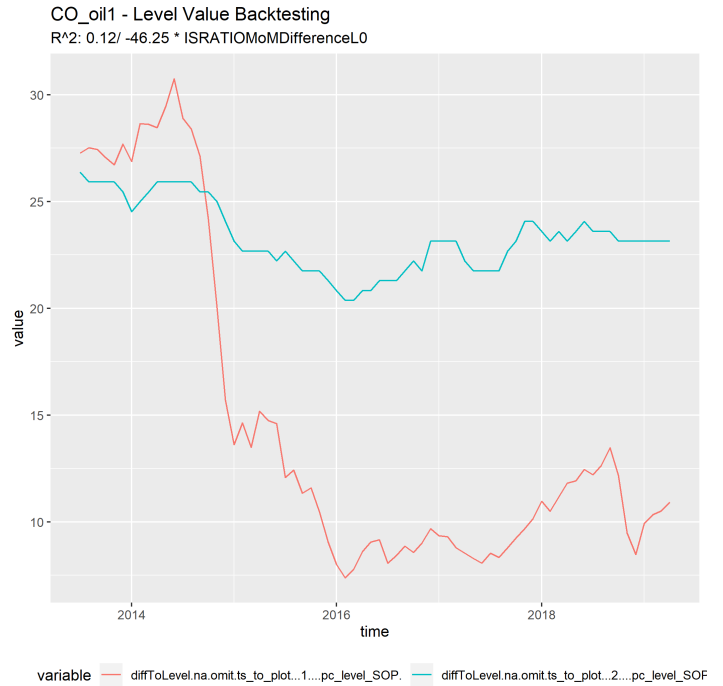


4.1.9 Model - 9

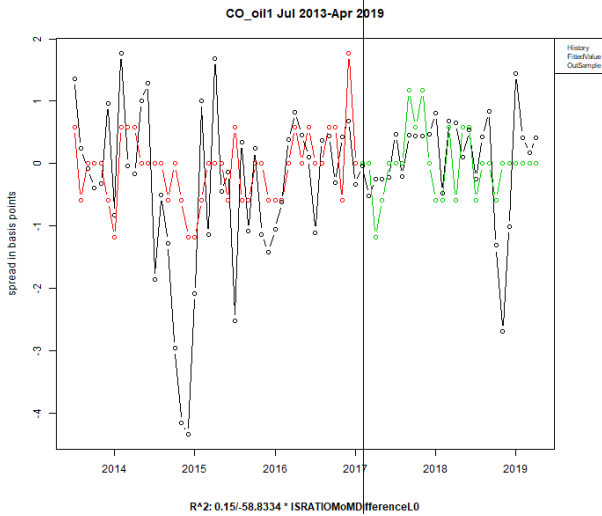
Backtesting on differenced values



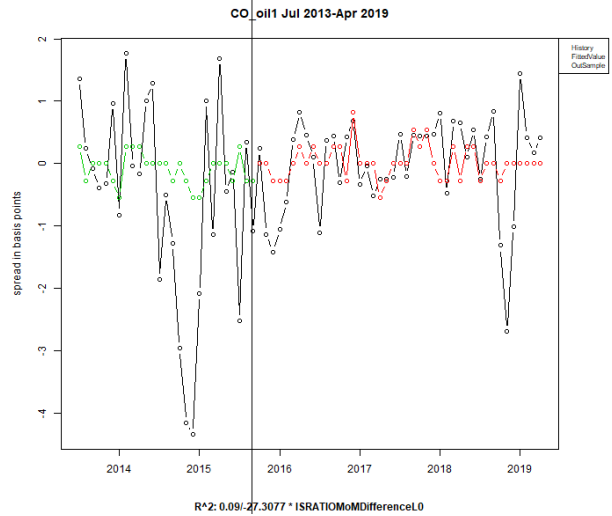
Backtesting on level values



Out-of-Sample testing

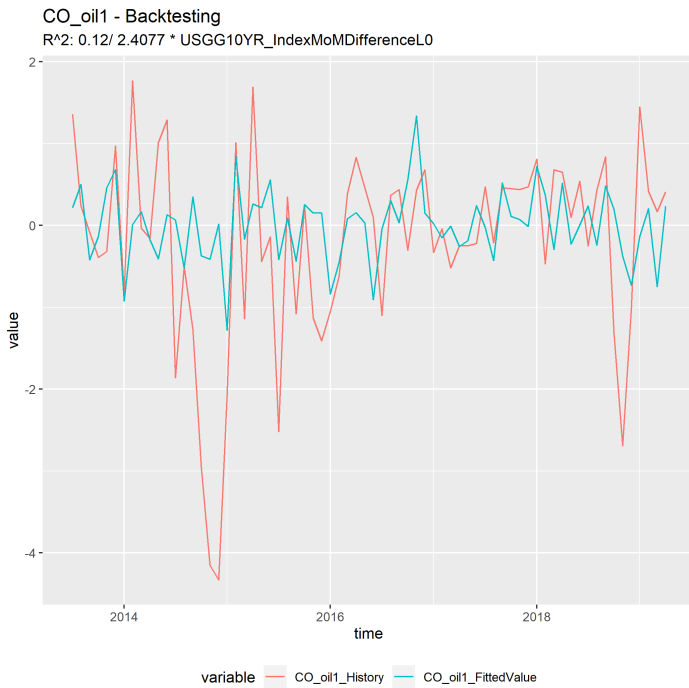


Out-of-Sample testing

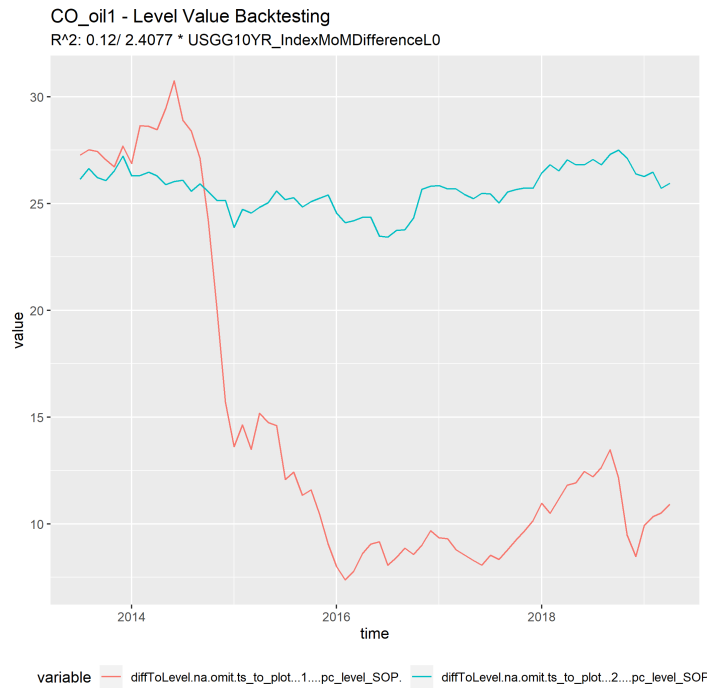


4.1.10 Model - 10

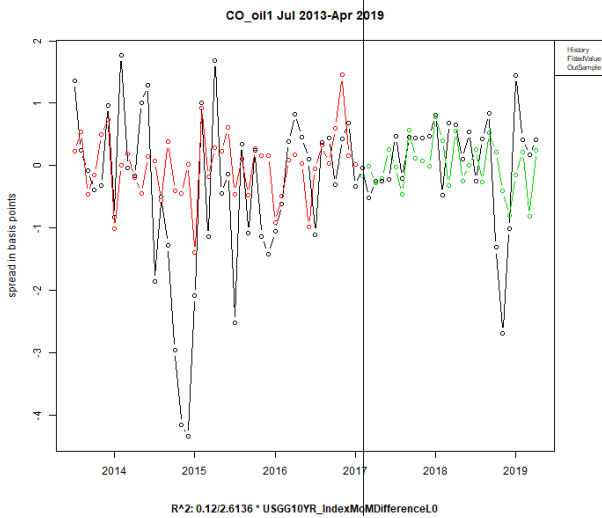
Backtesting on differenced values



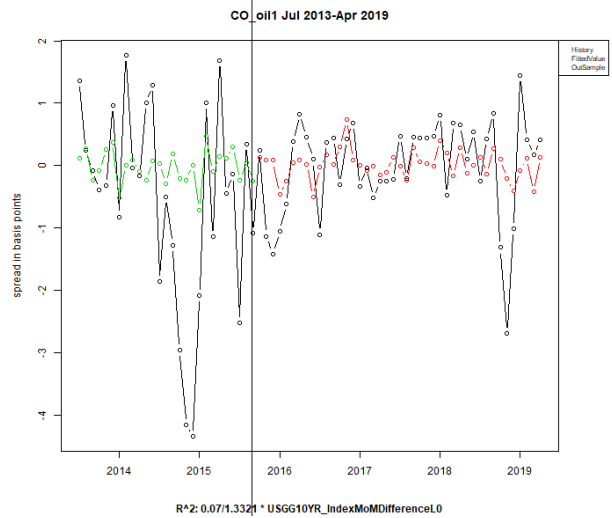
Backtesting on level values



Out-of-Sample testing

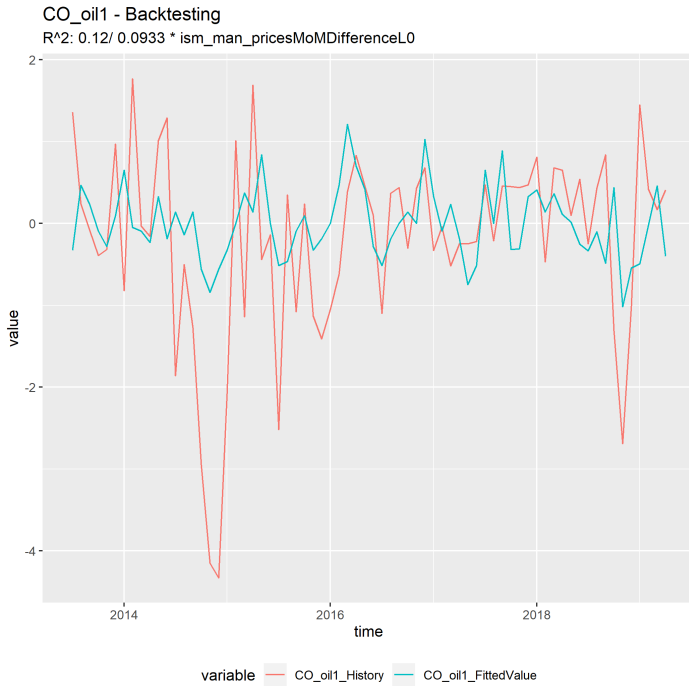


Out-of-Sample testing

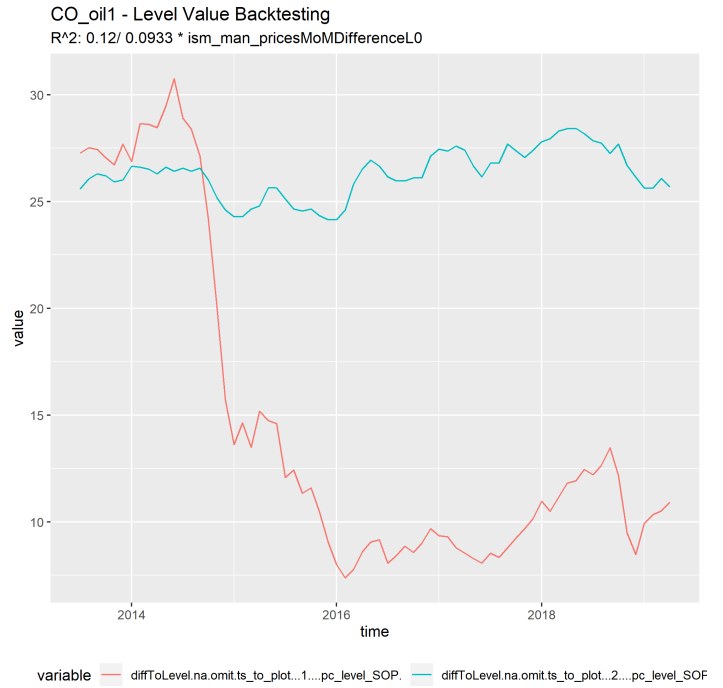


4.1.11 Model - 11

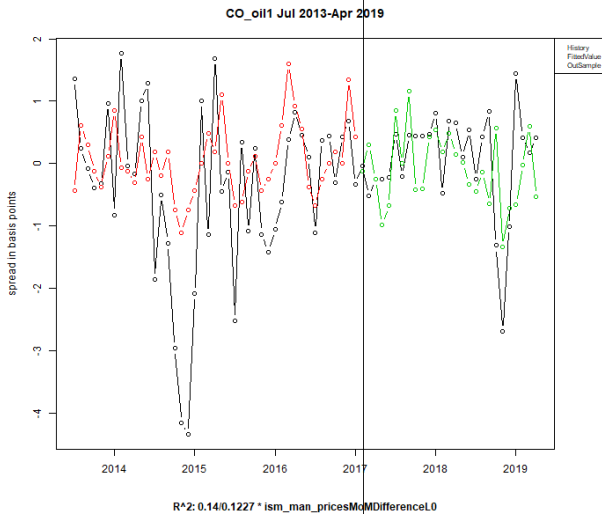
Backtesting on differenced values



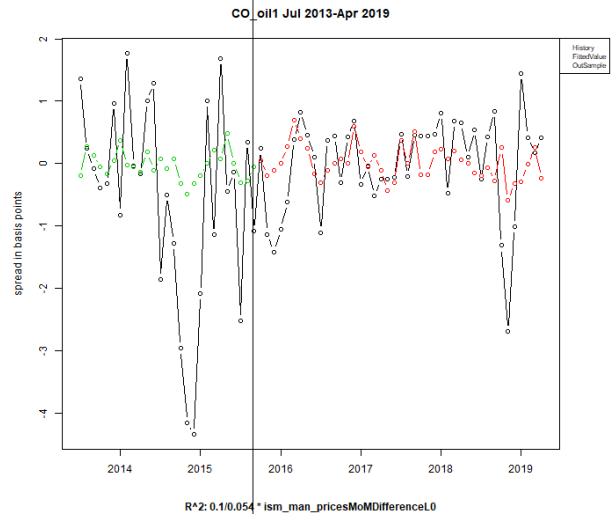
Backtesting on level values



Out-of-Sample testing

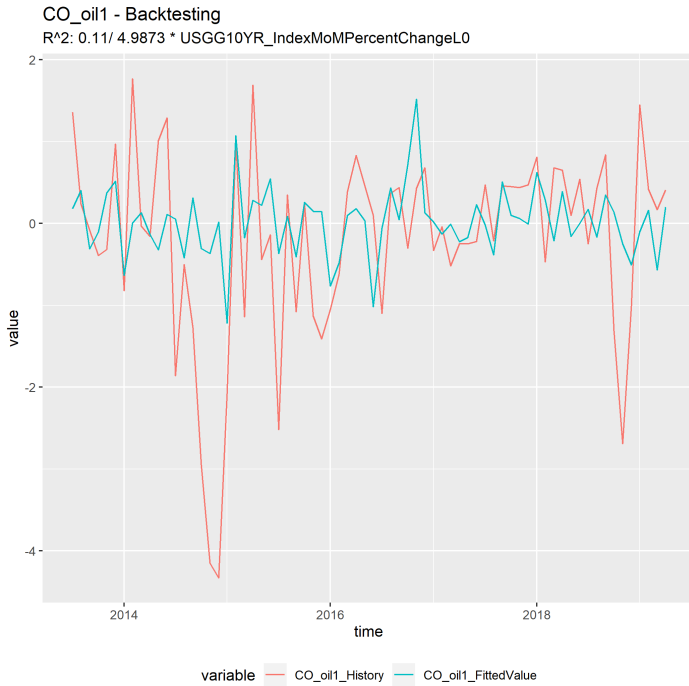


Out-of-Sample testing

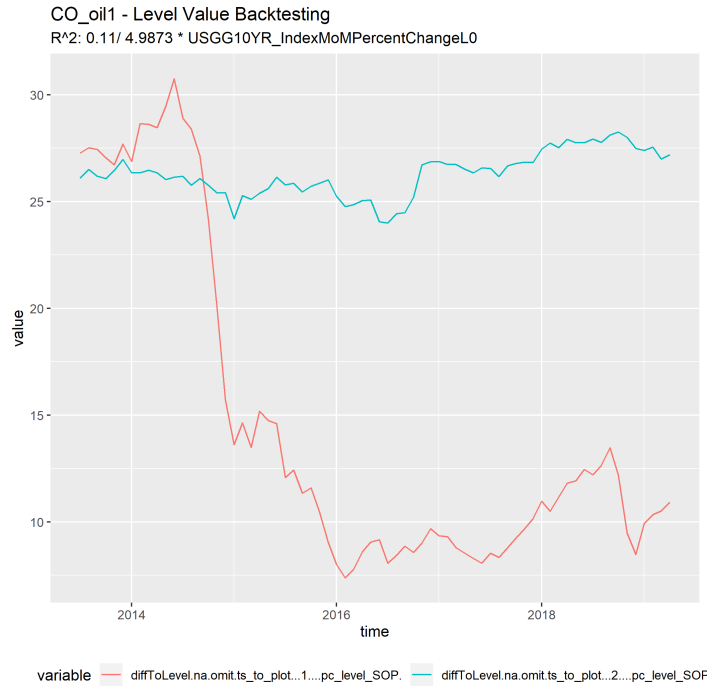


4.1.12 Model - 12

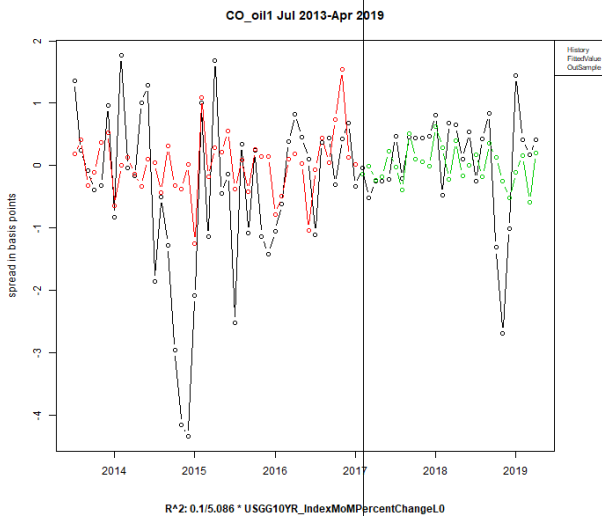
Backtesting on differenced values



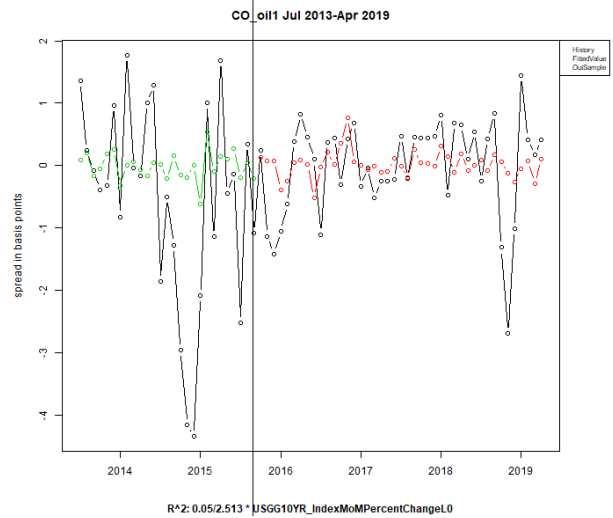
Backtesting on level values



Out-of-Sample testing



Out-of-Sample testing



4.2 Two Factor Model for CO_{oil}

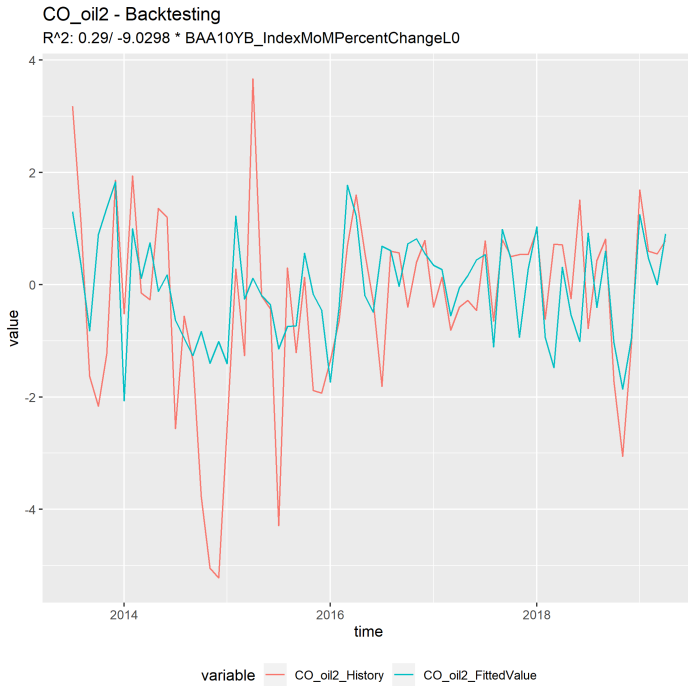
4.3 Single Factor Model for CO_{oil2}

	Variable	adjR2	AIC	Coef	Robst p	RMSE_OS	Criteria1	Criteria2	Criteria3
1	BAA10YB_IndexMoMPercentChangeL0	0.29	249.47	-9.03	0.00	0.95			
2	BAA10YB_IndexMoMDifferenceL0	0.25	252.98	-0.06	0.00	0.83			
3	USGG30YR_IndexMoMDifferenceL0	0.14	262.69	3.82	0.00	1.09			
4	ism_man_pricesMoMDifferenceL0	0.13	263.14	0.13	0.02	1.13			
5	ism_man_pricesMoMPercentChangeL0	0.13	263.51	6.81	0.03	1.02			
6	USGG30YR_IndexMoMPercentChangeL0	0.13	263.52	10.42	0.00	1.06			
7	WPSID61MoMDifferenceL0	0.12	263.89	0.44	0.04	1.19			
8	MANEMPMoMPercentChangeL0	0.11	264.92	-267.39	0.03	1.26			
9	MANEMPMoMDifferenceL0	0.11	265.06	-0.02	0.03	1.27			
10	USGG10YR_IndexMoMDifferenceL0	0.11	265.11	3.13	0.00	0.99			

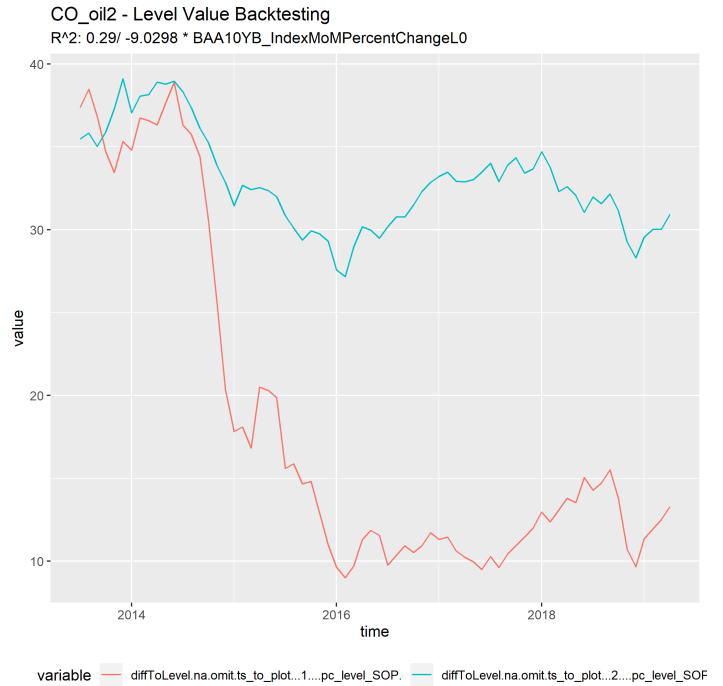
Table 4: Single Factor Analysis- CO_{oil2}

4.3.1 Model - 1

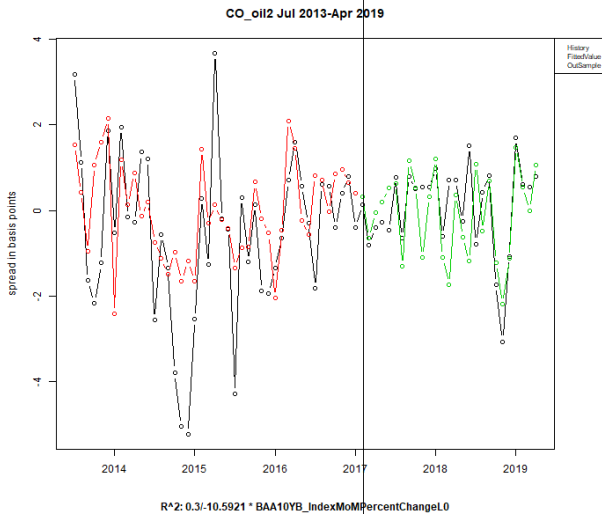
Backtesting on differenced values



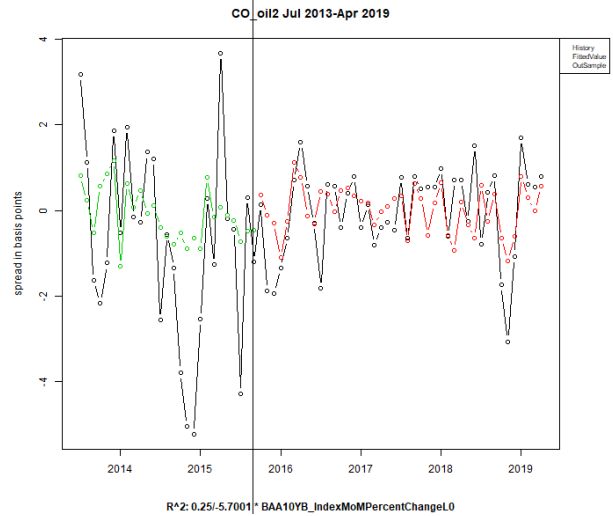
Backtesting on level values



Out-of-Sample testing

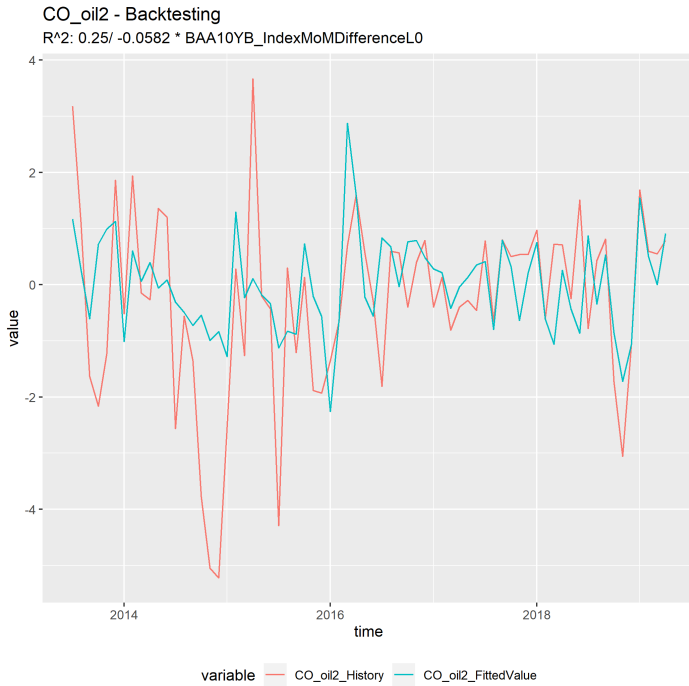


Out-of-Sample testing

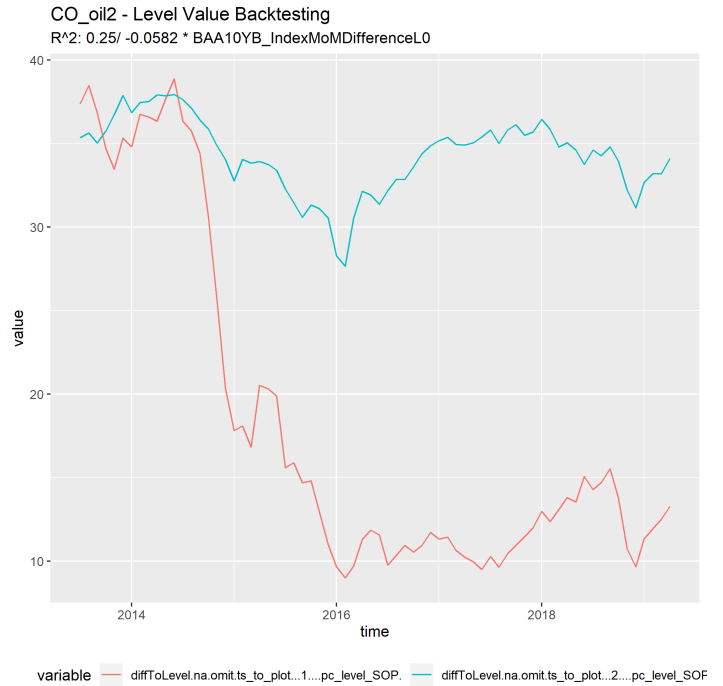


4.3.2 Model - 2

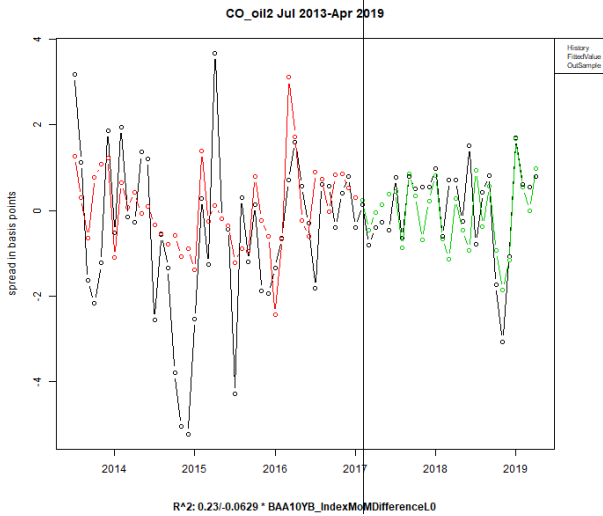
Backtesting on differenced values



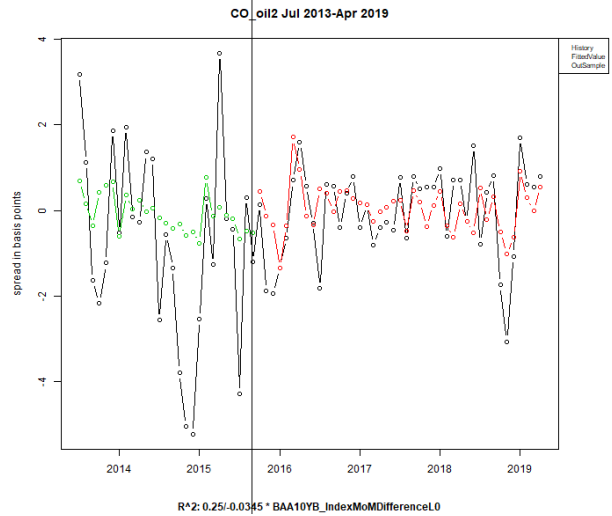
Backtesting on level values



Out-of-Sample testing

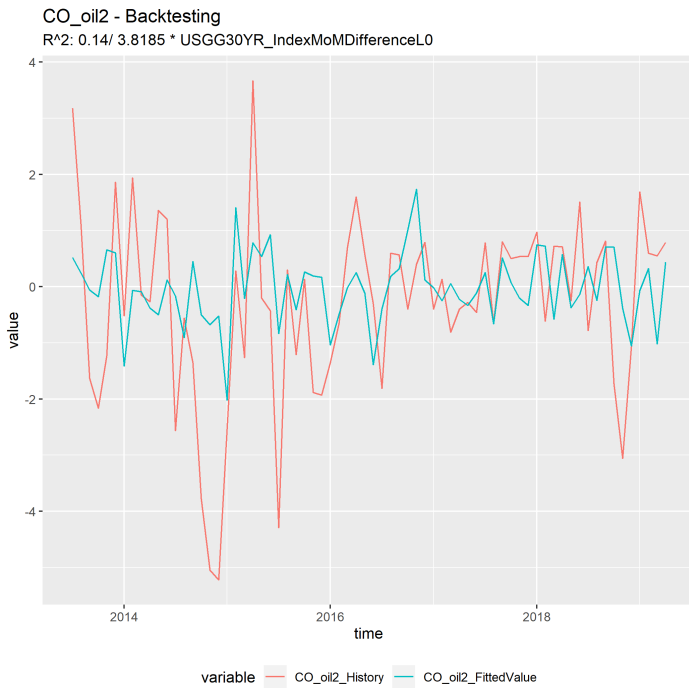


Out-of-Sample testing

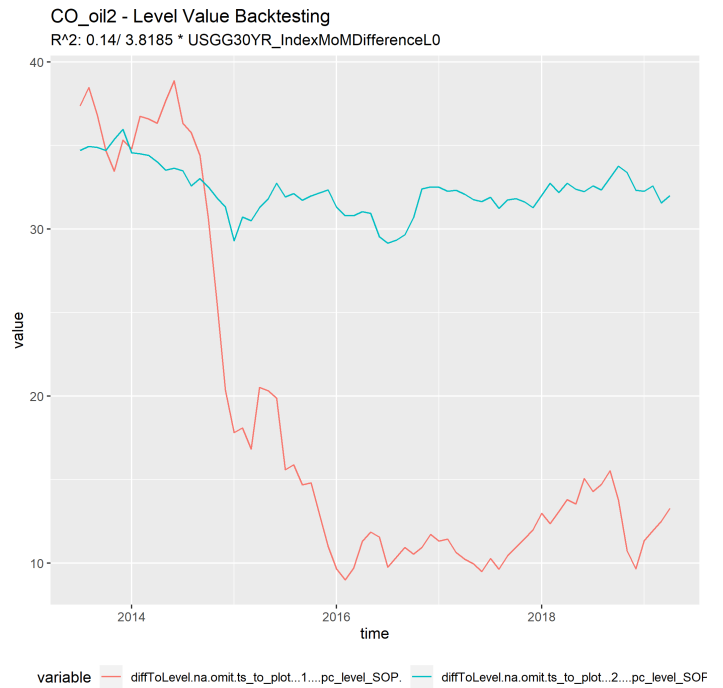


4.3.3 Model - 3

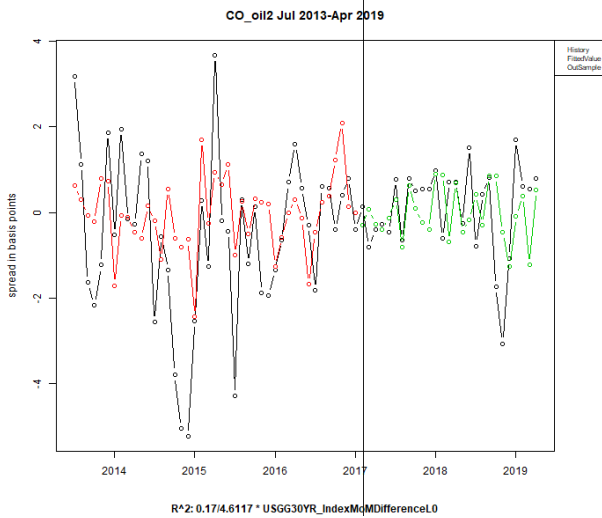
Backtesting on differenced values



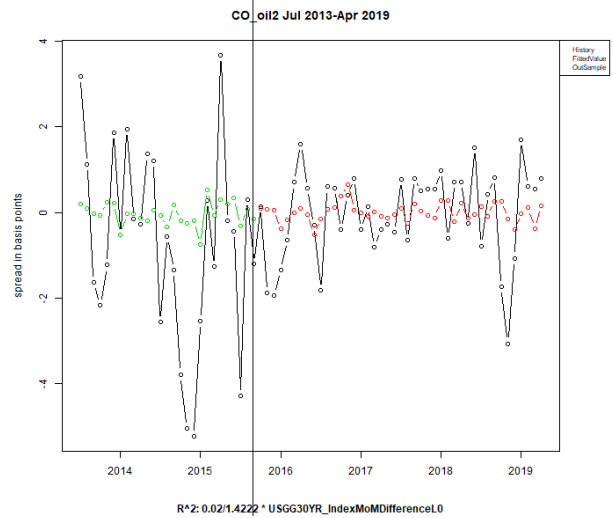
Backtesting on level values



Out-of-Sample testing

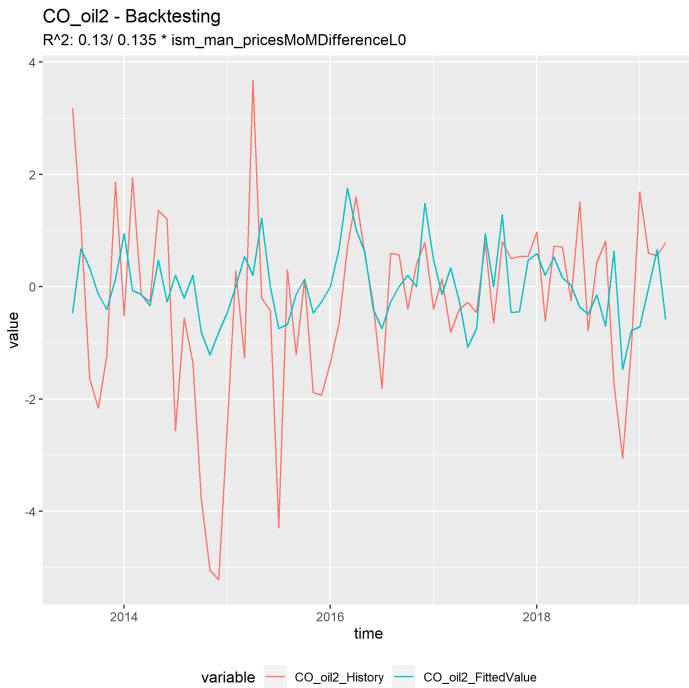


Out-of-Sample testing

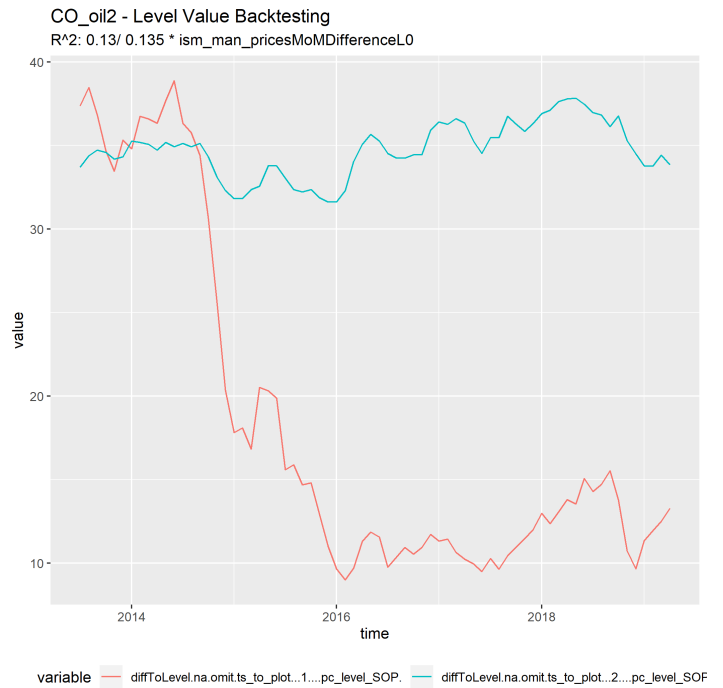


4.3.4 Model - 4

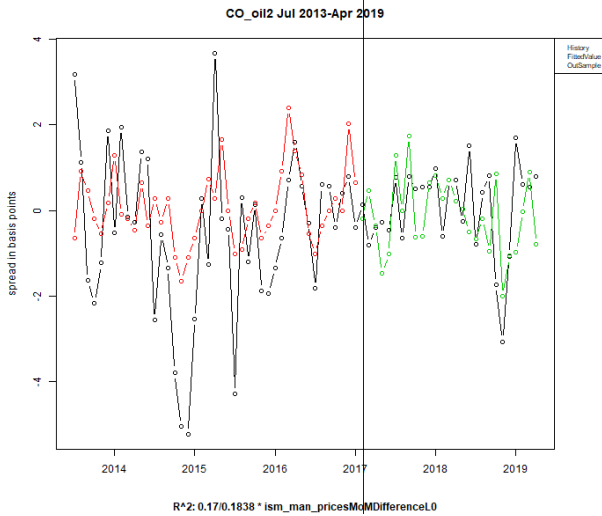
Backtesting on differenced values



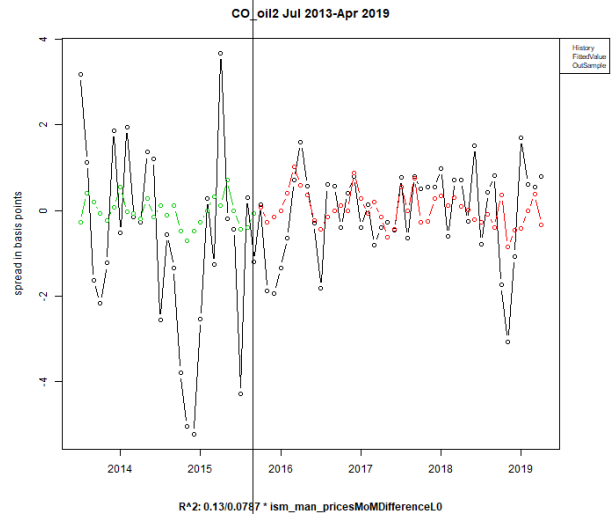
Backtesting on level values



Out-of-Sample testing

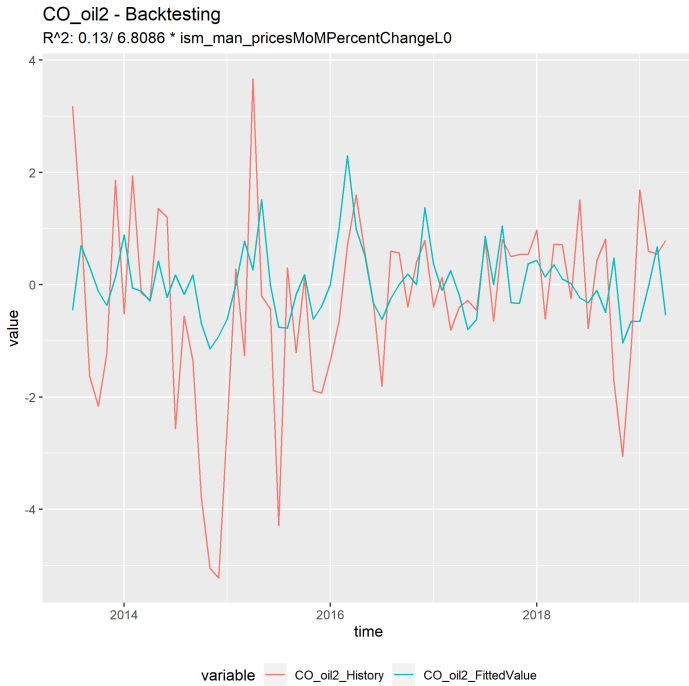


Out-of-Sample testing

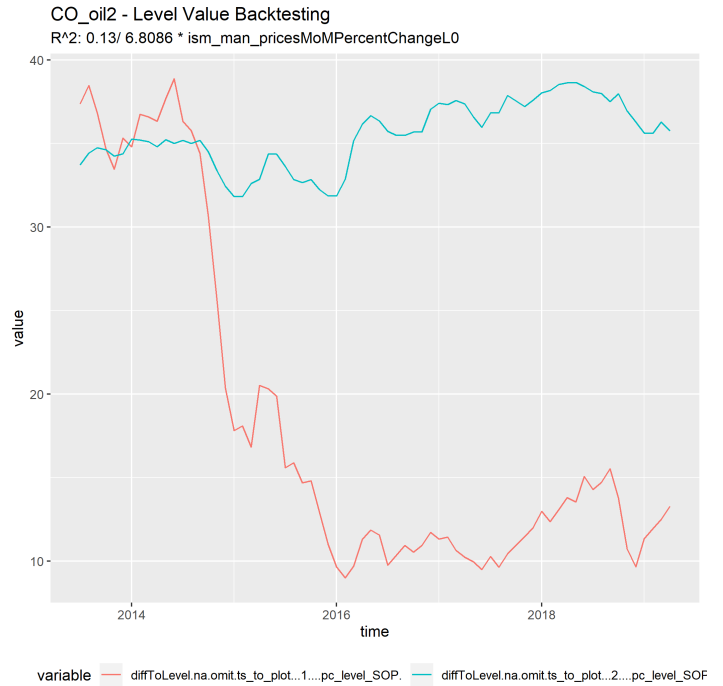


4.3.5 Model - 5

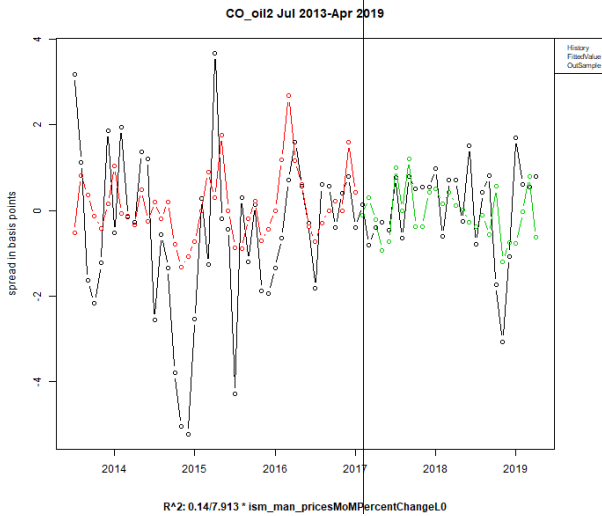
Backtesting on differenced values



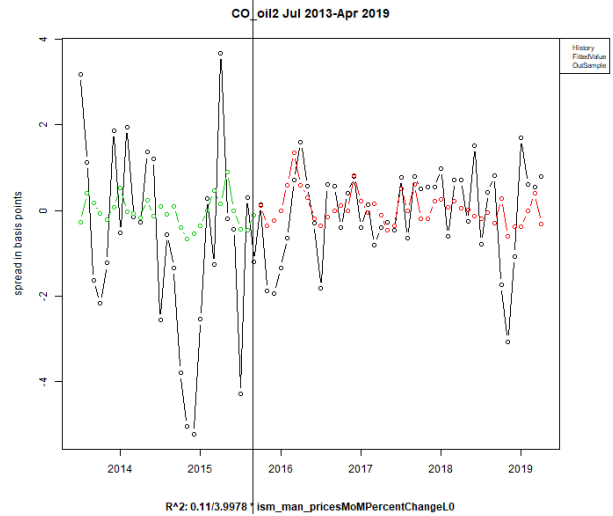
Backtesting on level values



Out-of-Sample testing

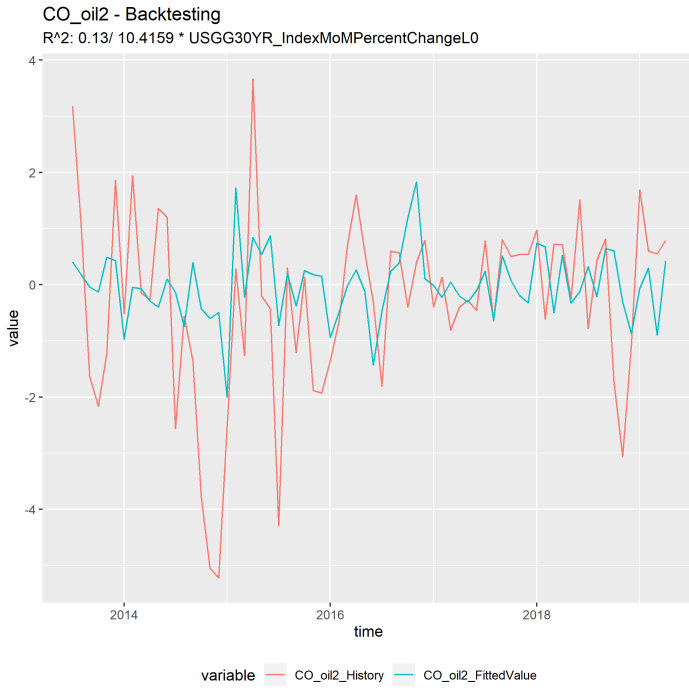


Out-of-Sample testing

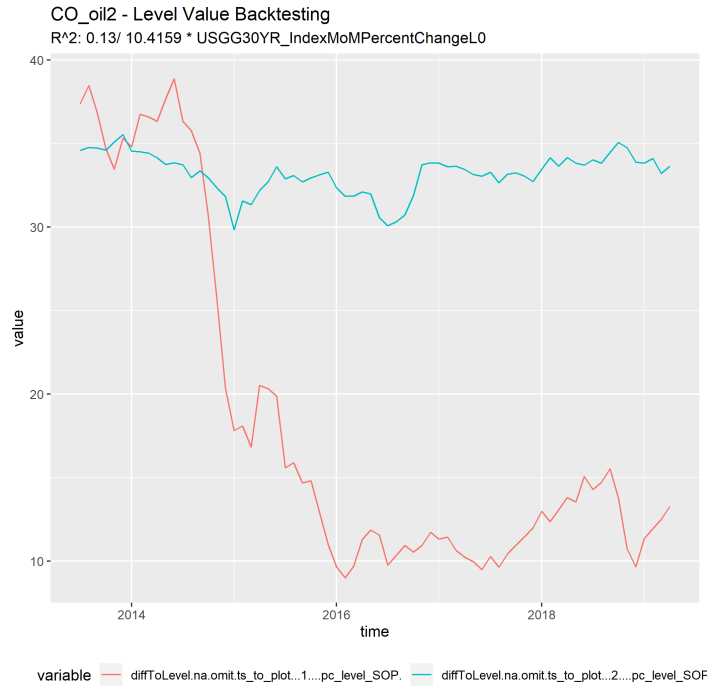


4.3.6 Model - 6

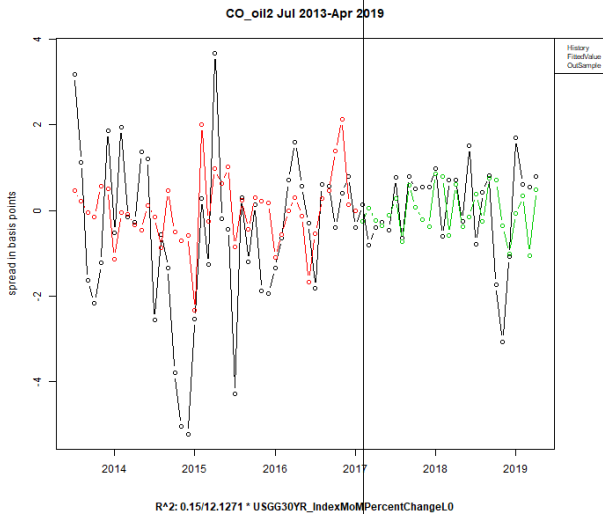
Backtesting on differenced values



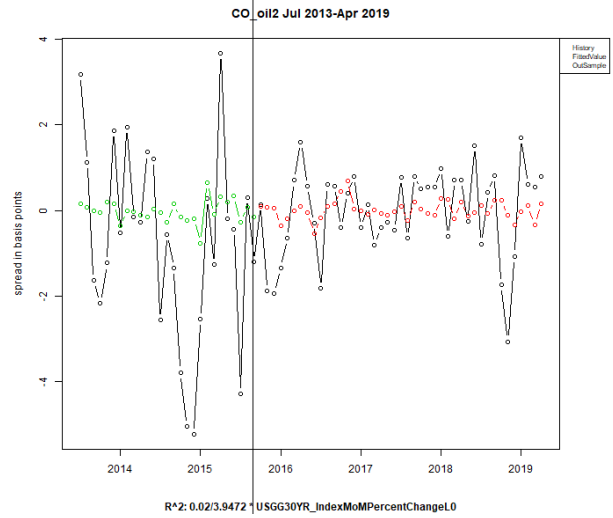
Backtesting on level values



Out-of-Sample testing

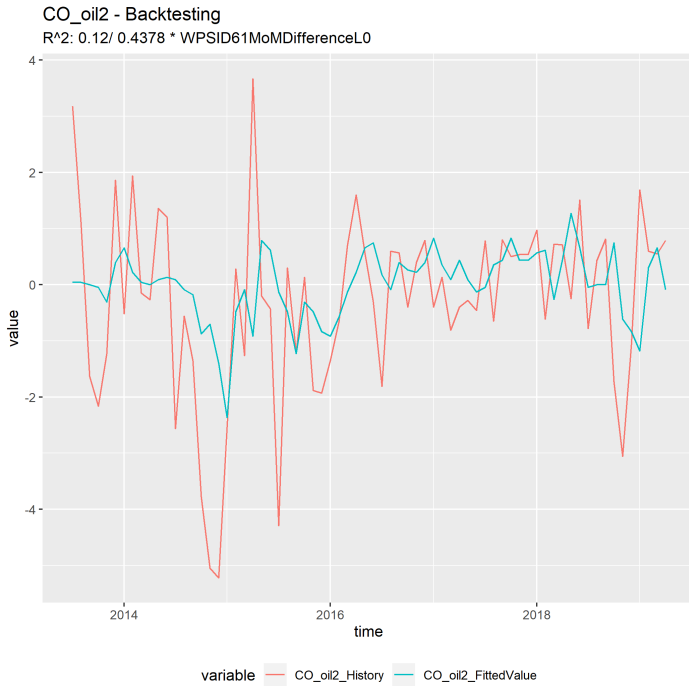


Out-of-Sample testing

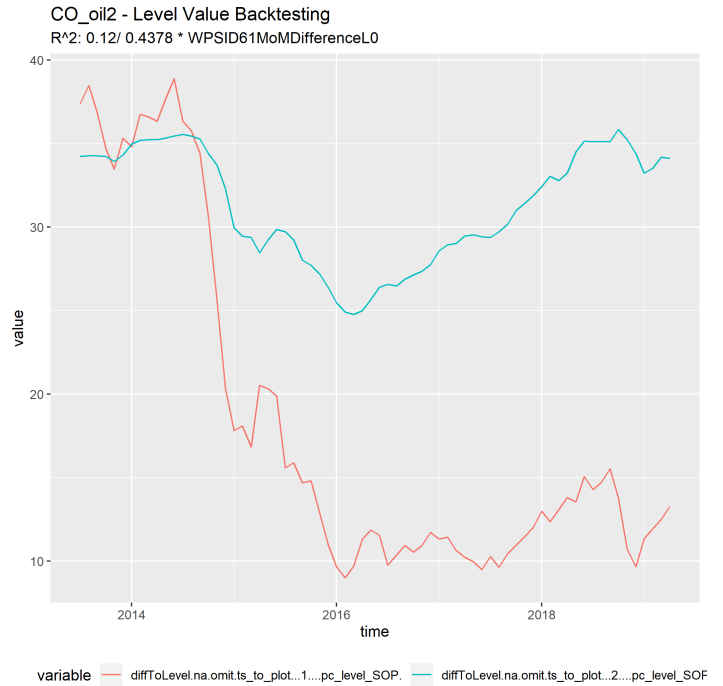


4.3.7 Model - 7

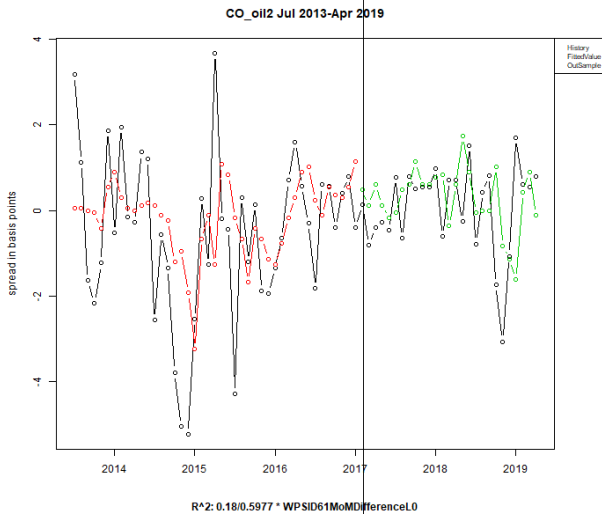
Backtesting on differenced values



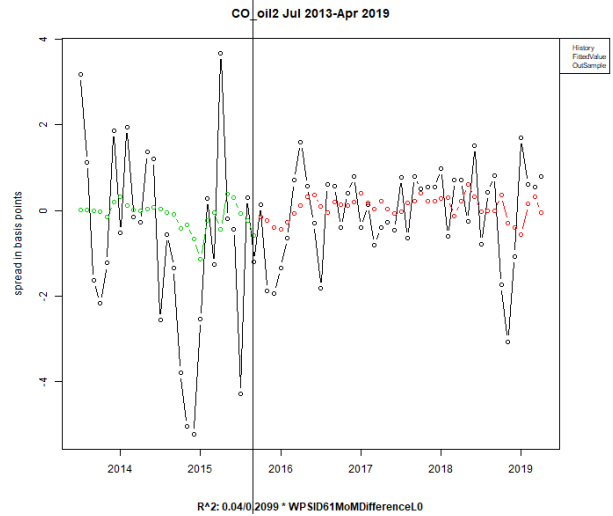
Backtesting on level values



Out-of-Sample testing

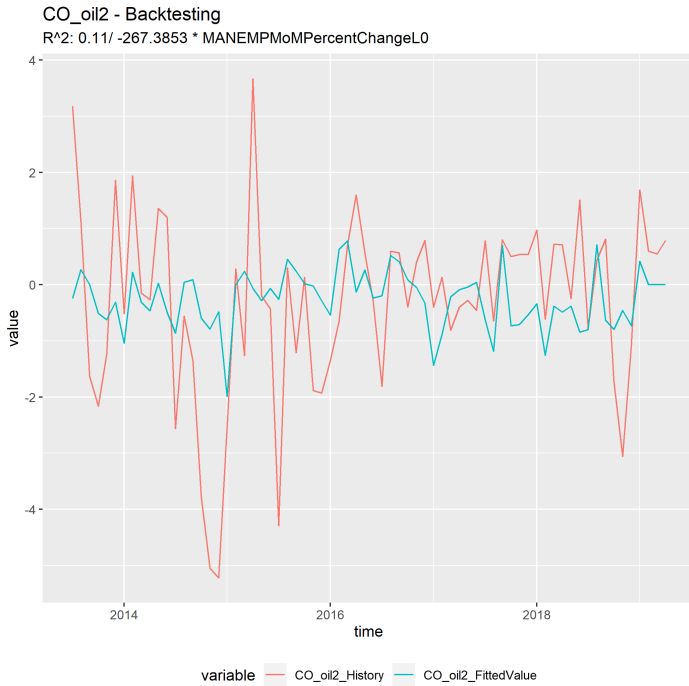


Out-of-Sample testing

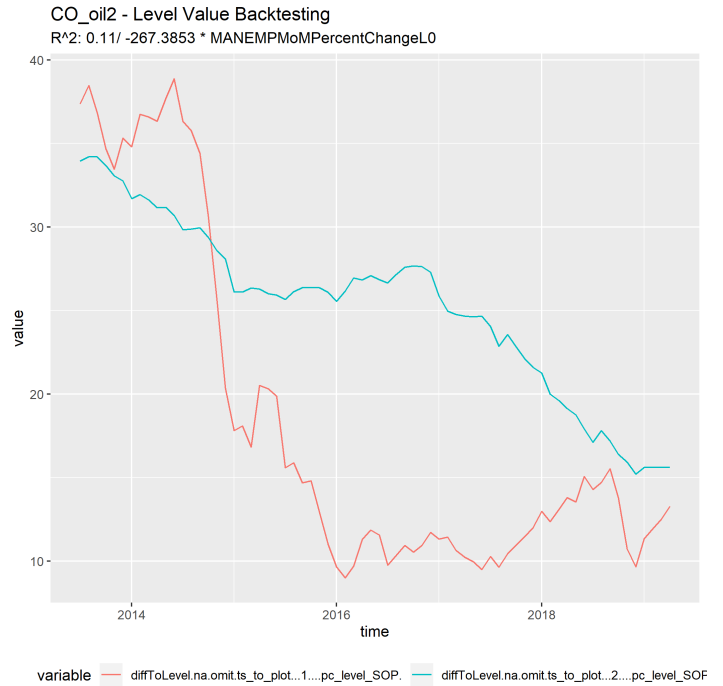


4.3.8 Model - 8

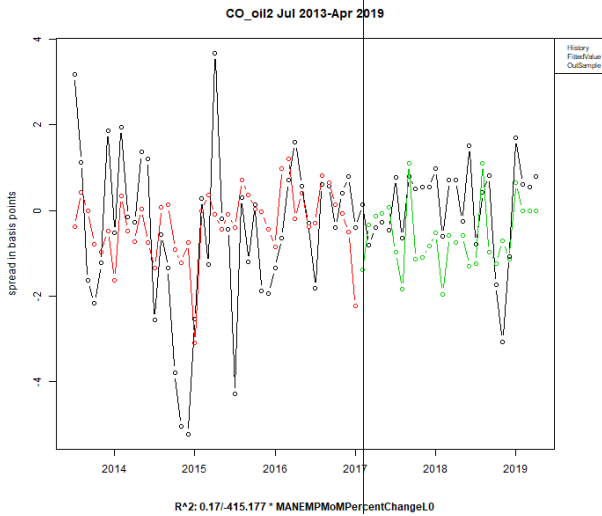
Backtesting on differenced values



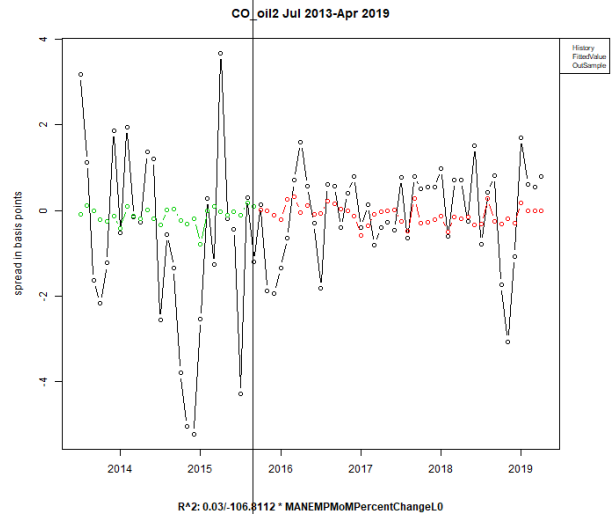
Backtesting on level values



Out-of-Sample testing

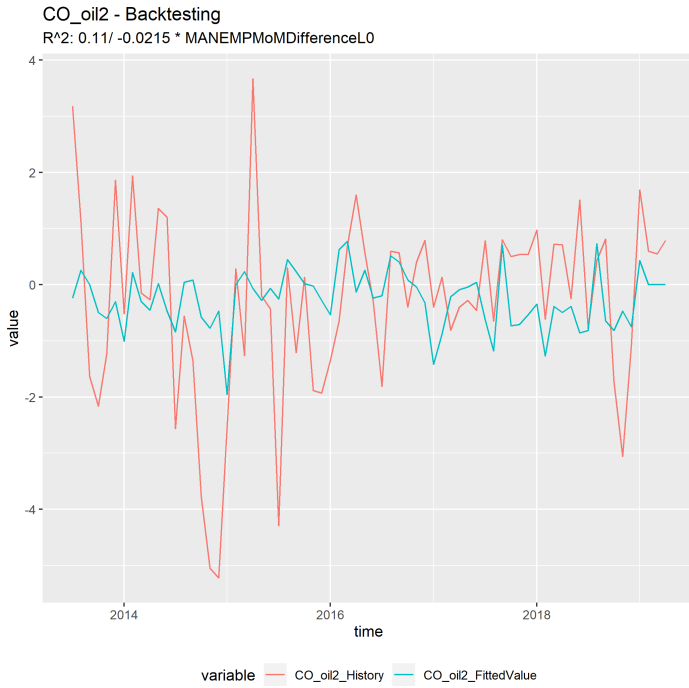


Out-of-Sample testing

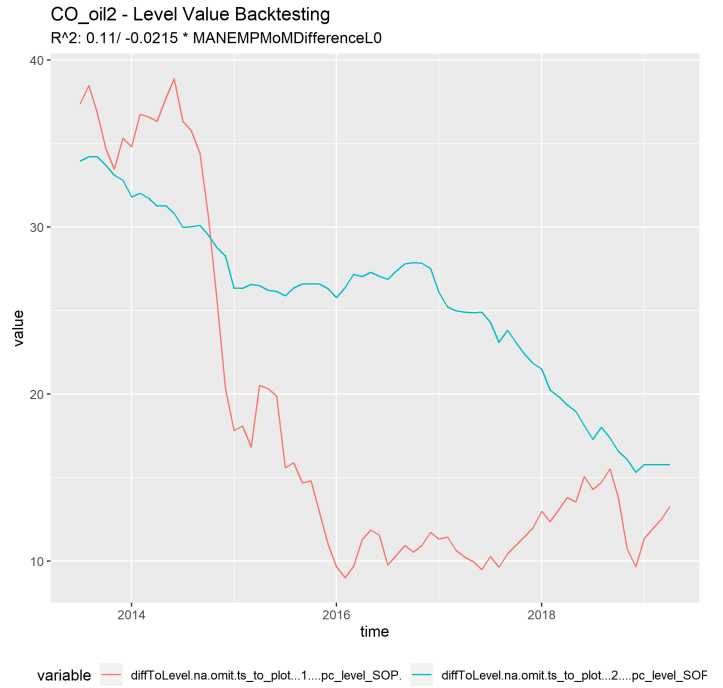


4.3.9 Model - 9

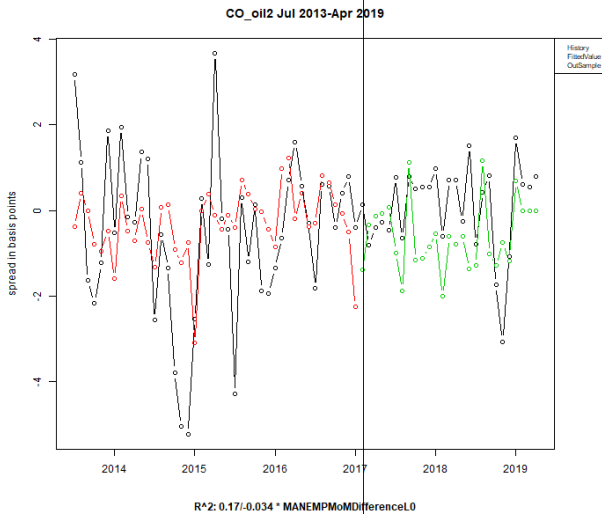
Backtesting on differenced values



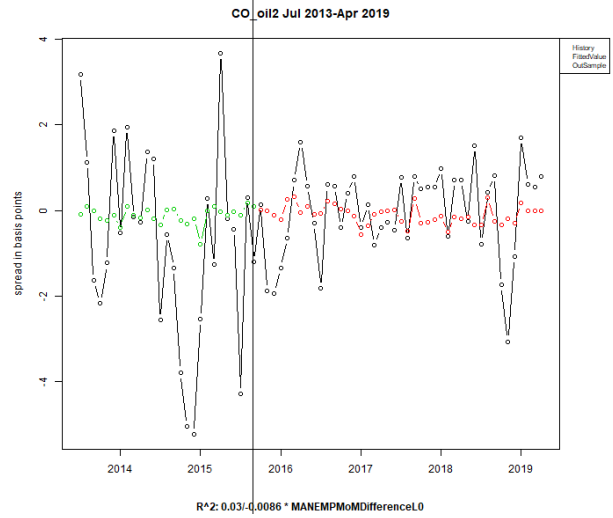
Backtesting on level values



Out-of-Sample testing

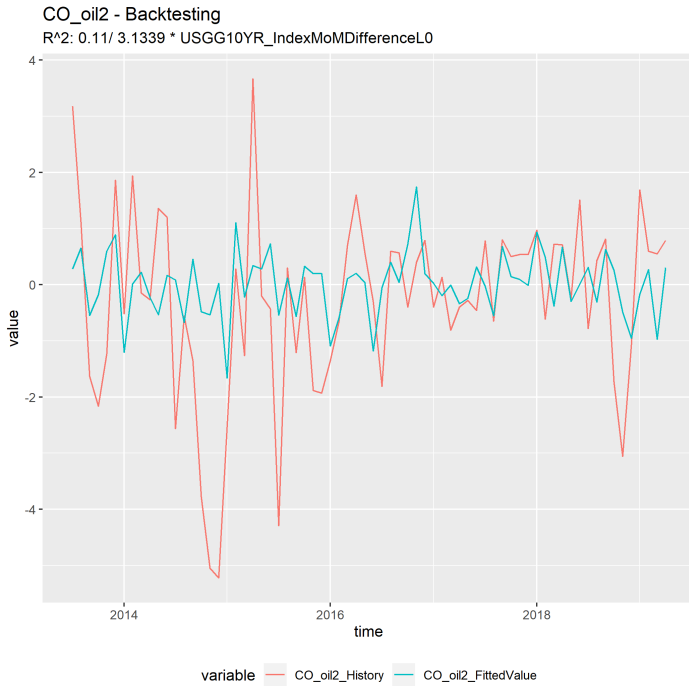


Out-of-Sample testing

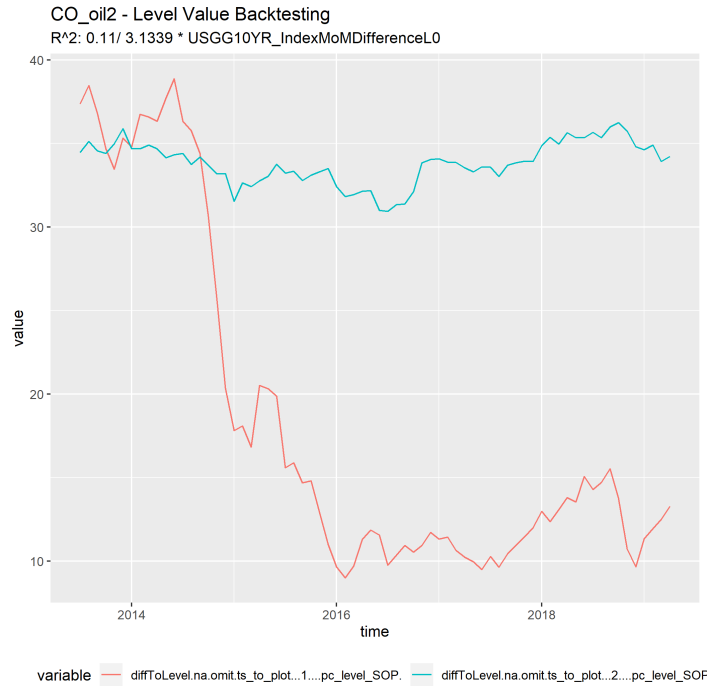


4.3.10 Model - 10

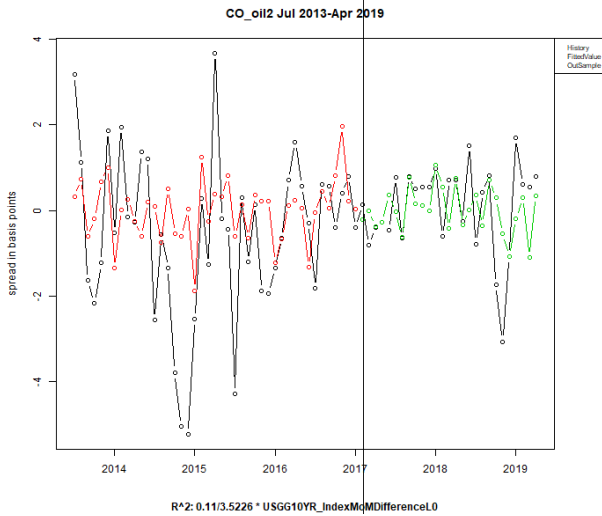
Backtesting on differenced values



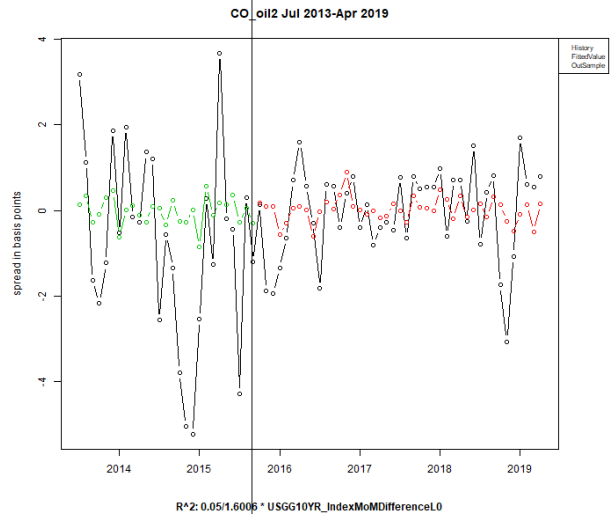
Backtesting on level values



Out-of-Sample testing



Out-of-Sample testing



4.4 Two Factor Model for CO_{oil2}

4.5 Single Factor Model for CO_nat_{gas}

	Variable	adjR2	AIC	Coef	Robst p	RMSE_OS	Criteria1	Criteria2	Criteria3
1	IPB51222SMoMDifferenceL0	0.13	430.20	0.38	0.03	3.53			
2	IPB51222SMoMPercentChangeL0	0.12	430.79	37.79	0.03	3.51			
3	UEMP15T26MoMDifferenceL0	0.00	439.61	0.01	0.38	3.66			

Table 5: Single Factor Analysis-CO_nat_{gas}

4.5.1 Model - 1

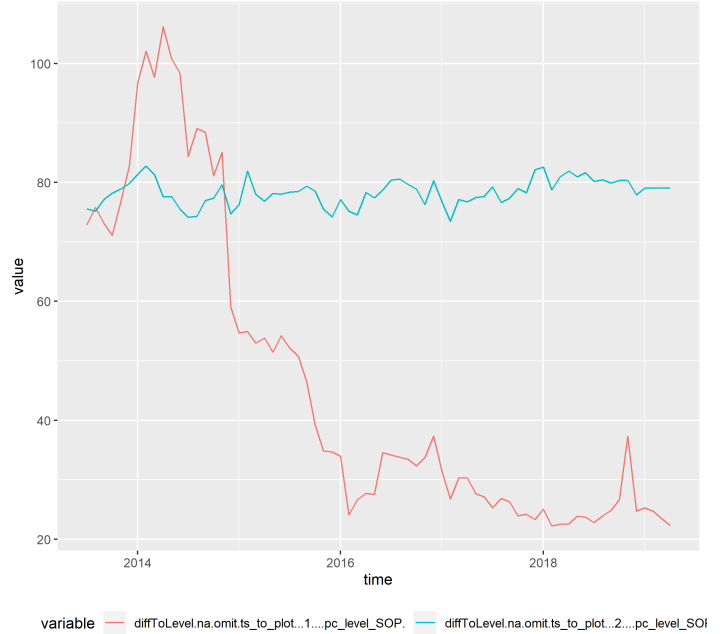
Backtesting on differenced values

CO_nat_gas - Backtesting
R²: 0.13/ 0.3824 * IPB51222SMoMDifferenceL0



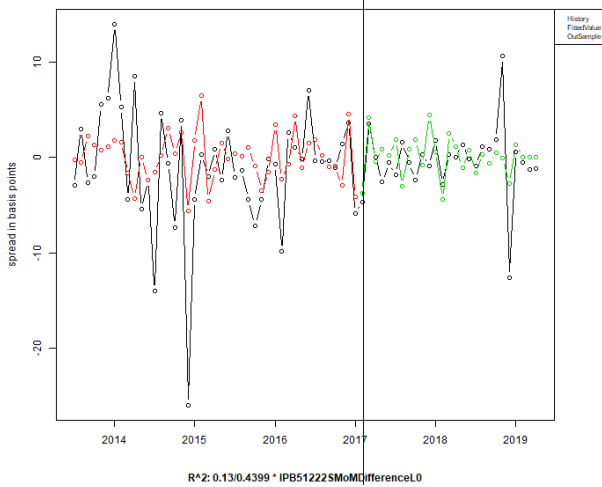
Backtesting on level values

CO_nat_gas - Level Value Backtesting
R²: 0.13/ 0.3824 * IPB51222SMoMDifferenceL0



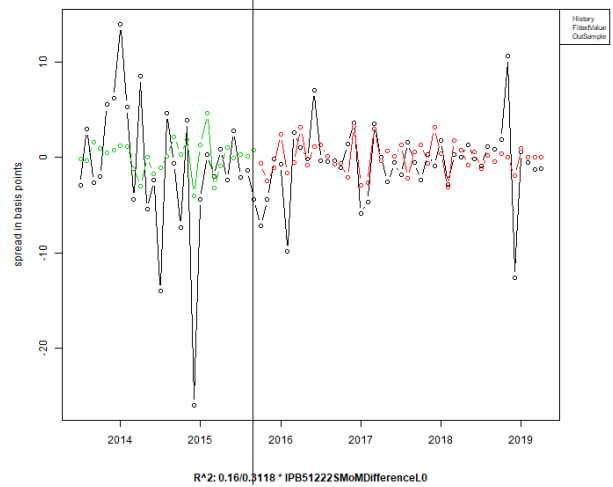
Out-of-Sample testing

CO_nat_gas Jul 2013-Apr 2019



Out-of-Sample testing

CO_nat_gas Jul 2013-Apr 2019



4.5.2 Model - 2

Backtesting on differenced values

CO_nat_gas - Backtesting

R²: 0.12/ 37.7925 * IPB51222SMoMPercentChangeL0



Backtesting on level values

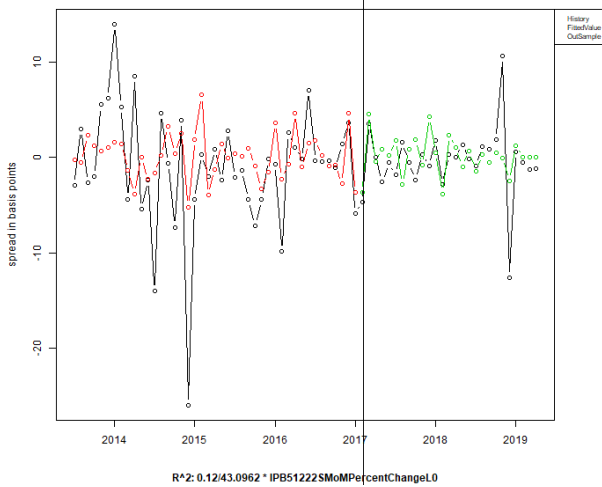
CO_nat_gas - Level Value Backtesting

R²: 0.12/ 37.7925 * IPB51222SMoMPercentChangeL0



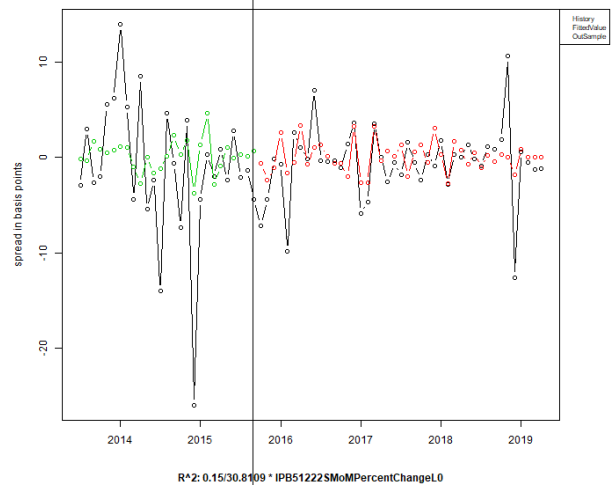
Out-of-Sample testing

CO_nat_gas Jul 2013-Apr 2019



Out-of-Sample testing

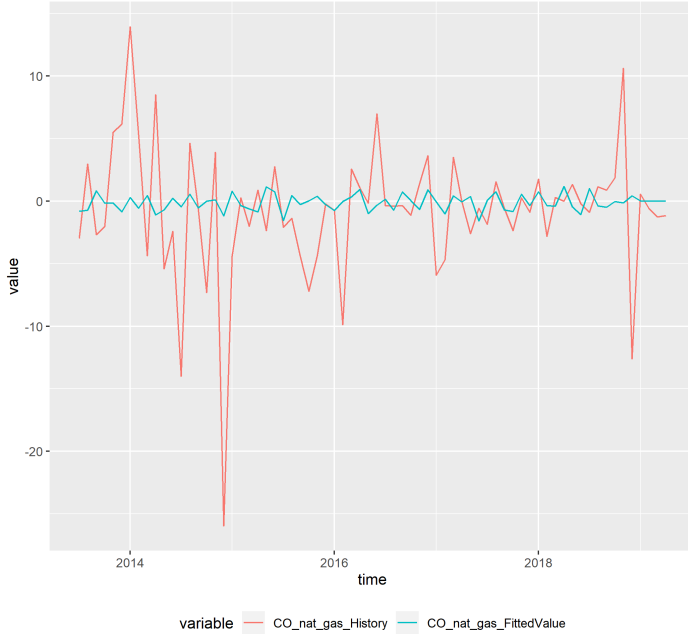
CO_nat_gas Jul 2013-Apr 2019



4.5.3 Model - 3

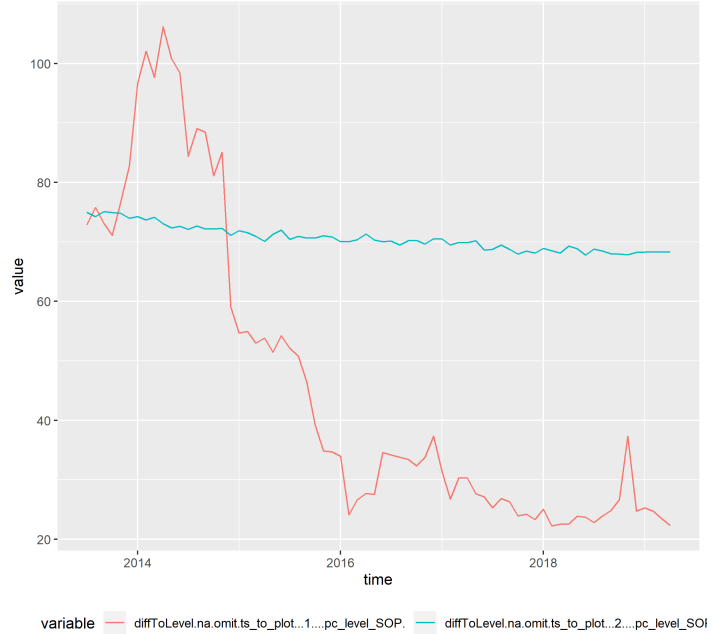
Backtesting on differenced values

CO_nat_gas - Backtesting
R²: 0/ 0.0075 * UEMP15T26MoMDifferenceL0



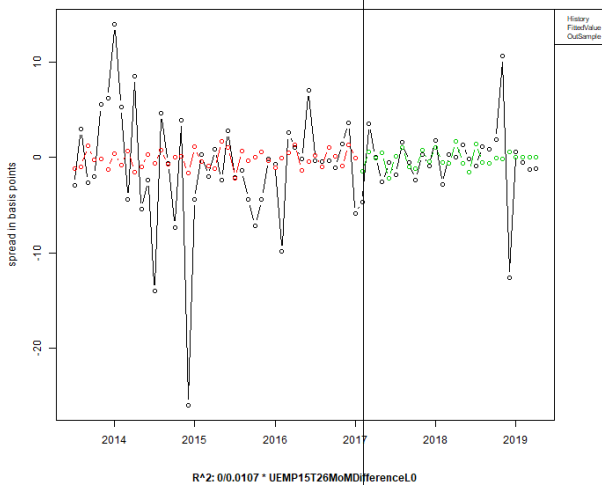
Backtesting on level values

CO_nat_gas - Level Value Backtesting
R²: 0/ 0.0075 * UEMP15T26MoMDifferenceL0



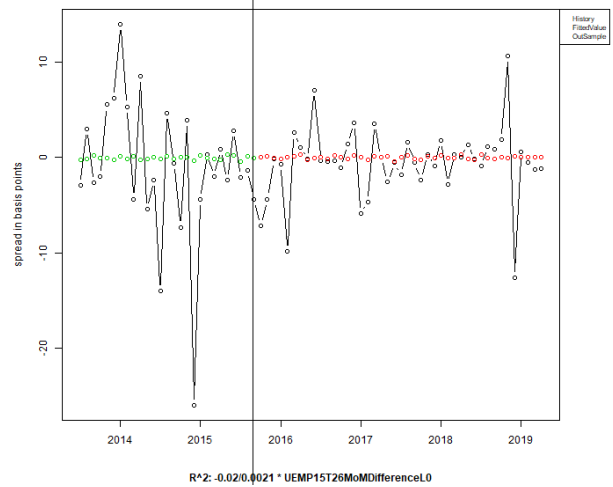
Out-of-Sample testing

CO_nat_gas Jul 2013-Apr 2019



Out-of-Sample testing

CO_nat_gas Jul 2013-Apr 2019



4.6 Two Factor Model for CO_{natgas}

4.7 Single Factor Model for CO_gasoline

	Variable	adjR2	AIC	Coef	Robst p	RMSE_OS	Criteria1	Criteria2	Criteria3
1	BAA10YB_IndexMoMPercentChangeL0	0.37	326.29	-18.86	0.00	2.46			
2	BAA10YB_IndexMoMDifferenceL0	0.32	331.77	-0.12	0.00	2.26			
3	USGG30YR_IndexMoMDifferenceL0	0.14	348.13	7.11	0.00	2.83			
4	USGG30YR_IndexMoMPercentChangeL0	0.14	348.30	19.96	0.00	2.77			
5	USGG10YR_IndexMoMDifferenceL0	0.11	350.37	5.91	0.00	2.67			
6	USGG10YR_IndexMoMPercentChangeL0	0.11	350.63	12.70	0.00	2.60			
7	ISRATIOMoMPercentChangeL0	0.10	351.12	-146.83	0.01	2.57			
8	VIX_IndexMoMPercentChangeL0	0.10	351.16	-3.87	0.01	2.27			

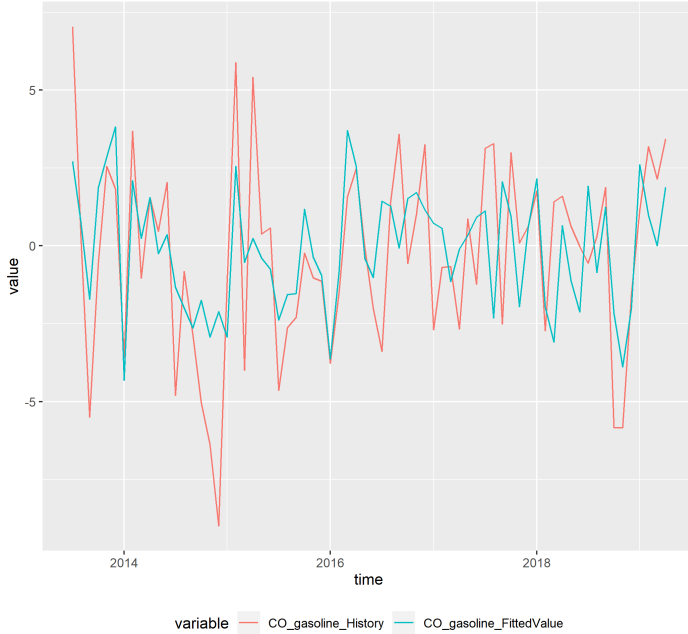
Table 6: Single Factor Analysis-CO_gasoline

4.7.1 Model - 1

Backtesting on differenced values

CO_gasoline - Backtesting

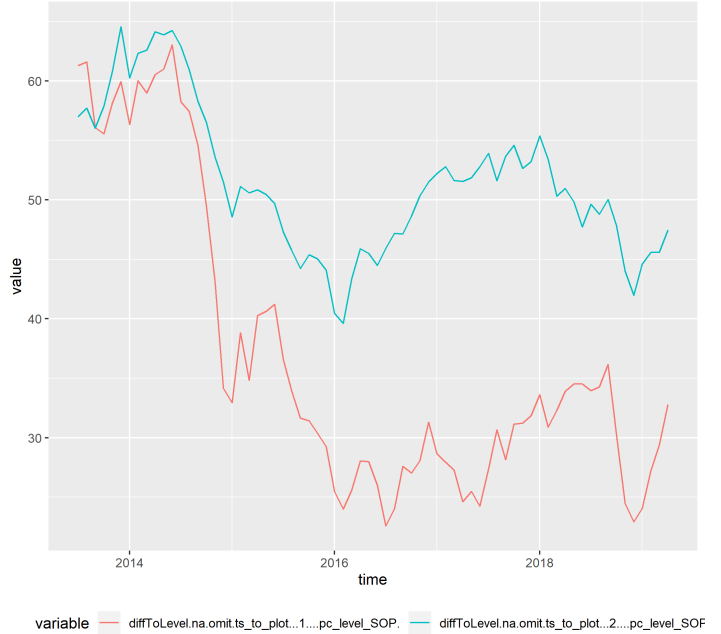
$R^2: 0.37 / -18.8629 * BAA10YB_IndexMoMPercentChangeL0$



Backtesting on level values

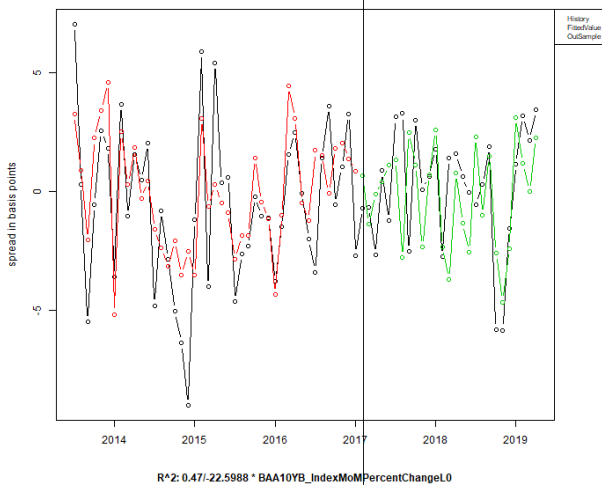
CO_gasoline - Level Value Backtesting

$R^2: 0.37 / -18.8629 * BAA10YB_IndexMoMPercentChangeL0$



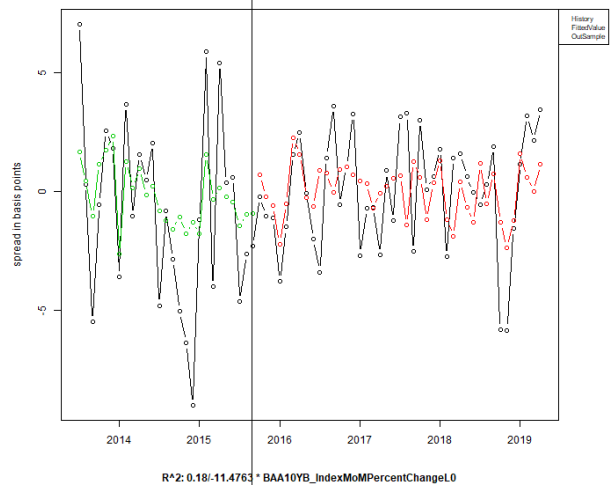
Out-of-Sample testing

CO_gasoline Jul 2013-Apr 2019



Out-of-Sample testing

CO_gasoline Jul 2013-Apr 2019

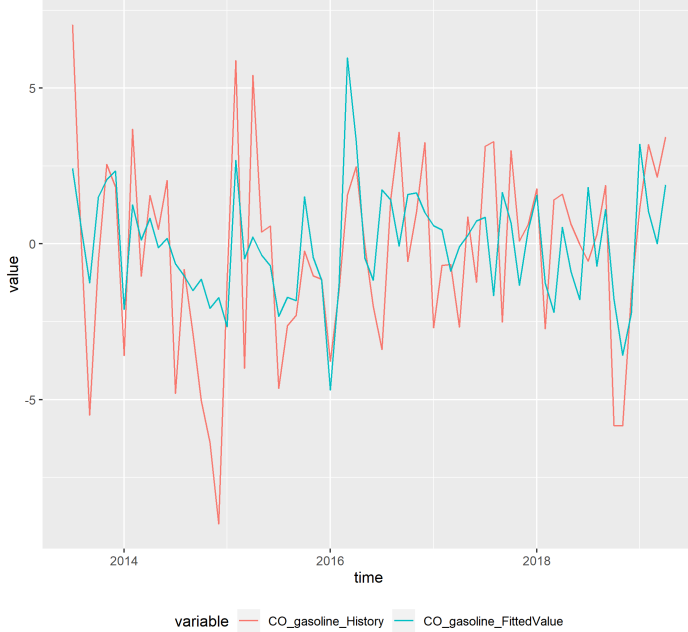


4.7.2 Model - 2

Backtesting on differenced values

CO_gasoline - Backtesting

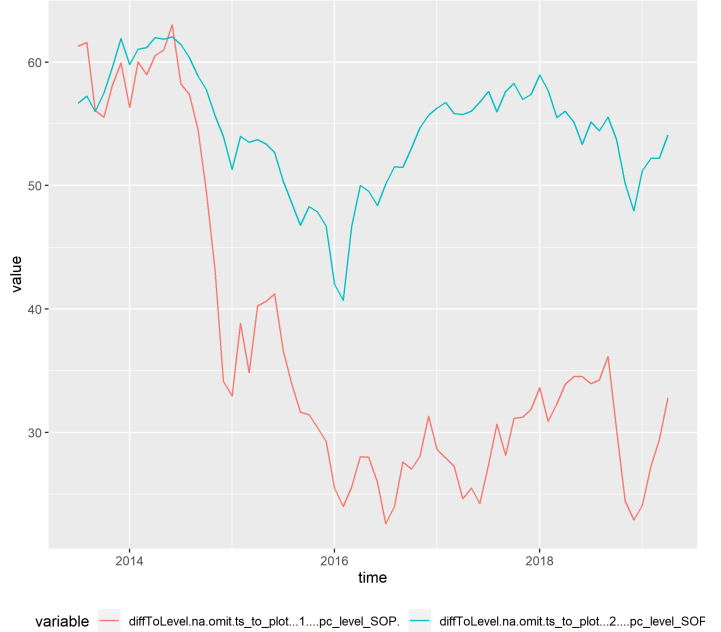
$R^2: 0.32 / -0.1208 * BAA10YB_IndexMoMDifferenceL0$



Backtesting on level values

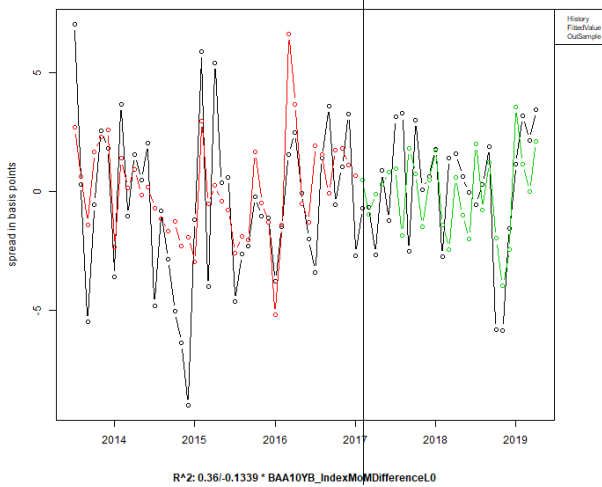
CO_gasoline - Level Value Backtesting

$R^2: 0.32 / -0.1208 * BAA10YB_IndexMoMDifferenceL0$



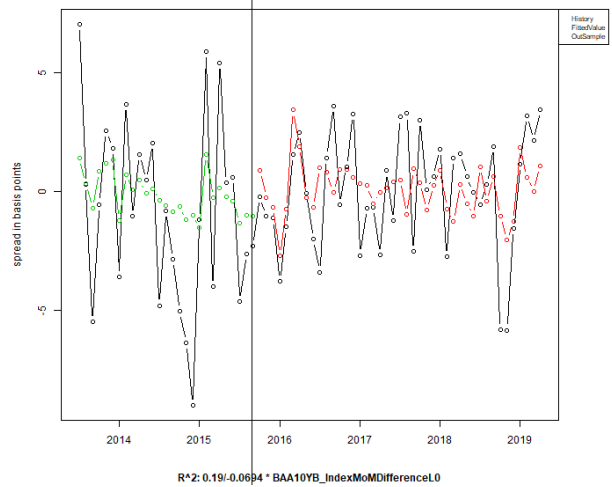
Out-of-Sample testing

CO_gasoline Jul 2013-Apr 2019



Out-of-Sample testing

CO_gasoline Jul 2013-Apr 2019

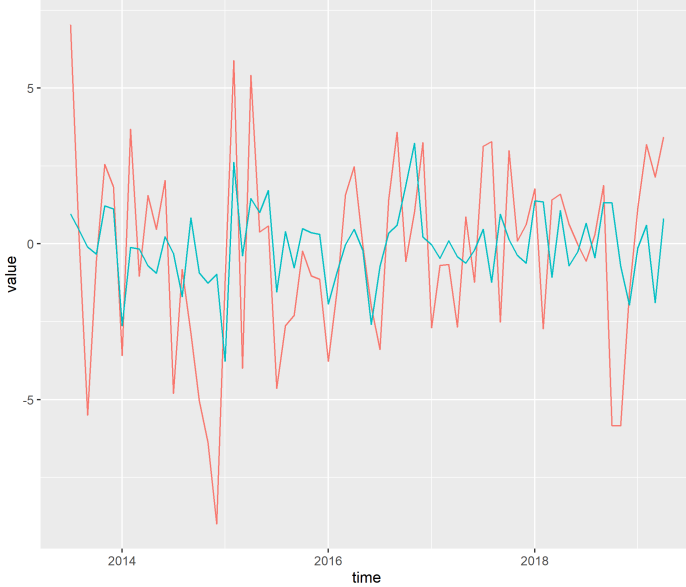


4.7.3 Model - 3

Backtesting on differenced values

CO_gasoline - Backtesting

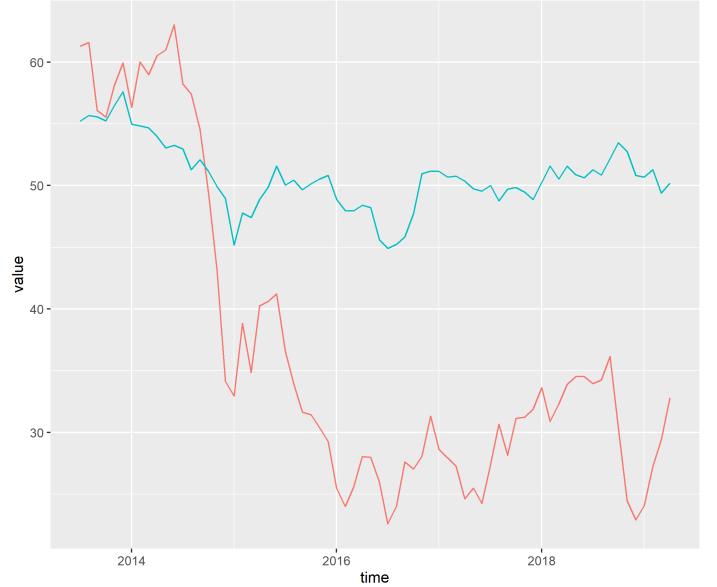
$R^2: 0.14 / 7.1116 * USGG30YR_IndexMoMDifferenceL0$



Backtesting on level values

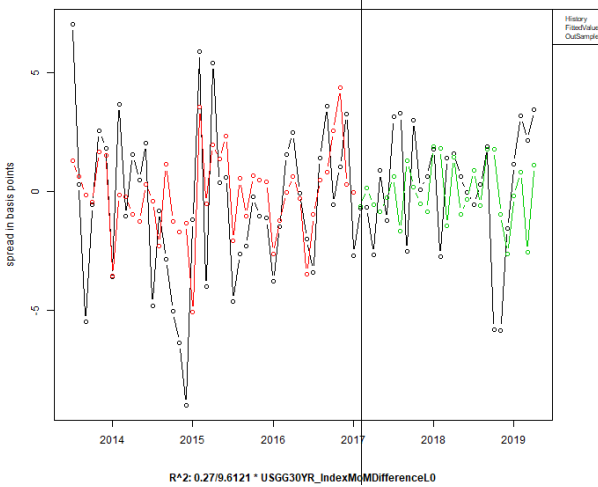
CO_gasoline - Level Value Backtesting

$R^2: 0.14 / 7.1116 * USGG30YR_IndexMoMDifferenceL0$



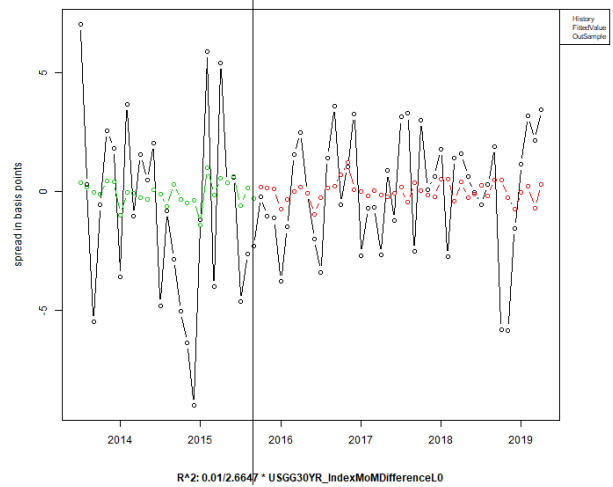
Out-of-Sample testing

CO_gasoline Jul 2013-Apr 2019



Out-of-Sample testing

CO_gasoline Jul 2013-Apr 2019

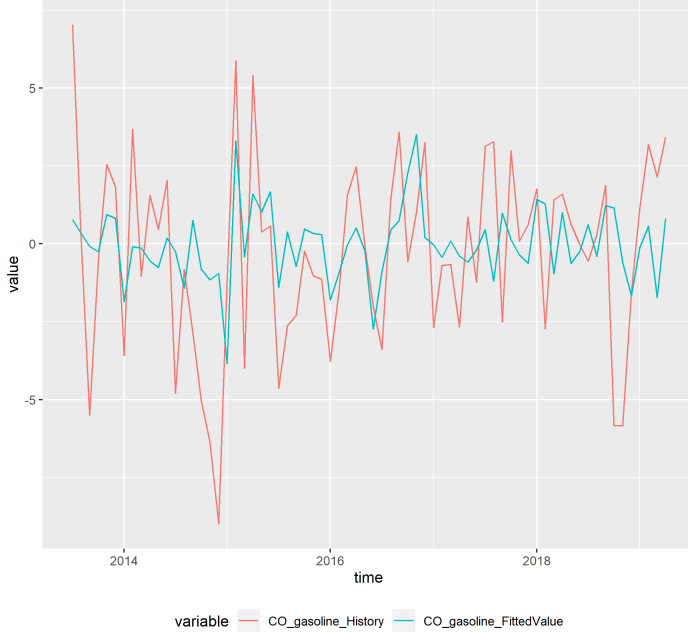


4.7.4 Model - 4

Backtesting on differenced values

CO_gasoline - Backtesting

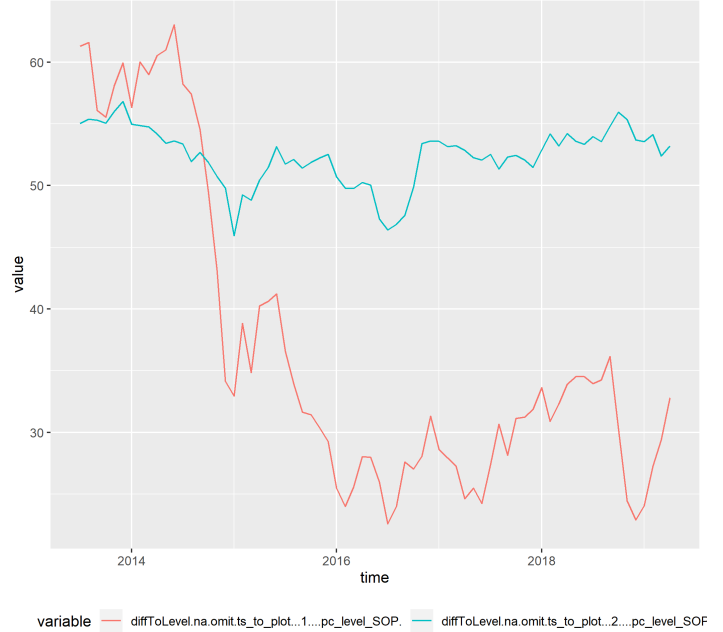
R²: 0.14/ 19.9606 * USGG30YR_IndexMoMPercentChangeL0



Backtesting on level values

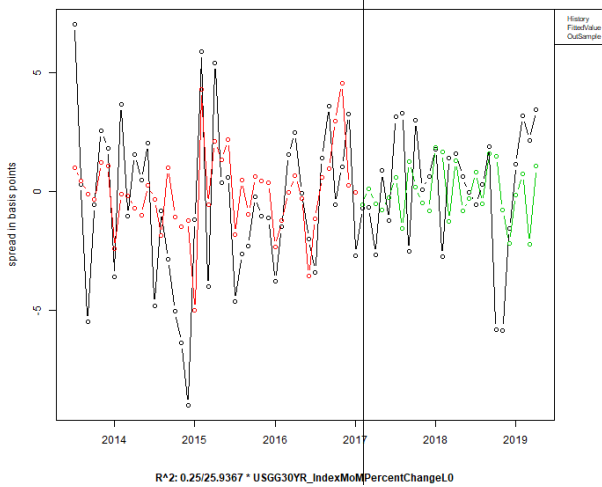
CO_gasoline - Level Value Backtesting

R²: 0.14/ 19.9606 * USGG30YR_IndexMoMPercentChangeL0



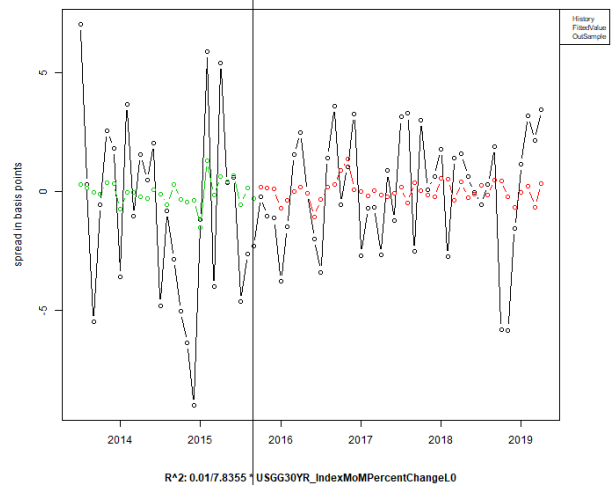
Out-of-Sample testing

CO_gasoline Jul 2013-Apr 2019



Out-of-Sample testing

CO_gasoline Jul 2013-Apr 2019



4.7.5 Model - 5

Backtesting on differenced values

CO_gasoline - Backtesting

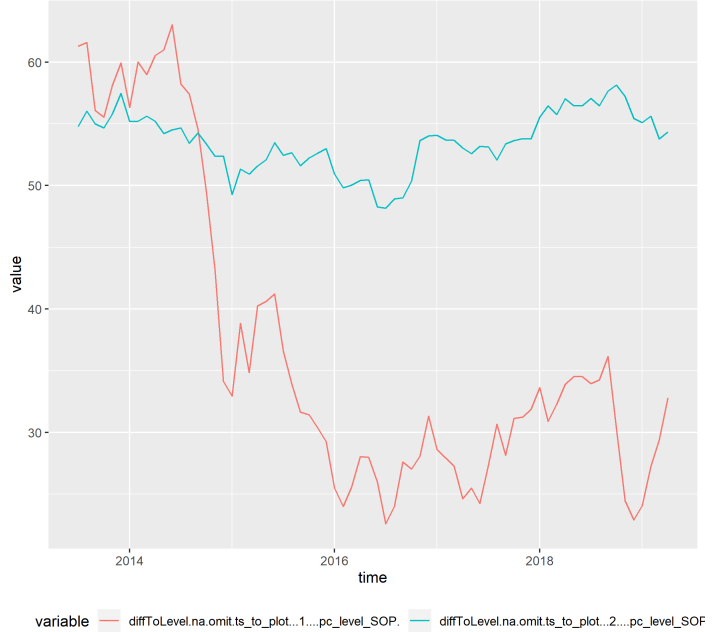
$R^2: 0.11 / 5.9053 * USGG10YR_IndexMoMDifferenceL0$



Backtesting on level values

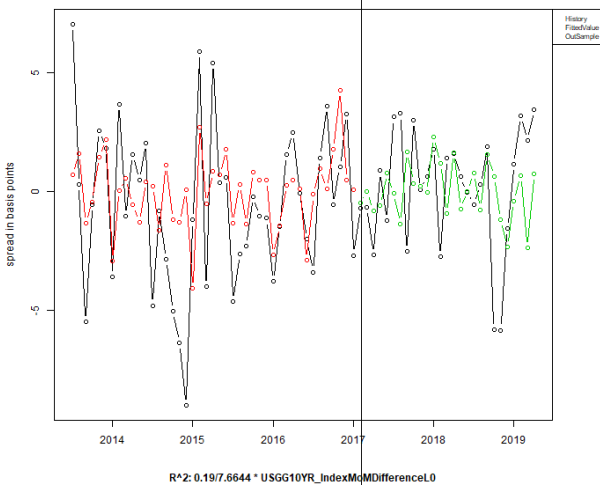
CO_gasoline - Level Value Backtesting

$R^2: 0.11 / 5.9053 * USGG10YR_IndexMoMDifferenceL0$



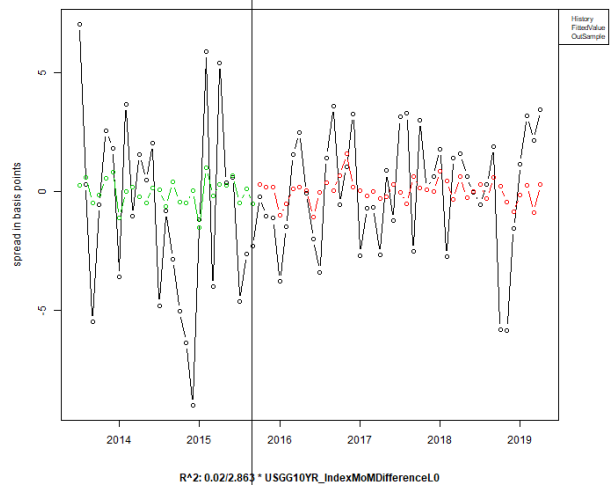
Out-of-Sample testing

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Out-of-Sample testing

CO_gasoline Jul 2013-Apr 2019

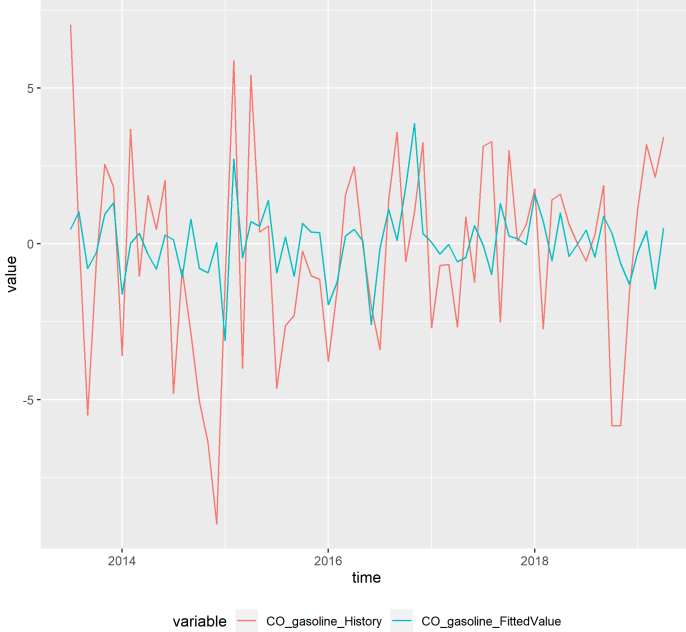


4.7.6 Model - 6

Backtesting on differenced values

CO_gasoline - Backtesting

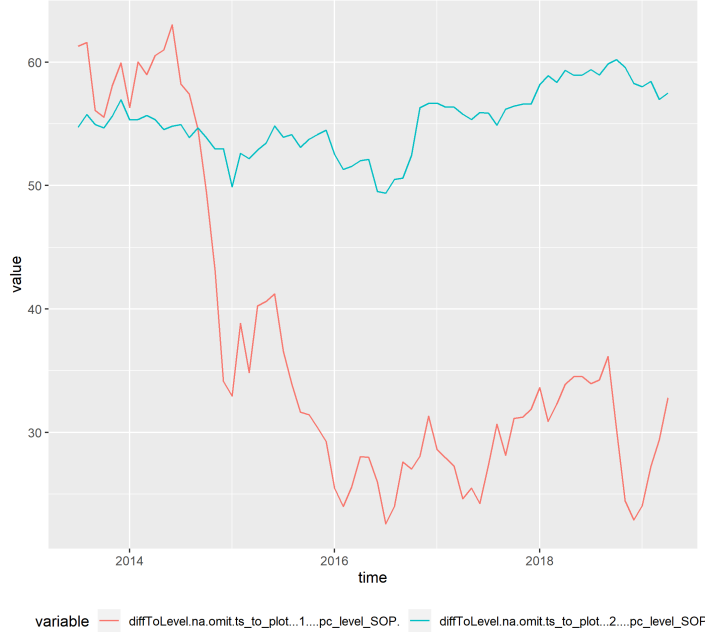
R²: 0.11/ 12.6966 * USGG10YR_IndexMoMPercentChangeL0



Backtesting on level values

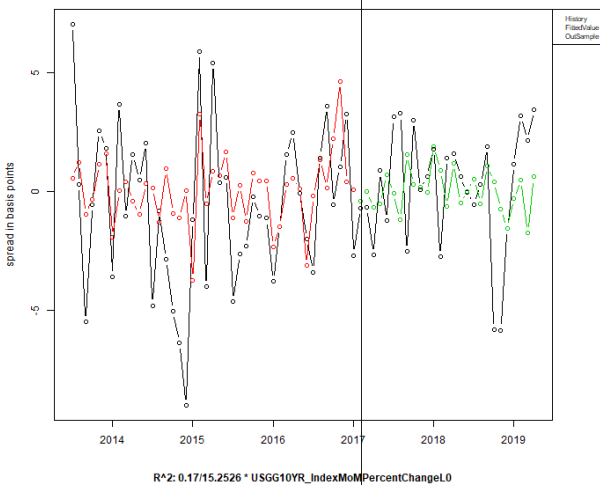
CO_gasoline - Level Value Backtesting

R²: 0.11/ 12.6966 * USGG10YR_IndexMoMPercentChangeL0



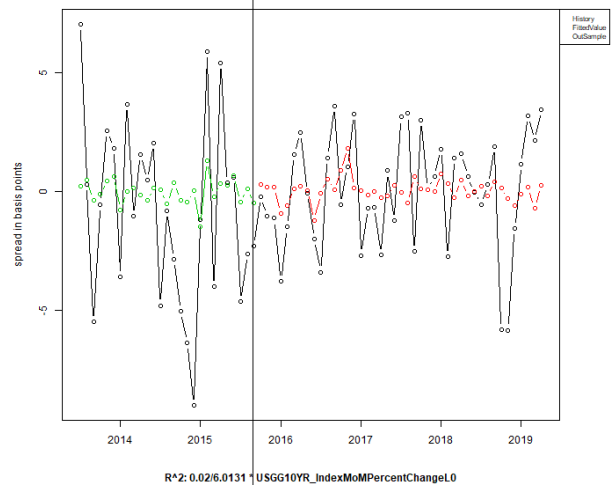
Out-of-Sample testing

CO_gasoline Jul 2013-Apr 2019



Out-of-Sample testing

CO_gasoline Jul 2013-Apr 2019

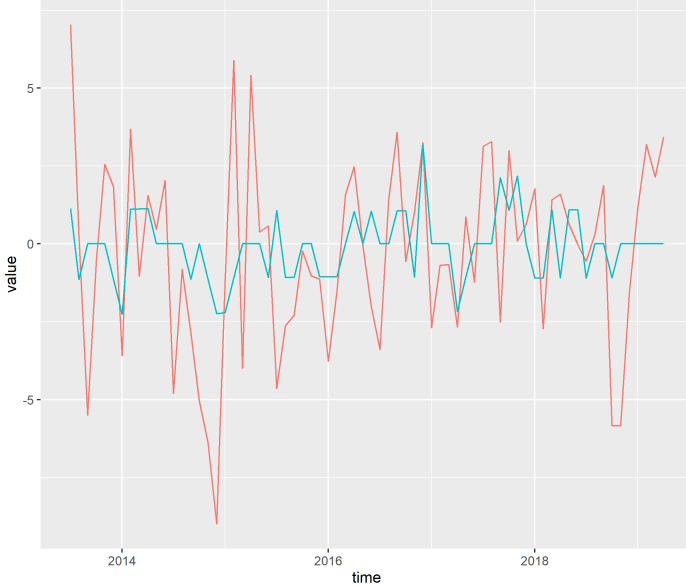


4.7.7 Model - 7

Backtesting on differenced values

CO_gasoline - Backtesting

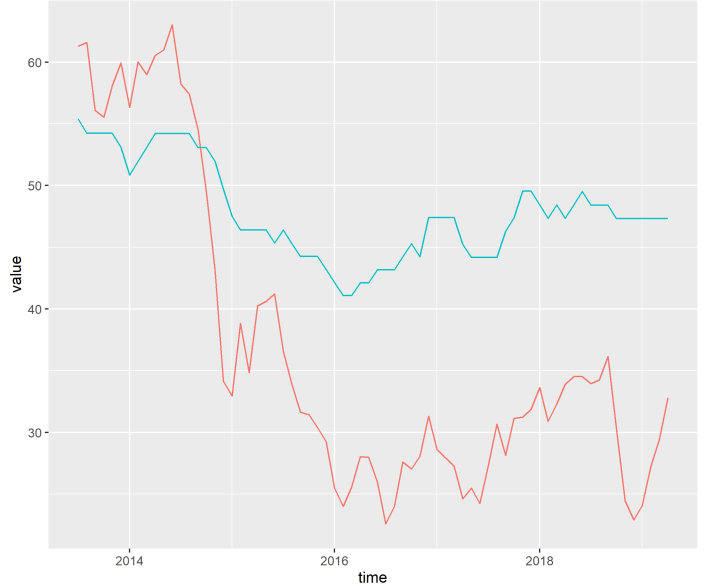
R²: 0.1/ -146.8314 * ISRATIOMoMPercentChangeL0



Backtesting on level values

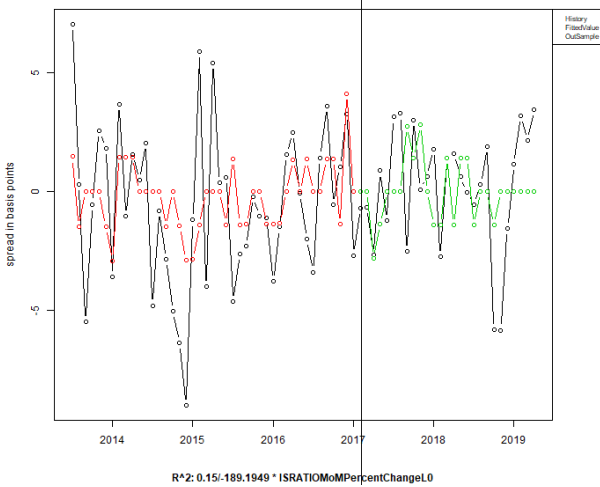
CO_gasoline - Level Value Backtesting

R²: 0.1/ -146.8314 * ISRATIOMoMPercentChangeL0



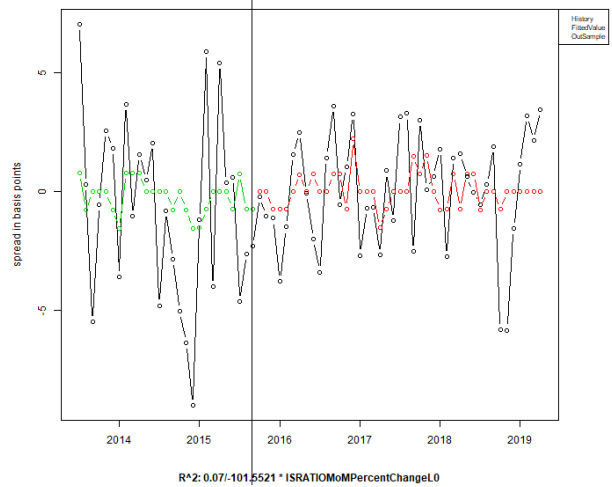
Out-of-Sample testing

CO_gasoline Jul 2013-Apr 2019



Out-of-Sample testing

CO_gasoline Jul 2013-Apr 2019

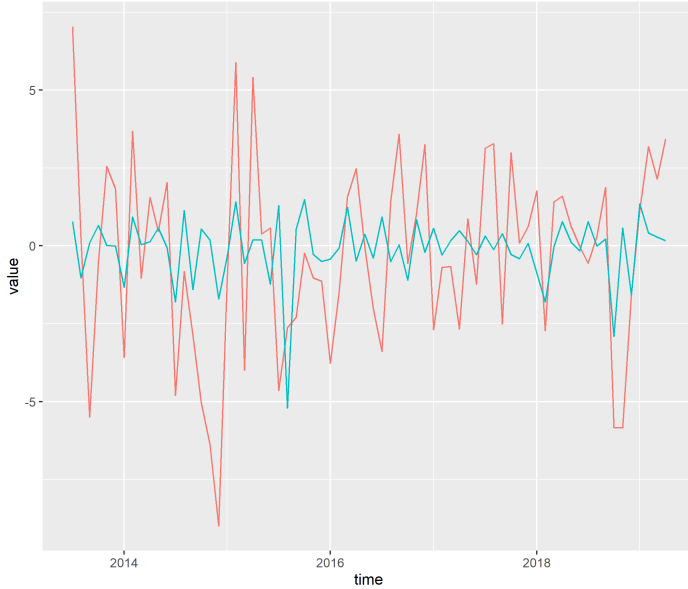


4.7.8 Model - 8

Backtesting on differenced values

CO_gasoline - Backtesting

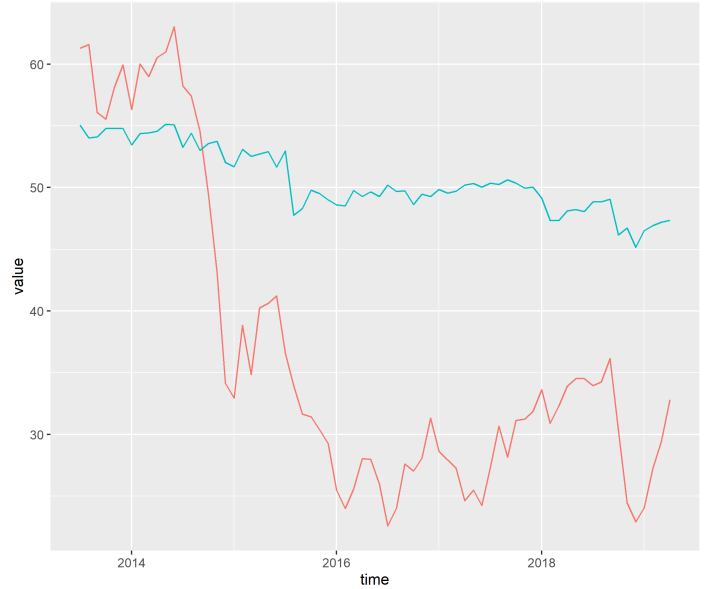
$R^2: 0.1 / -3.8705 * VIX_IndexMoMPercentChangeL0$



Backtesting on level values

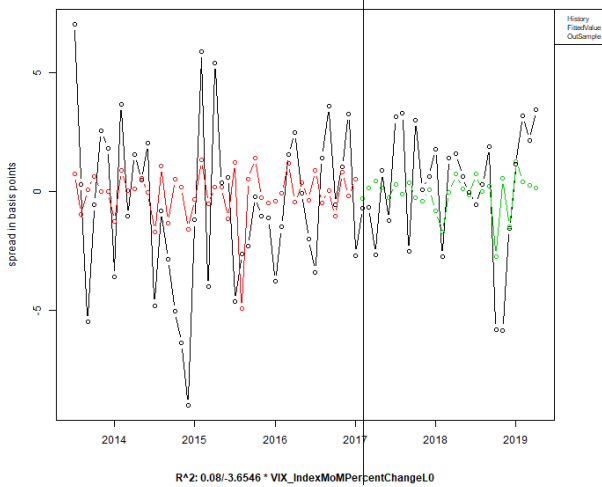
CO_gasoline - Level Value Backtesting

$R^2: 0.1 / -3.8705 * VIX_IndexMoMPercentChangeL0$



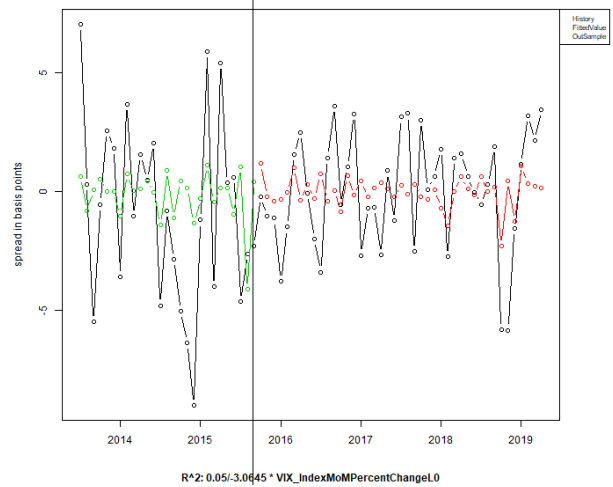
Out-of-Sample testing

CO_gasoline Jul 2013-Apr 2019



Out-of-Sample testing

CO_gasoline Jul 2013-Apr 2019



4.8 Two Factor Model for $\text{CO}_{gasoline}$

4.9 Single Factor Model for CO_g*old*

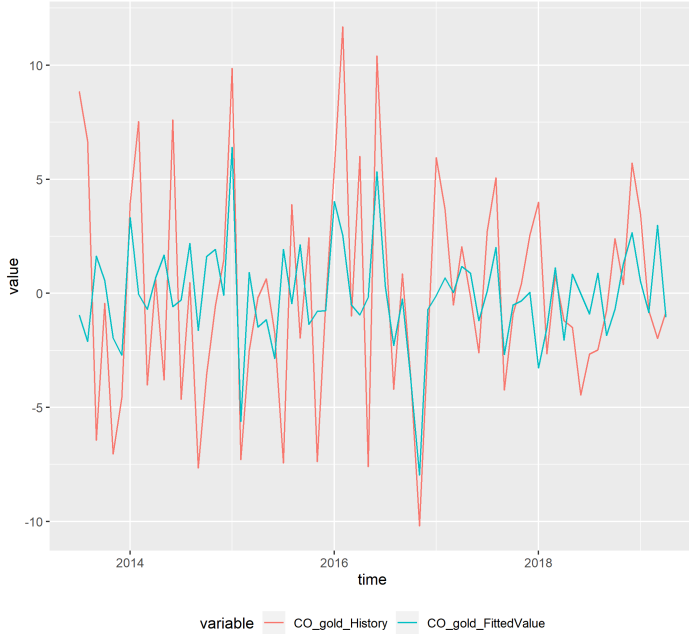
	Variable	adjR2	AIC	Coef	Robst p	RMSE_OS	Criteria1	Criteria2	Criteria3
1	USGG10YR_IndexMoMPercentChangeL0	0.21	401.44	-26.18	0.00	2.67			
2	USGG30YR_IndexMoMPercentChangeL0	0.18	404.50	-33.98	0.00	2.79			
3	USGG10YR_IndexMoMDifferenceL0	0.17	405.01	-10.86	0.00	2.76			
4	USGG30YR_IndexMoMDifferenceL0	0.16	406.16	-11.38	0.00	2.80			
5	USGG2YR_IndexMoMDifferenceL0	0.13	408.52	-14.40	0.00	3.50			
6	USGG2YR_IndexMoMPercentChangeL0	0.11	409.78	-10.74	0.01	2.66			

Table 7: Single Factor Analysis-CO_g*old*

4.9.1 Model - 1

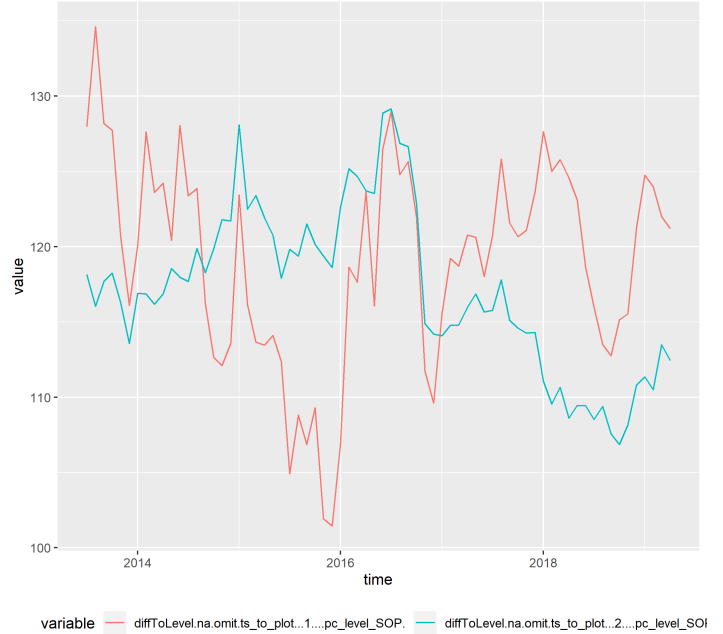
Backtesting on differenced values

CO_gold - Backtesting
R²: 0.21/-26.1767 * USGG10YR_IndexMoMPercentChangeL0



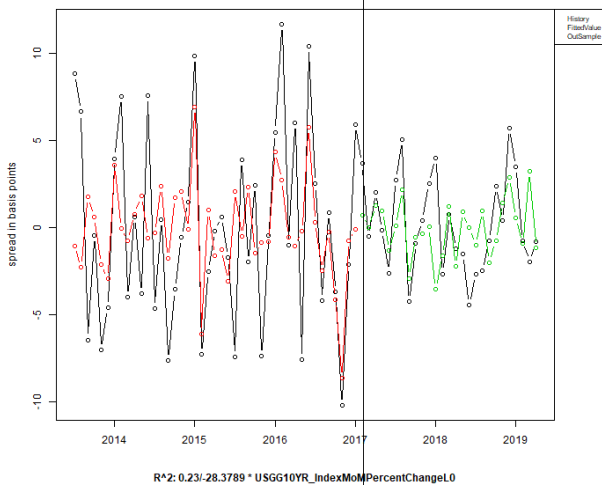
Backtesting on level values

CO_gold - Level Value Backtesting
R²: 0.21/-26.1767 * USGG10YR_IndexMoMPercentChangeL0



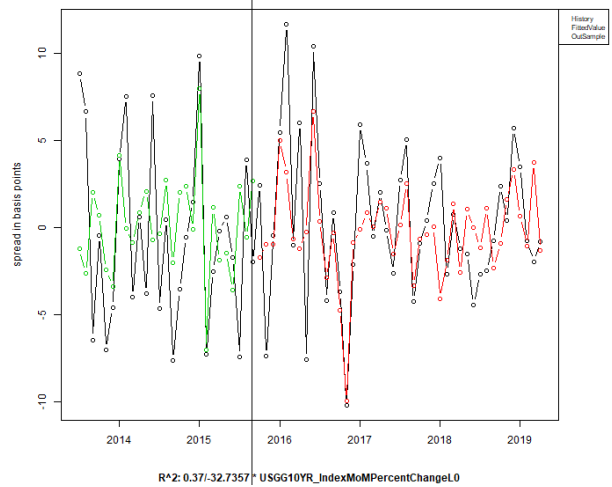
Out-of-Sample testing

CO_gold Jul 2013-Apr 2019



Out-of-Sample testing

CO_gold Jul 2013-Apr 2019

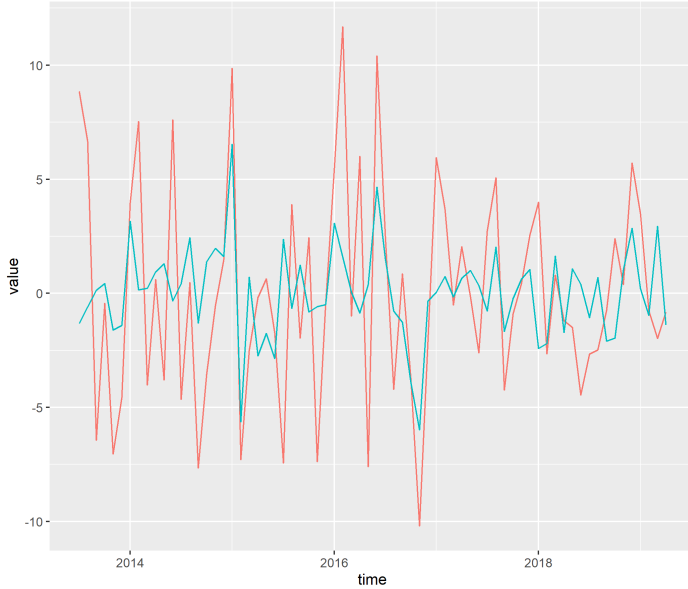


4.9.2 Model - 2

Backtesting on differenced values

CO_gold - Backtesting

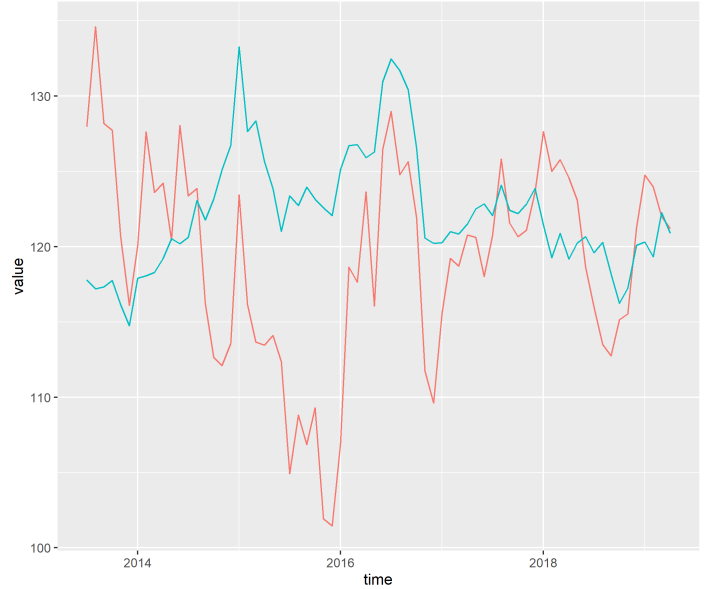
$R^2: 0.18 / -33.9817 * USGG30YR_IndexMoMPercentChangeL0$



Backtesting on level values

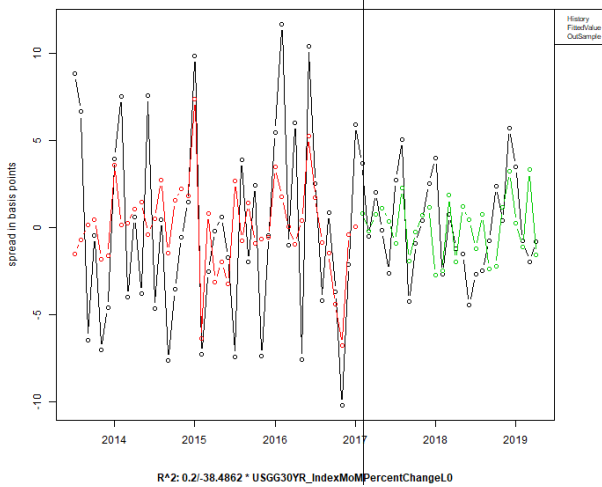
CO_gold - Level Value Backtesting

$R^2: 0.18 / -33.9817 * USGG30YR_IndexMoMPercentChangeL0$



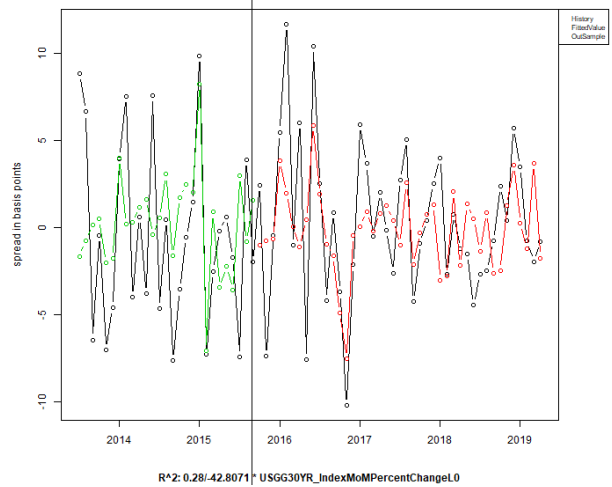
Out-of-Sample testing

CO_gold Jul 2013-Apr 2019



Out-of-Sample testing

CO_gold Jul 2013-Apr 2019

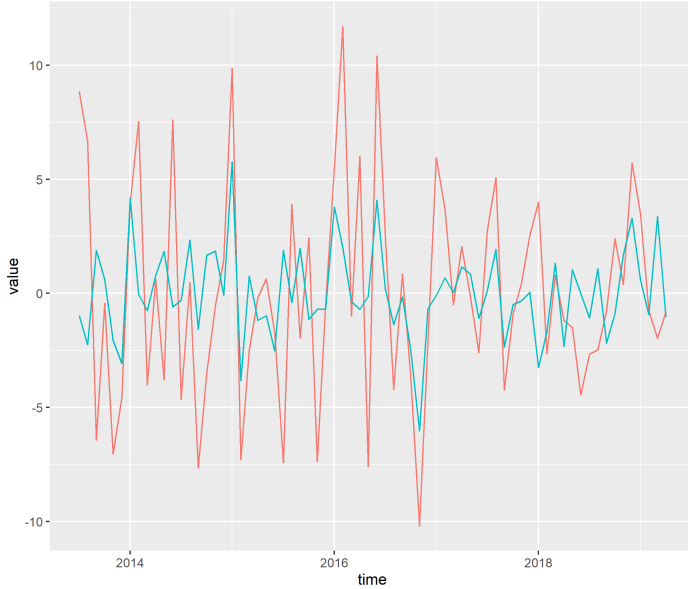


4.9.3 Model - 3

Backtesting on differenced values

CO_gold - Backtesting

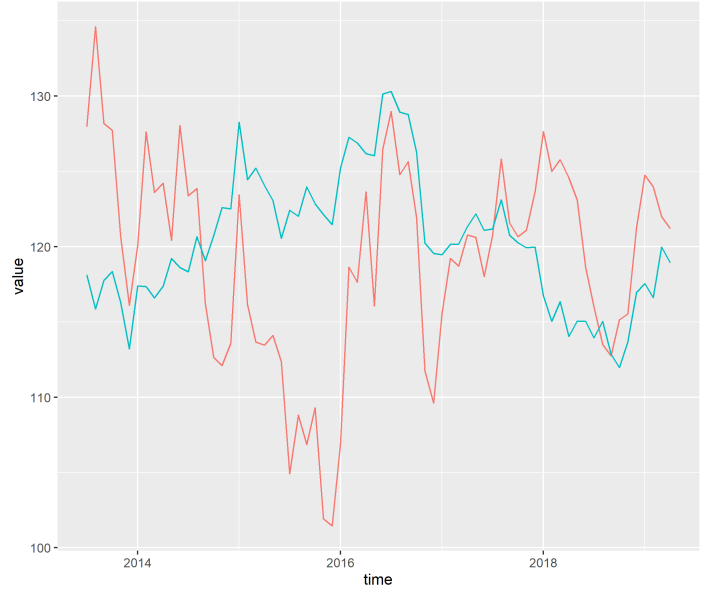
R²: 0.17/ -10.8564 * USGG10YR_IndexMoMDifferenceL0



Backtesting on level values

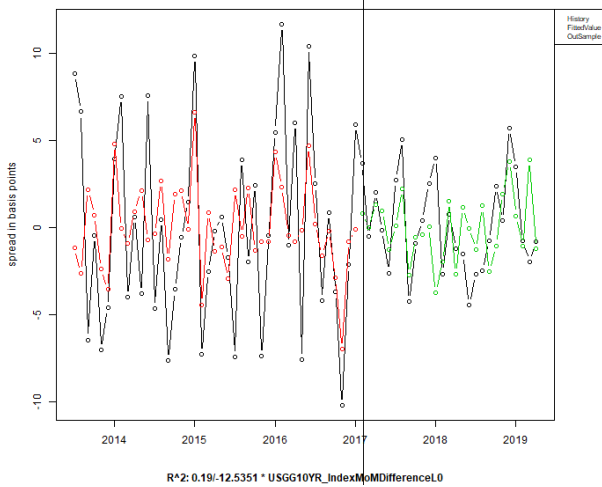
CO_gold - Level Value Backtesting

R²: 0.17/ -10.8564 * USGG10YR_IndexMoMDifferenceL0



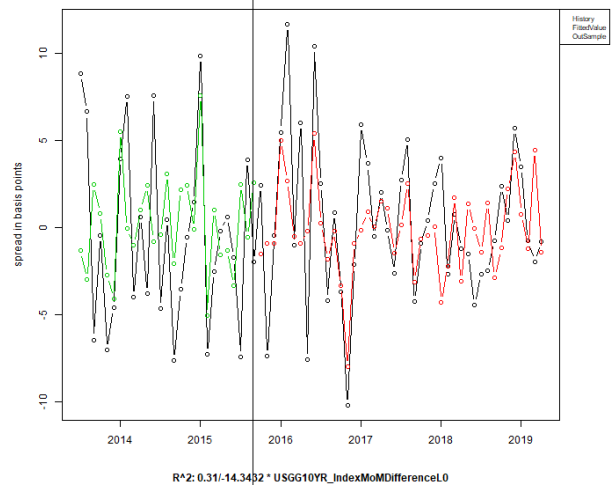
Out-of-Sample testing

CO_gold Jul 2013-Apr 2019



Out-of-Sample testing

CO_gold Jul 2013-Apr 2019

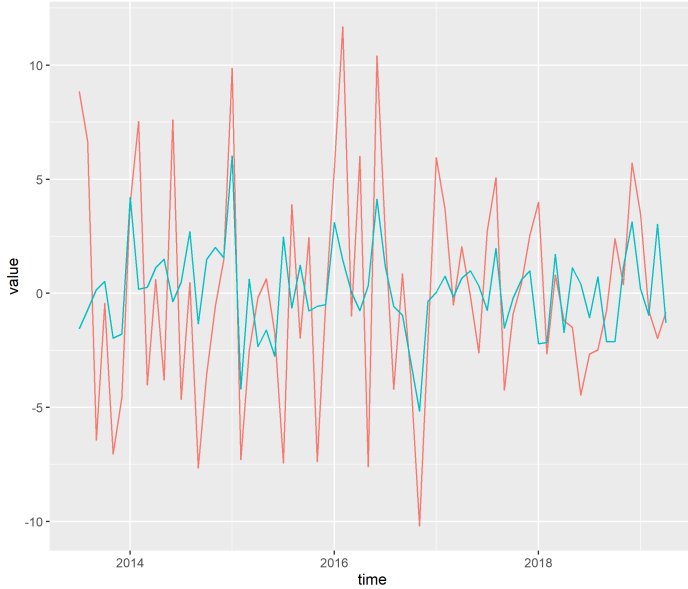


4.9.4 Model - 4

Backtesting on differenced values

CO_gold - Backtesting

R²: 0.16/ -11.3799 * USGG30YR_IndexMoMDifferenceL0

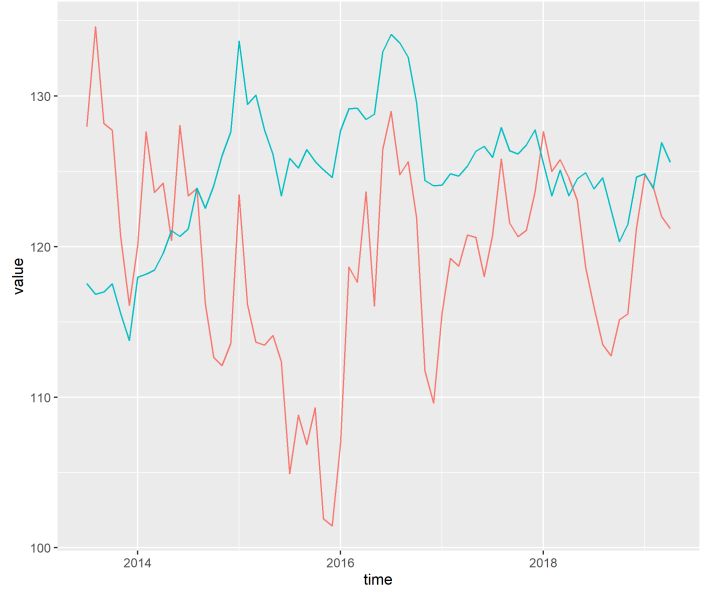


variable — CO_gold_History — CO_gold_FittedValue

Backtesting on level values

CO_gold - Level Value Backtesting

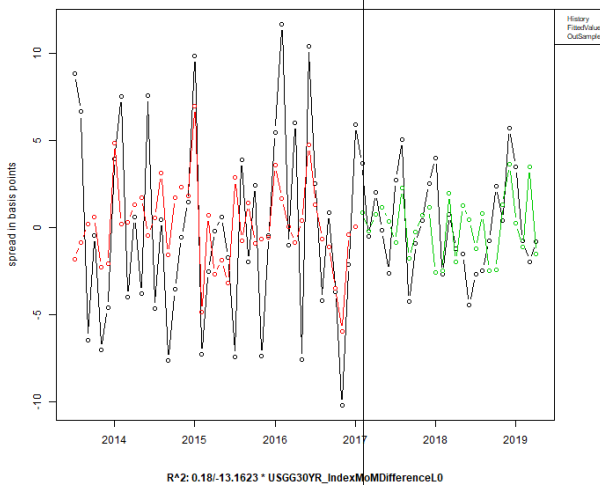
R²: 0.16/ -11.3799 * USGG30YR_IndexMoMDifferenceL0



variable — diffToLevel.na.omit.ts...1....pc_level_SOP — diffToLevel.na.omit.ts...2....pc_level_SOI

Out-of-Sample testing

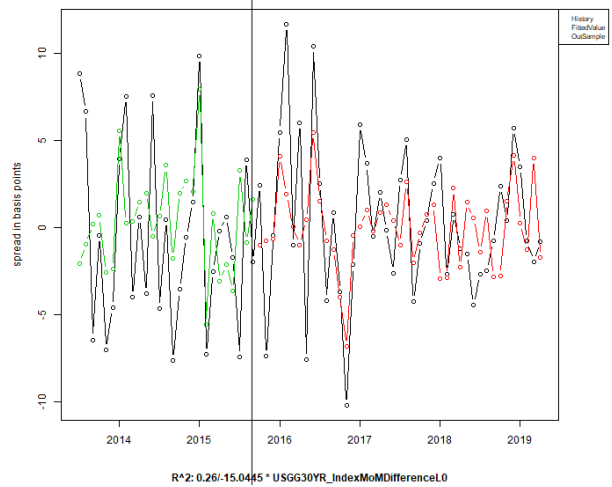
CO_gold Jul 2013-Apr 2019



R²: 0.18/-13.1623 * USGG30YR_IndexMoMDifferenceL0

Out-of-Sample testing

CO_gold Jul 2013-Apr 2019



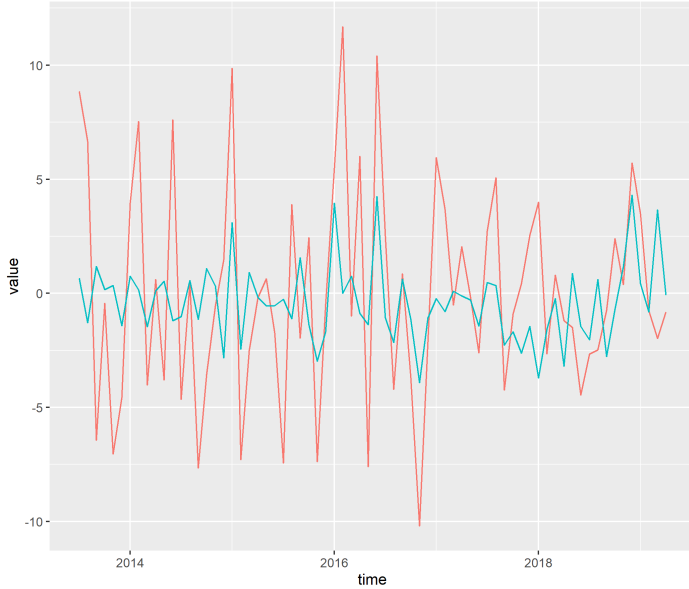
R²: 0.26/-15.0445 * USGG30YR_IndexMoMDifferenceL0

4.9.5 Model - 5

Backtesting on differenced values

CO_gold - Backtesting

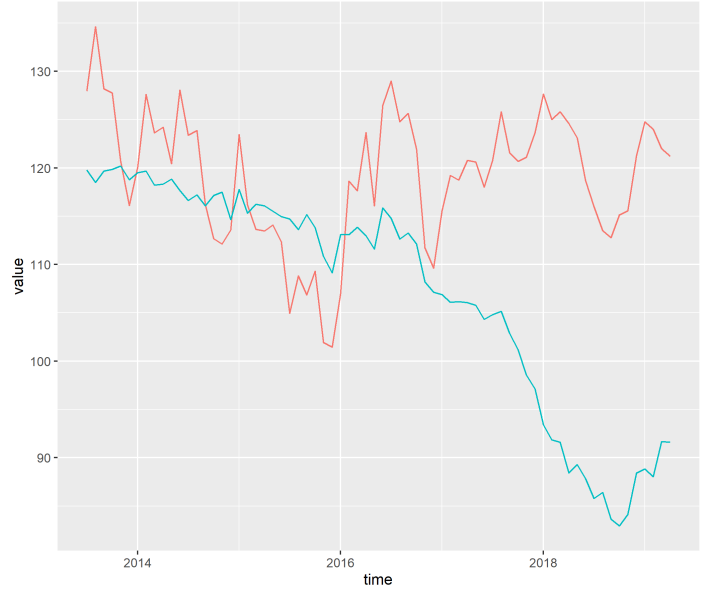
$R^2: 0.13 / -14.3967 * USGG2YR_IndexMoMDifferenceL0$



Backtesting on level values

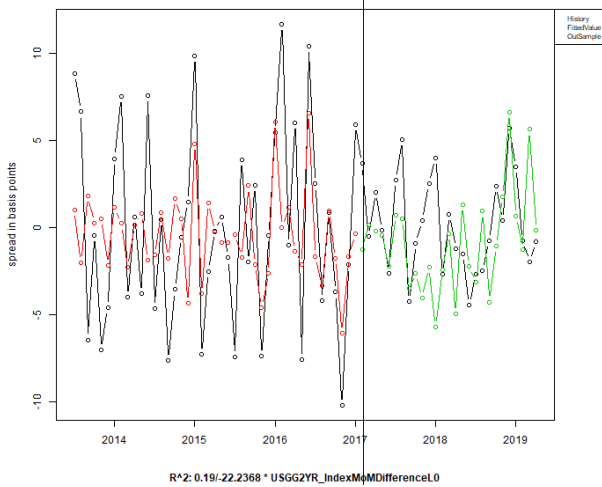
CO_gold - Level Value Backtesting

$R^2: 0.13 / -14.3967 * USGG2YR_IndexMoMDifferenceL0$



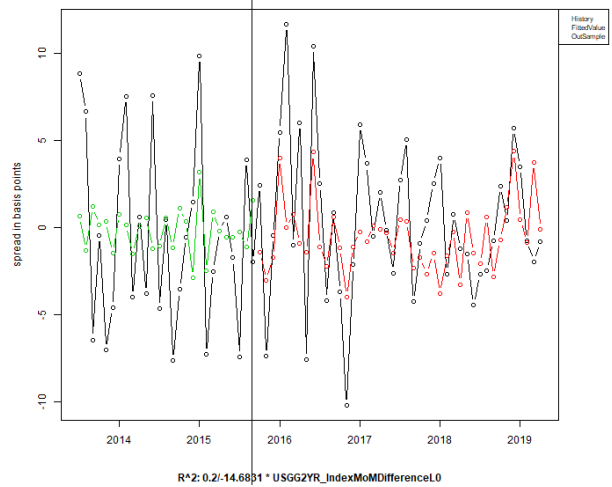
Out-of-Sample testing

CO_gold Jul 2013-Apr 2019



Out-of-Sample testing

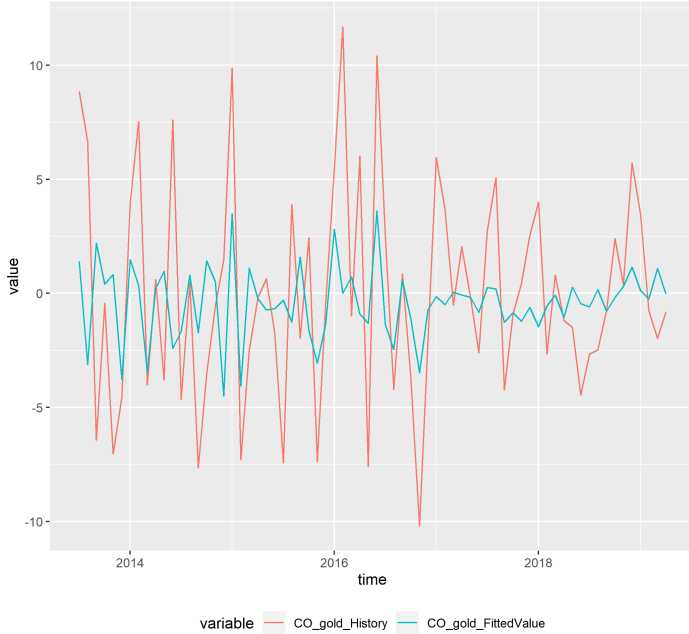
CO_gold Jul 2013-Apr 2019



4.9.6 Model - 6

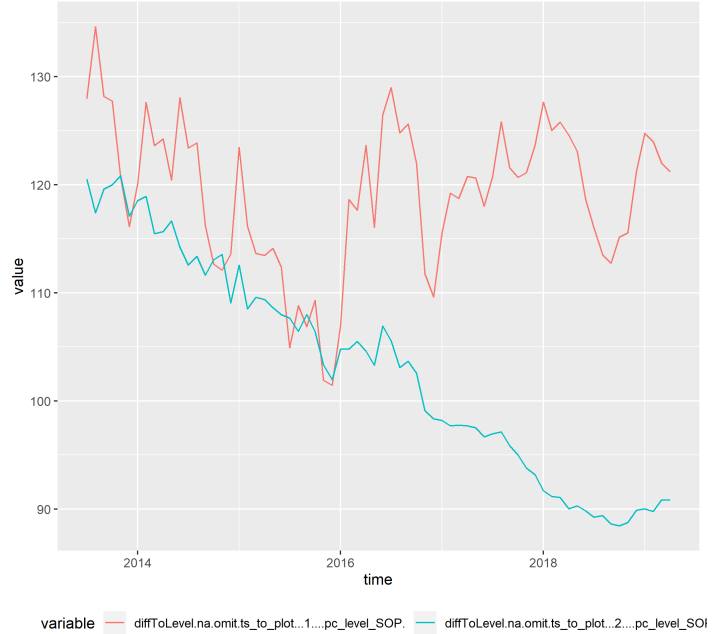
Backtesting on differenced values

CO_gold - Backtesting
R²: 0.11 / -10.743 * USGG2YR_IndexMoMPercentChangeL0



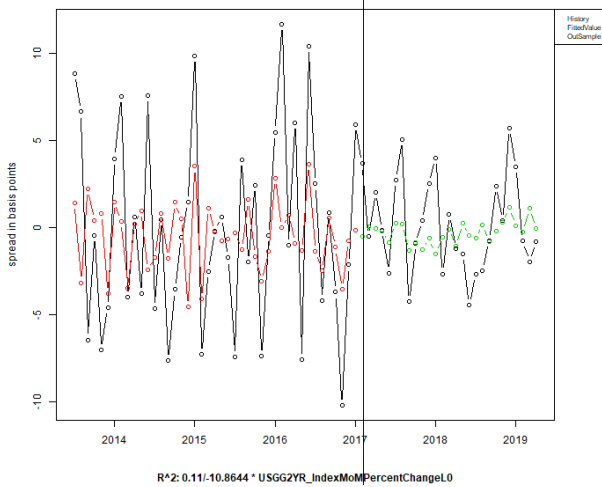
Backtesting on level values

CO_gold - Level Value Backtesting
R²: 0.11 / -10.743 * USGG2YR_IndexMoMPercentChangeL0



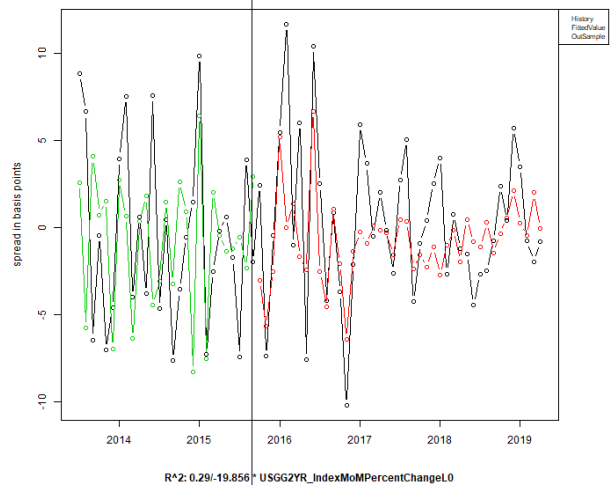
Out-of-Sample testing

CO_gold Jul 2013-Apr 2019



Out-of-Sample testing

CO_gold Jul 2013-Apr 2019



4.10 Two Factor Model for $CO_{g\text{old}}$

4.11 Single Factor Model for CO_{silver}

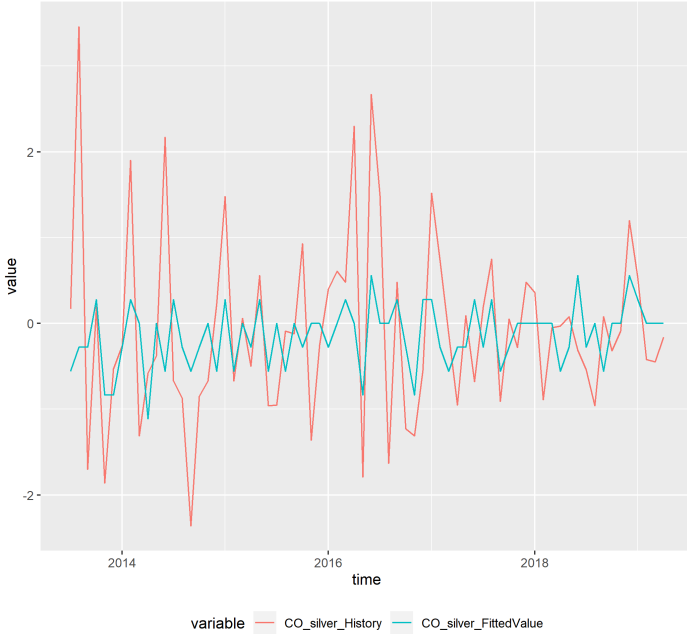
	Variable	adjR2	AIC	Coef	Robst p	RMSE_OS	Criteria1	Criteria2	Criteria3
1	UNRATEMoMDifferenceL0	0.12	202.69	2.78	0.00	0.57			
2	UNRATEMoMPercentChangeL0	0.12	202.77	14.45	0.00	0.66			

Table 8: Single Factor Analysis-CO_{silver}

4.11.1 Model - 1

Backtesting on differenced values

CO_silver - Backtesting
 $R^2: 0.12 / 2.7831 * UNRATEMoMDifferenceL0$



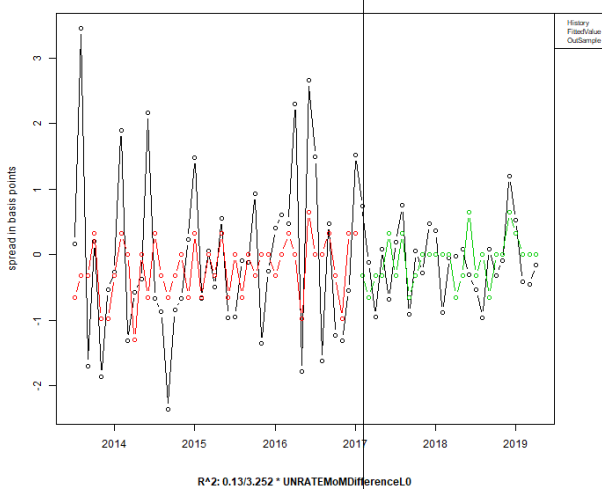
Backtesting on level values

CO_silver - Level Value Backtesting
 $R^2: 0.12 / 2.7831 * UNRATEMoMDifferenceL0$



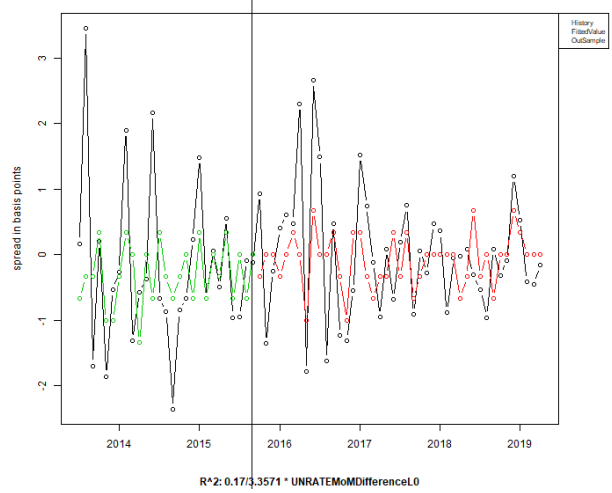
Out-of-Sample testing

CO_silver Jul 2013-Apr 2019



Out-of-Sample testing

CO_silver Jul 2013-Apr 2019



4.11.2 Model - 2

Backtesting on differenced values

CO_silver - Backtesting

$R^2: 0.12 / 14.4503 * \text{UNRATEMoMPercentChangeL0}$



Backtesting on level values

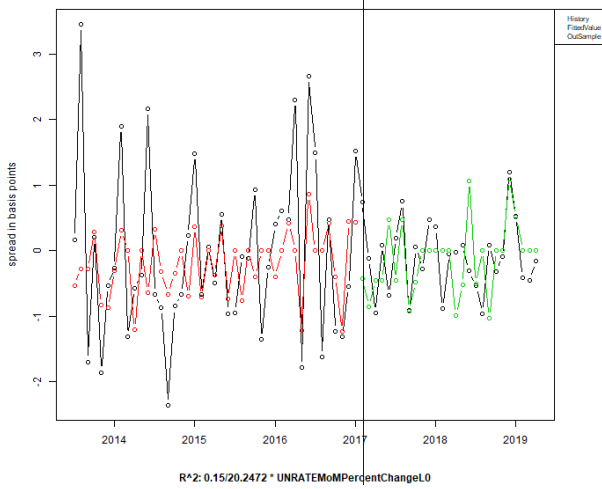
CO_silver - Level Value Backtesting

$R^2: 0.12 / 14.4503 * \text{UNRATEMoMPercentChangeL0}$



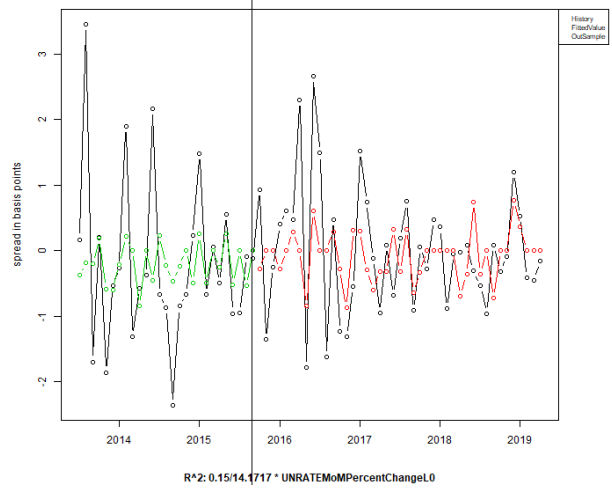
Out-of-Sample testing

CO_silver Jul 2013-Apr 2019



Out-of-Sample testing

CO_silver Jul 2013-Apr 2019



4.12 Two Factor Model for CO_silver

4.13 Single Factor Model for CO_{grains}

	Variable	adjR2	AIC	Coef	Robst p	RMSE_OS	Criteria1	Criteria2	Criteria3
1	AAIIBEAR_IndexMoMDifferenceL0	0.00	399.16	-0.06	0.19	5.67			

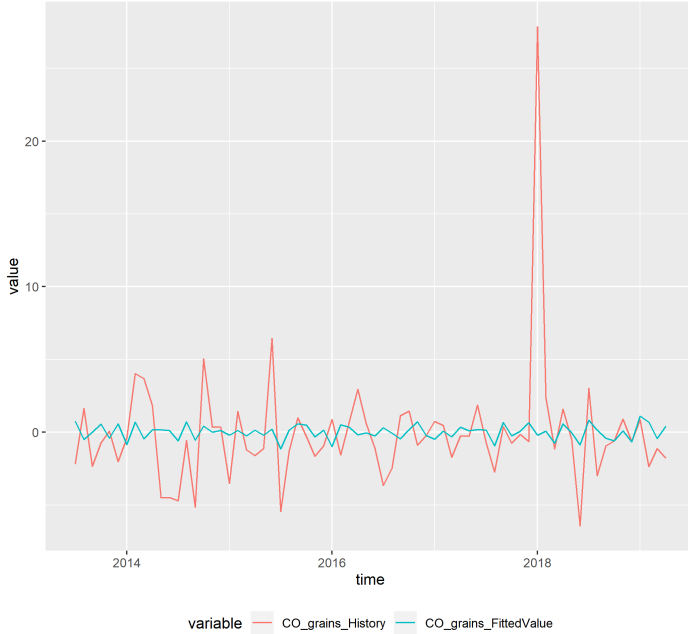
Table 9: Single Factor Analysis- CO_{grains}

4.13.1 Model - 1

Backtesting on differenced values

CO_grains - Backtesting

$R^2: 0 / -0.0601 * AAIBEAR_IndexMoMDifferenceL0$



Backtesting on level values

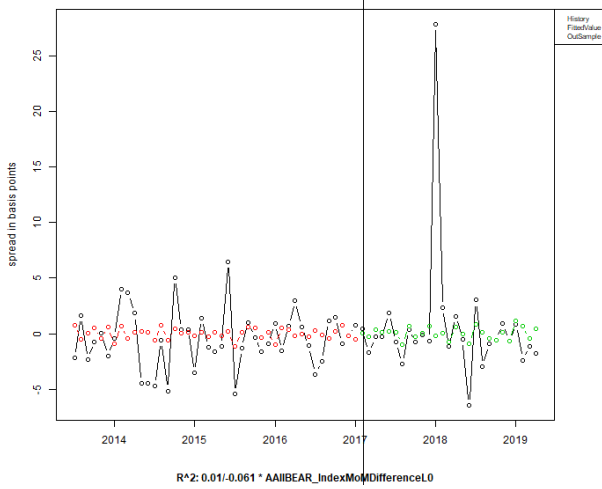
CO_grains - Level Value Backtesting

$R^2: 0 / -0.0601 * AAIBEAR_IndexMoMDifferenceL0$



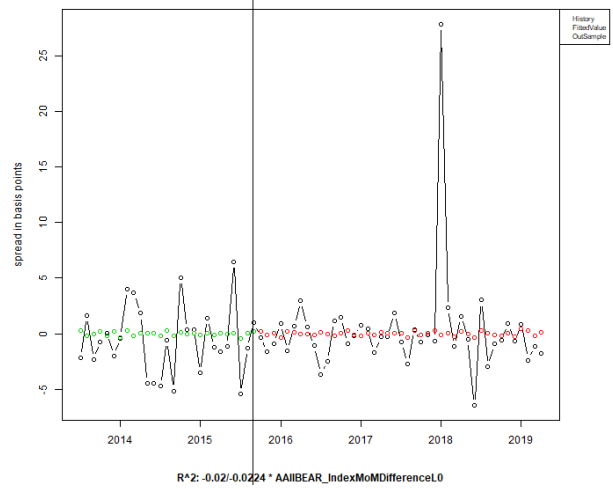
Out-of-Sample testing

CO_grains Jul 2013-Apr 2019



Out-of-Sample testing

CO_grains Jul 2013-Apr 2019



4.14 Two Factor Model for $\text{CO}_{g\text{rains}}$